

S1

Problem 1.12. Calculate the average volume per molecule for an ideal gas at room temperature and atmospheric pressure. Then take the cube root to get an estimate of the average distance between molecules. How does this distance compare to the size of a small molecule like N₂ or H₂O?

We are working with an ideal gas which means that we can use the ideal gas law

$$pV = nRT = nN_A \frac{R}{N_A} T = NkT$$

⇕

$$\frac{V}{N} = \frac{kT}{p} = \frac{1.3806488 \cdot 10^{-23} \frac{J}{K} \cdot 300K}{1 \text{ atm}} \approx 40.877828768813 \text{ nm}^3$$

$$\frac{1.3806488 \cdot 10^{-23} \frac{J}{K} \cdot 300K}{1 \text{ atm}} \approx 4.0877828768813 \cdot 10^{-26} \text{ m}^3$$

The average distance between molecules is

$$\sqrt[3]{40.877828768813 \text{ nm}^3} \approx 3.444789 \text{ nm}$$

A typical bond length is in the order of 100 pm = 0.1 nm. There is a factor 10 difference between the size of molecules and average distance

Problem 1.22. If you poke a hole in a container full of gas, the gas will start leaking out. In this problem you will make a rough estimate of the rate at which gas escapes through a hole. (This process is called **effusion**, at least when the hole is sufficiently small.)

- (a) Consider a small portion (area = A) of the inside wall of a container full of gas. Show that the number of molecules colliding with this surface in a time interval Δt is $PA\Delta t/(2m\bar{v}_x)$, where P is the pressure, m is the average molecular mass, and $\bar{v}_x$ is the average x velocity of those molecules that collide with the wall.
- (b) It's not easy to calculate $\bar{v}_x$, but a good enough approximation is $(\overline{v_x^2})^{1/2}$, where the bar now represents an average over all molecules in the gas. Show that $(\overline{v_x^2})^{1/2} = \sqrt{kT/m}$.
- (c) If we now take away this small part of the wall of the container, the molecules that *would* have collided with it will instead escape through the hole. Assuming that nothing *enters* through the hole, show that the number N of molecules inside the container as a function of time is governed by the differential equation

$$\frac{dN}{dt} = -\frac{A}{2V} \sqrt{\frac{kT}{m}} N.$$

Solve this equation (assuming constant temperature) to obtain a formula of the form $N(t) = N(0)e^{-t/\tau}$, where τ is the “characteristic time” for N (and P) to drop by a factor of e .

- a) The number of molecules colliding with the surface is

Let v_x be the velocity of the molecule towards the surface. The distance towards the surface that the molecules will be able to travel is

$$d = v_x \Delta t$$

The volume where collisions to the wall within a time interval Δt is possible is given by

$$V_{col} = Ad = Av_x \Delta t$$

Let the concentration of molecules per volume with the velocity v_x be given by $n_{v_x} = \frac{N_{v_x}}{V_{con}}$.

Where N_x is the total number of molecules in the container and V_{con} is the volume of the container. Assuming a symmetric distribution of v_x then half of the molecules will fly away from the surface and half will fly towards the surface. Then the number of collisions is given by

$$N_{col,v_x} = \frac{1}{2} \frac{N_{v_x}}{V_{con}} Av_x \Delta t$$

The total number of collisions must be the infinite sum of all N_{v_x} over v_x

$$N_{col} = \int N_{col,v_x} = \frac{1}{2} \frac{A\Delta t}{V_{con}} \int N_{v_x} v_x$$

Dead end. We need to revise. We want pressure in the expression so we can use equation 1.12 to find the average pressure on a wall exerted by a single molecule

$$P_1 = \frac{mv_x^2}{V_{con}}$$

This gives

$$N_{col,v_x} = \frac{1}{2} \frac{P_1 N_{v_x}}{mv_x^2} A v_x \Delta t = \frac{1}{2} \frac{P_1 N_{v_x}}{mv_x^2} A v_x \Delta t$$

Notice that the total pressure given by all molecules with velocity v_x is given by

$$P_{v_x} = P_1 N_{v_x}$$

This gives

$$N_{col,v_x} = \frac{P_{v_x} A \Delta t}{2mv_x}$$

We define the average velocity as

$$\bar{v}_x = \frac{1}{N_{col}} \sum v_x N_{col,v_x}$$

Where N_{col} is the total number of collisions. Thus, summing over the expression gives

$$\sum P_{v_x} = P, \quad \sum v_x N_{col,v_x} = \bar{v}_x N_{col}$$

And

$$N_{col} = \frac{PA\Delta t}{2m\bar{v}_x}$$

Another simpler method would be to calculate the average pressure directly

$$P = \frac{\overline{F_{tot}}}{A} = \frac{M\bar{a}}{A} = \frac{N_{col}m\left(\frac{\Delta v_x}{\Delta t}\right)}{A}$$

The average change in velocity after a collision assuming Δt is small enough to only allow one collision is

$$\overline{\Delta v_x} = \bar{v}_x - (-\bar{v}_x) = 2\bar{v}_x$$

This gives

$$N_{col} = \frac{PA\Delta t}{2m\bar{v}_x}$$

b) We want to show that

$$\sqrt{\overline{v_x^2}} = \sqrt{\frac{kT}{m}}$$

We start from

$$P_1 V = m v_x^2$$

For an ensemble of N number of molecules

$$PV = m v_{x1}^2 + m v_{x2}^2 + \dots = N m \overline{v_x^2}$$

From the ideal gas law

$$PV = NkT$$

Thus

$$m \overline{v_x^2} = kT$$

$\Downarrow$

$$\sqrt{\overline{v_x^2}} = \sqrt{\frac{kT}{m}}$$

c) We start from

$$N_{col} = \frac{PA\Delta t}{2m\overline{v_x}} \Leftrightarrow \frac{N_{col}}{\Delta t} = \frac{PA}{2m\overline{v_x}}$$

Without any loss of generality

$$N_{col} = \Delta N_{col} = N_{col} - 0$$

Now we let $\Delta t \rightarrow 0$ then the number of collisions will also be very small

$$\frac{dN_{col}}{dt} = \frac{PA}{2m\overline{v_x}}$$

The change in collisions is the negative change of the number of atoms in the container

$$\frac{dN}{dt} = -\frac{PA}{2m\overline{v_x}}$$

Using the ideal gas law and the newly derived

$$\sqrt{\overline{v_x^2}} = \sqrt{\frac{kT}{m}}, \quad P = \frac{NkT}{V}$$

We obtain

$$\frac{dN}{dt} = -\frac{\frac{NkT}{V}A}{2m\sqrt{\frac{kT}{m}}} = -\frac{NA}{2V} \frac{kT}{m} \sqrt{\frac{m}{kT}} = -\frac{NA}{2V} \sqrt{\frac{kT}{m}}$$

The solution to the differential equation is of the form

$$N(t) = N(0)e^{-\frac{t}{\tau}}$$

Where $\tau = \frac{2V}{A} \sqrt{\frac{m}{kT}}$

Problem 1.33. An ideal gas is made to undergo the cyclic process shown in Figure 1.10(a). For each of the steps *A*, *B*, and *C*, determine whether each of the following is positive, negative, or zero: (a) the work done on the gas; (b) the change in the energy content of the gas; (c) the heat added to the gas. Then determine the sign of each of these three quantities for the whole cycle. What does this process accomplish?

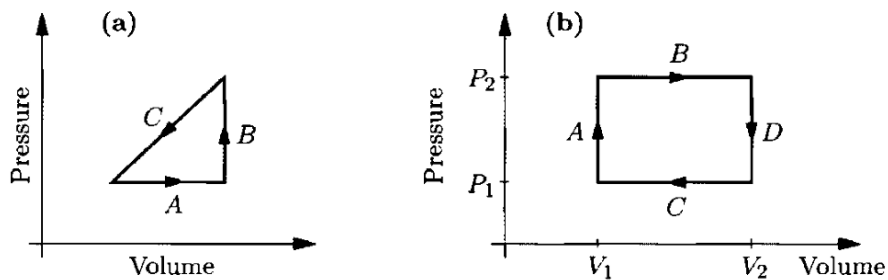


Figure 1.10. *PV* diagrams for Problems 1.33 and 1.34.

The definition here is

$$\Delta U = Q + W$$

- A) Is an isobaric process
- B) Is an isochoric process
- C) Is neither isobar or isochor

The work is determined by

$$W = - \int_{V_1}^{V_2} P dV$$

- A) Does negative work because it is an expansion. Work is done on surroundings
- B) Does zero work because of no change in volume
- C) Does positive work because it is a contraction. Work is done on the system

From MekTerm the internal energy is defined

$$u = \sum_i K_i + \sum_i U_i$$

For an ideal non-interacting gas

$$\sum_i U_i = 0$$

The translational kinetic energy is given from the equipartition theorem

$$u \propto nRT \propto PV$$

$$dU = \left(\frac{\partial U}{\partial P} \right)_V dP + \left(\frac{\partial U}{\partial V} \right)_P dV$$

- A) The energy increases
- B) The energy increases
- C) The energy decreases

The heat is given by

$$Q = u - W$$

- A) W is negative and u is positive so the heat increases
- B) W is zero and u is positive so heat increases
- C) U is negative and W is positive so heat decreases

For the whole cycle ΔU must be zero because we return to the same state.

The area contained within the cycle is positive so overall work, W, must be positive

Since ΔU must be zero then the heat Q must be negative

Problem 1.34. An ideal diatomic gas, in a cylinder with a movable piston, undergoes the rectangular cyclic process shown in Figure 1.10(b). Assume that the temperature is always such that rotational degrees of freedom are active, but vibrational modes are “frozen out.” Also assume that the only type of work done on the gas is quasistatic compression-expansion work.

- (a) For each of the four steps *A* through *D*, compute the work done on the gas, the heat added to the gas, and the change in the energy content of the gas. Express all answers in terms of P_1 , P_2 , V_1 , and V_2 . (Hint: Compute ΔU before Q , using the ideal gas law and the equipartition theorem.)
- (b) Describe in words what is physically being done during each of the four steps; for example, during step *A*, heat is added to the gas (from an external flame or something) while the piston is held fixed.
- (c) Compute the net work done on the gas, the net heat added to the gas, and the net change in the energy of the gas during the entire cycle. Are the results as you expected? Explain briefly.

- a) First we calculate the work done

$$W_A = W_D = - \int_{V_1}^{V_2} P dV = 0$$

$$W_C = - \int_{V_2}^{V_1} P_1 dV = P_1(V_2 - V_1)$$

$$W_B = -P_2(V_2 - V_1)$$

Next we calculate the energy from the equipartition theorem. There are 3 translational modes but only 2 rotational modes for a linear molecule

$$U = \frac{5}{2}NkT = \frac{5}{2}PV$$

Therefore

$$\Delta U_A = \frac{5}{2}V_1(P_2 - P_1)$$

$$\Delta U_B = \frac{5}{2}P_2(V_2 - V_1)$$

$$\Delta U_C = \frac{5}{2}P_1(V_1 - V_2)$$

$$\Delta U_D = \frac{5}{2}V_2(P_1 - P_2)$$

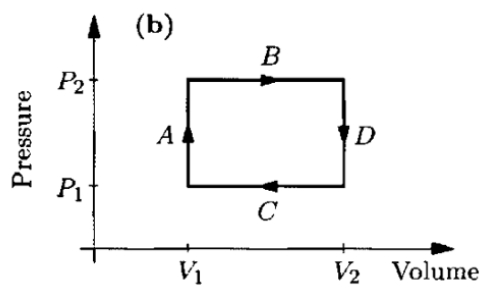
The heat is given by $Q = \Delta U - W$

$$Q_A = \frac{5}{2}V_1(P_2 - P_1)$$

$$Q_B = \frac{5}{2}P_2(V_2 - V_1) + P_2(V_2 - V_1) = \frac{7}{2}P_2(V_2 - V_1)$$

$$Q_C = \frac{5}{2}P_1(V_1 - V_2) - P_1(V_2 - V_1) = \frac{7}{2}P_1(V_1 - V_2)$$

$$Q_D = \frac{5}{2}V_2(P_1 - P_2)$$



- b) In process A the piston is held steady while the system is heated. In process B the system is still heated but the piston is allowed to move instead. In process D the system is cooled at constant volume. In process C the system is cooled further while allowing the piston to move

c) The net work done on the gas is

$$\sum W = P_1(V_2 - V_1) - P_2(V_2 - V_1) = (P_1 - P_2)(V_2 - V_1) < 0$$

The net internal energy must be zero because we return to the same state which means that

$$Q = \Delta U - W = -W > 0$$

Problem 2.12. The natural logarithm function, $\ln$, is defined so that $e^{\ln x} = x$ for any positive number x .

(a) Sketch a graph of the natural logarithm function.

(b) Prove the identities

$$\ln ab = \ln a + \ln b \quad \text{and} \quad \ln a^b = b \ln a.$$

(c) Prove that $\frac{d}{dx} \ln x = \frac{1}{x}$.

(d) Derive the useful approximation

$$\ln(1+x) \approx x,$$

which is valid when $|x| \ll 1$. Use a calculator to check the accuracy of this approximation for $x = 0.1$ and $x = 0.01$.

b) Proving the identities

$$\ln(ab) = \ln(e^{\ln a} e^{\ln b}) = \ln(e^{\ln(a) + \ln(b)}) = \ln(a) + \ln(b)$$

$$\ln(a^b) = \ln((e^{\ln(a)})^b) = \ln(e^{b \ln(a)}) = b \ln(a)$$

c) Proving the identity

$$1 = \frac{dx}{dx} = \frac{de^{\ln(x)}}{dx} = e^{\ln(x)} \frac{d \ln(x)}{dx} = x \frac{d \ln(x)}{dx}$$

$\Downarrow$

$$\frac{d \ln(x)}{dx} = \frac{1}{x}$$

d) Approximating

Using the Taylor series

$$\ln(x+1) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

As $|x| \ll 1$ then

$$\ln(x+1) \approx x$$

Checking for $x = 0.1$ and $x = 0.01$

$$\ln(0.1 + 1) \approx 0.09531018$$

$$\ln(0.01 + 1) \approx 0.009950331$$

Problem A1

Problem A1

This problem deals with combinatorial mathematics, illustrated by card distributions. This type of mathematics is essential for the evaluation of entropy in many physical systems. A deck of cards consists of $N = 52$ different cards (distinguishable). Before a game of cards, the cards are shuffled to obtain a random sequence (distribution).

a) How many distributions are there with N cards? Calculate this number for $N = 52$.

$$52! \approx 8.065818 \cdot 10^{67}$$

b) In statistical physics, the quantity $N!$ plays an important role. A rather precise value is obtained with Stirling's formula for large N :

$$N! \simeq N^N e^{-N} \sqrt{2\pi N}$$

How accurate is this formula for $N = 52$?

$$52^{52} e^{-52} \sqrt{2\pi \cdot 52} \approx 8.052902 \cdot 10^{67}$$

Deviation is

$$\frac{|8.052902 \cdot 10^{67} - 8.065818 \cdot 10^{67}|}{8.065818 \cdot 10^{67}} \approx 0.001601325 = 0.16\%$$

c) In statistical physics, we are often interested in the logarithm of $N!$. Show that Stirling's formula is equivalent to

$$\ln(N!) \simeq N \ln(N) - N,$$

where we ignore terms of order $\ln(N)$. What is the accuracy of this estimate for $N = 52$?

$$\begin{aligned} \ln(N!) &\approx \ln(N^N e^{-N} \sqrt{2\pi N}) \\ &= \ln(N^N) + \ln(e^{-N}) + \ln(\sqrt{2\pi N}) \\ &= N \ln(N) - N + \frac{1}{2} \ln(2\pi N) \\ &\approx N \ln(N) - N \end{aligned}$$

The accuracy is

$$\begin{aligned} \ln(52!) &\approx 156.3608 \\ 52 \ln(52) - 52 &\approx 153.4647 \\ \frac{|153.4647 - 156.3608|}{156.3608} &\approx 1.852191\% \end{aligned}$$

d) A hand in bridge consists of $m = 13$ cards and their sequence is unimportant. The number of different hands is given by the binomial formula

$$\binom{N}{m} = \frac{N!}{m!(N-m)!}$$

Repeat the argument for this formula.

$N!$ is the number of combinations in the entire deck

$\frac{N!}{(N-m)!}$ is the number of combinations that can be drawn from 13 cards

We divide by an extra $m!$ because a set of m cards in different order is also counted

e) How many different hands can be dealt in bridge where $N = 52$ and $m = 13$?

$$\frac{52!}{13! \cdot (52-13)!} \approx 6.350136 \cdot 10^{11}$$

f) Now we wish to divide N cards into three piles, one with m cards, one with l cards, and one with $N - m - l$ cards. Piles with 0 cards are at first not considered. In how many ways can this be done? We now include the possibility that some of the three numbers could be zero. Does this change the result?

Let n and k be natural numbers and let $r_1, \dots, r_k$ be numbers in $\{0, 1, 2, \dots, n\}$ such that

$$n = r_1 + r_2 + \dots + r_k$$

The multinomial coefficient describes the number of ways that an amount of n in k disjunctive sets, $M_1, M_2, \dots, M_k$ can be divided such that

$$\#M_1 = r_1, \#M_2 = r_2, \dots, \#M_k = r_k$$

The multinomial coefficient is

$$\binom{n}{r_1 r_2 \dots r_k} = \frac{n!}{r_1! r_2! \dots r_k!}$$

Therefore the answer is

$$\frac{N!}{l! m! (N-m-l)!}$$

$0! = 1$ so it doesn't change the result

g) Generalize this result to an arbitrary number of piles, M , with n_i cards in the i 'th pile and $\sum_{i=1}^M n_i = N$.

$$\frac{N!}{n_1! n_2! \dots n_M!}$$

h) A game of bridge depends on all four hands. How many different card distributions are there?

$$\frac{52!}{(13!)^4} \approx 5.364474 \cdot 10^{28}$$

Problem A2

Problem A2

Compute the limits $x \rightarrow 0^+$ and $x \rightarrow \infty$ of the following expressions. Note that $x \rightarrow 0^+$ means that x approaches zero through positive values. Note also that in some cases it may be easier to rewrite the expression before taking the limit.

Hint: Remind yourself of L'Hôpital's rule.

a)

$$e^{-\frac{1}{x}}$$

$$\lim_{x \rightarrow 0^+} e^{-\frac{1}{x}} = 0$$

$$\lim_{x \rightarrow \infty} e^{-\frac{1}{x}} = 1$$

b)

$$xe^{-x}$$

$$\lim_{x \rightarrow 0^+} xe^{-x} = 0$$

$$\lim_{x \rightarrow \infty} xe^{-x} = \lim_{x \rightarrow \infty} \frac{x}{e^x} \stackrel{LH}{=} \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$$

c)

$$\frac{e^{-\frac{1}{x}}}{x^2}$$

$$\lim_{x \rightarrow 0^+} \frac{e^{-\frac{1}{x}}}{x^2} = \lim_{x \rightarrow 0^+} \frac{x^{-2}}{e^{\frac{1}{x}}} \stackrel{LH}{=} \lim_{x \rightarrow 0^+} \frac{-2x^{-3}}{-\frac{1}{x^2} e^{\frac{1}{x}}} = \lim_{x \rightarrow 0^+} \frac{2x^{-1}}{e^{\frac{1}{x}}} \stackrel{LH}{=} \lim_{x \rightarrow 0^+} \frac{-2x^{-2}}{-\frac{1}{x^2} e^{\frac{1}{x}}} = \lim_{x \rightarrow 0^+} \frac{2}{e^{\frac{1}{x}}} = 0$$

$$\lim_{x \rightarrow \infty} \frac{e^{-\frac{1}{x}}}{x^2} = 0$$

d)

$$\frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\lim_{x \rightarrow 0^+} \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1 - 1}{1 + 1} = 0$$

$$\lim_{x \rightarrow \infty} \frac{e^x - e^{-x}}{e^x + e^{-x}} = \lim_{x \rightarrow \infty} \frac{e^x}{e^x} = 1$$

e)

$$\frac{\epsilon_1 e^{-\frac{\epsilon_1}{x}} + \epsilon_2 e^{-\frac{\epsilon_2}{x}}}{e^{-\frac{\epsilon_1}{x}} + e^{-\frac{\epsilon_2}{x}}}, \quad \text{where } \epsilon_2 > \epsilon_1,$$

Notice that $e^{-\frac{\epsilon_1}{x}} > e^{-\frac{\epsilon_2}{x}}$ and therefore as $x \rightarrow 0$ then $e^{-\frac{\epsilon_1}{x}} \gg e^{-\frac{\epsilon_2}{x}}$ and the difference in the rate of growth will only exaggerate the difference. Therefore

$$\lim_{x \rightarrow 0^+} \frac{\epsilon_1 e^{-\frac{\epsilon_1}{x}} + \epsilon_2 e^{-\frac{\epsilon_2}{x}}}{e^{-\frac{\epsilon_1}{x}} + e^{-\frac{\epsilon_2}{x}}} = \lim_{x \rightarrow 0^+} \frac{\epsilon_1 e^{-\frac{\epsilon_1}{x}}}{e^{-\frac{\epsilon_1}{x}}} = \epsilon_1$$

$$\lim_{x \rightarrow \infty} \frac{\epsilon_1 e^{-\frac{\epsilon_1}{x}} + \epsilon_2 e^{-\frac{\epsilon_2}{x}}}{e^{-\frac{\epsilon_1}{x}} + e^{-\frac{\epsilon_2}{x}}} = \frac{\epsilon_1 + \epsilon_2}{2}$$

f)

$$\ln(e^x + e^{-x}) - x$$

$$\lim_{x \rightarrow 0^+} \ln(e^x + e^{-x}) - x = \ln(2)$$

$$\lim_{x \rightarrow \infty} \ln(e^x + e^{-x}) - x = \lim_{x \rightarrow \infty} \ln(e^x) - x = \lim_{x \rightarrow \infty} x - x = 0$$

g)

$$\ln(e^x + e^{-x}) + \frac{-xe^x + xe^{-x}}{e^x + e^{-x}}$$

$$\lim_{x \rightarrow 0^+} \ln(e^x + e^{-x}) + \frac{-xe^x + xe^{-x}}{e^x + e^{-x}} = \ln(2) + \lim_{x \rightarrow 0^+} \frac{-x + x}{2} = \ln(2)$$

$$\lim_{x \rightarrow \infty} \ln(e^x + e^{-x}) + \frac{-xe^x + xe^{-x}}{e^x + e^{-x}} = \lim_{x \rightarrow \infty} \ln(e^x) + \frac{-xe^x + xe^{-x}}{e^x}$$

From b) we know that $\lim_{x \rightarrow \infty} xe^{-x} = 0$ which gives

$$\lim_{x \rightarrow \infty} \ln(e^x) + \frac{-xe^x + xe^{-x}}{e^x} = \lim_{x \rightarrow \infty} \ln(e^x) + \frac{-xe^x}{e^x} = \lim_{x \rightarrow \infty} x - x = 0$$

S2

Problem 2.2. Suppose you flip 20 fair coins.

- (a) How many possible outcomes (microstates) are there?
- (b) What is the probability of getting the sequence HTHHTTTHTHHHTHHHHTHT (in exactly that order)?
- (c) What is the probability of getting 12 heads and 8 tails (in any order)?

- a) The number of microstates is $2^{20} = 1048576$
- b) The probability is $\frac{1}{2^{20}} \approx 9.536743 \cdot 10^{-7}$
- c) The probability of getting 12 heads and 8 tails can be found using the binomial distribution

$$\begin{aligned}
 P(X = j) &= \binom{n}{m} p^j (1-p)^{n-j} \\
 &= \frac{20!}{8! (20-8)!} \cdot \left(\frac{1}{2}\right)^8 \cdot \left(1 - \frac{1}{2}\right)^{20-8} \\
 &= \frac{20!}{8! \cdot 12!} \cdot \frac{1}{2^{20}} \approx 0.1201344 = 12\%
 \end{aligned}$$

Problem 2.8. Consider a system of two Einstein solids, A and B , each containing 10 oscillators, sharing a total of 20 units of energy. Assume that the solids are weakly coupled, and that the total energy is fixed.

- (a) How many different *macrostates* are available to this system?
- (b) How many different *microstates* are available to this system?
- (c) Assuming that this system is in thermal equilibrium, what is the probability of finding all the energy in solid A ?
- (d) What is the probability of finding exactly half of the energy in solid A ?
- (e) Under what circumstances would this system exhibit irreversible behavior?

- a) Solid A can have anywhere from 0 energy to 20 energy. Therefore there must be 21 macrostates
- b) The number of microstates can be found with the binomial distribution

$$\Omega(N, q) = \binom{q + N - 1}{q}$$

In this case it doesn't matter if the oscillators are divided by two solids. We don't care about a specific macrostate, and so the question is the same as how many microstates are in an Einstein solid with 20 oscillators with a total energy of 20

$$\Omega(20, 20) = \binom{20 + 20 - 1}{20} = \frac{39!}{20! \cdot 19!} \approx 6.892326 \cdot 10^{10}$$

- c) We now want to calculate the number of microstates in the macrostate where $q_A = 20$

We use the formulas

$$\Omega_{tot} = \Omega_A(N_A, q_A) \Omega_B(N_B, q - q_A)$$

$$\Omega_A(N_A, q_A) = \frac{(q_A + N_A - 1)!}{q_A! (N_A - 1)!}, \quad \Omega_B(N_B, q - q_A) = \frac{(q - q_A + N_B - 1)!}{(q - q_A)! (N_B - 1)!}$$

Finding the microstates in A and B

$$\Omega_A(10, 20) = \frac{(20 + 10 - 1)!}{20! \cdot (10 - 1)!} \approx 1.0015 \cdot 10^7$$

$$\Omega_B(10, 0) = \frac{(0 + 10 - 1)!}{0! \cdot (10 - 1)!} = 1$$

Therefore the probability is

$$\frac{\Omega_A(10, 20)}{\Omega(20, 10)} = \frac{1.0015 \cdot 10^7}{6.892326 \cdot 10^{10}} \approx 1.453065 \cdot 10^{-4}$$

d) We now do the same but with $q_A = 10$

$$\Omega_A(10, 10) = \Omega_B(10, 10) = \frac{(10 + 10 - 1)!}{10! \cdot (10 - 1)!} = \frac{19!}{10! \cdot 9!}$$

Therefore

$$\Omega_{tot} = \Omega_A \Omega_B = \frac{19!}{10! \cdot 9!} \cdot \frac{19!}{10! \cdot 9!} \approx 8.533695 \cdot 10^9$$

And the probability is

$$\frac{\Omega_A(10, 10) \Omega_B(10, 10)}{\Omega(20, 10)} = \frac{8.533695 \cdot 10^9}{6.892326 \cdot 10^{10}} \approx 0.1238144$$

- e) If the initial energy state was anything other than the macrostate $q_A = 10$ then the energy would irreversibly flow from the solid with more energy quanta to the solid with less energy quanta

Problem 2.9. Use a computer to reproduce the table and graph in Figure 2.4: two Einstein solids, each containing three harmonic oscillators, with a total of six units of energy. Then modify the table and graph to show the case where one Einstein solid contains six harmonic oscillators and the other contains four harmonic oscillators (with the total number of energy units still equal to six). Assuming that all microstates are equally likely, what is the most probable macrostate, and what is its probability? What is the least probable macrostate, and what is its probability?

Problem 2.13. Fun with logarithms.

- Simplify the expression $e^{a \ln b}$. (That is, write it in a way that doesn't involve logarithms.)
- Assuming that $b \ll a$, prove that $\ln(a + b) \approx (\ln a) + (b/a)$. (Hint: Factor out the a from the argument of the logarithm, so that you can apply the approximation of part (d) of the previous problem.)

a) $e^{a \ln(b)} = e^{\ln(b^a)} = b^a$

b) $\ln(a + b) = \ln\left(a\left(1 + \frac{b}{a}\right)\right) = \ln(a) + \ln\left(1 + \frac{b}{a}\right)$

For a function $\ln(1 + x)$ where $|x| \ll 1$ then $\ln(1 + x) \approx x$

$$\ln(a) + \ln\left(1 + \frac{b}{a}\right) \approx \ln(a) + \frac{b}{a}$$

Problem 2.14. Write $e^{10^{23}}$ in the form 10^x , for some x .

$$e^{10^{23}} = 10^{\log(e^{10^{23}})} = 10^{10^{23} \cdot \log(e)}$$

$$10^{23} \cdot \log(e) \approx 4.342945 \cdot 10^{22}$$

Type equation here.

Problem 2.18. Use Stirling's approximation to show that the multiplicity of an Einstein solid, for any large values of N and q , is approximately

$$\Omega(N, q) \approx \frac{\left(\frac{q+N}{q}\right)^q \left(\frac{q+N}{N}\right)^N}{\sqrt{2\pi q(q+N)/N}}.$$

The square root in the denominator is merely large, and can often be neglected. However, it is needed in Problem 2.22. (Hint: First show that $\Omega = \frac{N}{q+N} \frac{(q+N)!}{q! N!}$. Do not neglect the $\sqrt{2\pi N}$ in Stirling's approximation.)

Remember when I said that you guys should be familiar with Stirling's approximation? Good luck

$$N! = N^N e^{-N} \sqrt{2\pi N}, \quad N \gg 1$$

$$\ln(N!) = N \ln(N) - N$$

Let the math do the talking

$$\Omega(N, q) = \frac{(q+N-1)!}{q! (N-1)!} = \frac{N}{q+N} \cdot \frac{(q+N)!}{q! N!}$$

We can do this because

$$(k-1)! = \frac{k!}{k}$$

Then we use Stirling's approximation

$$\Omega(N, q) \approx \frac{N}{q+N} \cdot \frac{(q+N)^{q+N} e^{-(q+N)} \sqrt{2\pi(q+N)}}{q^q e^{-q} \sqrt{2\pi q} \cdot N^N e^{-N} \sqrt{2\pi N}}$$

$$\begin{aligned}
&= \frac{N}{q+N} \cdot \frac{(q+N)^q (q+N)^N e^{-q} e^{-N} \sqrt{2\pi(q+N)}}{q^q e^{-q} \sqrt{2\pi q} \cdot N^N e^{-N} \sqrt{2\pi N}} \\
&= \frac{N}{q+N} \cdot \frac{(q+N)^q (q+N)^N \sqrt{(q+N)}}{q^q \sqrt{2\pi q} \cdot N^N \sqrt{N}} \\
&= \frac{N}{q+N} \cdot \frac{\sqrt{(q+N)}}{\sqrt{N}} \cdot \frac{(q+N)^q}{q^q} \cdot \frac{(q+N)^N}{N^N} \cdot \frac{1}{\sqrt{2\pi q}} \\
&= \sqrt{\frac{N}{2\pi q(q+N)}} \cdot \left(\frac{q+N}{q}\right)^q \cdot \left(\frac{q+N}{N}\right)^N
\end{aligned}$$

DONE

Problem 2.22. This problem gives an alternative approach to estimating the width of the peak of the multiplicity function for a system of two large Einstein solids.

- Consider two identical Einstein solids, each with N oscillators, in thermal contact with each other. Suppose that the total number of energy units in the combined system is exactly $2N$. How many different macrostates (that is, possible values for the total energy in the first solid) are there for this combined system?
- Use the result of Problem 2.18 to find an approximate expression for the total number of microstates for the combined system. (Hint: Treat the combined system as a single Einstein solid. Do *not* throw away factors of “large” numbers, since you will eventually be dividing two “very large” numbers that are nearly equal. *Answer:* $2^{4N}/\sqrt{8\pi N}$.)
- The most likely macrostate for this system is (of course) the one in which the energy is shared equally between the two solids. Use the result of Problem 2.18 to find an approximate expression for the multiplicity of this macrostate. (*Answer:* $2^{4N}/(4\pi N)$.)
- You can get a rough idea of the “sharpness” of the multiplicity function by comparing your answers to parts (b) and (c). Part (c) tells you the height of the peak, while part (b) tells you the total area under the entire graph. As a very crude approximation, pretend that the peak’s shape is rectangular. In this case, how wide would it be? Out of all the macrostates, what fraction have reasonably large probabilities? Evaluate this fraction numerically for the case $N = 10^{23}$.

- Analogous to problem 2.8A. The number of macrostates is $2N+1$
- Again analogous to problem 2.8B. We essentially have to calculate

$$\Omega(2N, 2N) \approx \sqrt{\frac{2N}{2\pi 2N(2N+2N)}} \cdot \left(\frac{2N+2N}{2N}\right)^{2N} \cdot \left(\frac{2N+2N}{2N}\right)^{2N}$$

$$= \sqrt{\frac{1}{8\pi N}} \cdot 2^{2N} \cdot 2^{2N} = \frac{2^{4N}}{\sqrt{8\pi N}}$$

c) We calculate the microstates for $\Omega_{tot} = \Omega_A(N, N)\Omega_B(N, N)$

$$\begin{aligned}\Omega_A(N, N) &\approx \sqrt{\frac{N}{2\pi N(N+N)}} \cdot \left(\frac{N+N}{N}\right)^N \cdot \left(\frac{N+N}{N}\right)^N \\ &= \sqrt{\frac{1}{4\pi N}} \cdot 2^N \cdot 2^N \\ &= \frac{2^{2N}}{\sqrt{4\pi N}}\end{aligned}$$

Therefore

$$\Omega_{tot} = \left(\frac{2^{2N}}{\sqrt{4\pi N}}\right)^2 = \frac{2^{4N}}{4\pi N}$$

d) The area of our “rectangle” must be $\Omega(2N, 2N)$ and the height of our rectangle must be Ω_{tot}

Therefore the width of our rectangle must be

$$\frac{\Omega(2N, 2N)}{\Omega_{tot}} = \frac{\frac{2^{4N}}{\sqrt{8\pi N}}}{\frac{2^{4N}}{4\pi N}} = \frac{4\pi N}{\sqrt{8\pi N}} = \sqrt{\frac{16\pi^2 N^2}{8\pi N}} = \sqrt{2\pi N}$$

This width compared to our total number of macrostates is

$$\frac{\sqrt{2\pi N}}{2N + 1}$$

For $N = 10^{23}$ this is

$$\frac{\sqrt{2\pi \cdot 10^{23}}}{2 \cdot 10^{23} + 1} \approx 3.963327 \cdot 10^{-12}$$

Technically by defining the height of our rectangle to be the peak, we will underestimate the number of macrostates with a significantly large distribution. But not by a factor of 10^{12} , so the peak is relatively sharp

Problem 2.24. For a single *large* two-state paramagnet, the multiplicity function is very sharply peaked about $N_{\uparrow} = N/2$.

- Use Stirling's approximation to estimate the height of the peak in the multiplicity function.
- Use the methods of this section to derive a formula for the multiplicity function in the vicinity of the peak, in terms of $x \equiv N_{\uparrow} - (N/2)$. Check that your formula agrees with your answer to part (a) when $x = 0$.
- How wide is the peak in the multiplicity function?
- Suppose you flip 1,000,000 coins. Would you be surprised to obtain 501,000 heads and 499,000 tails? Would you be surprised to obtain 510,000 heads and 490,000 tails? Explain.

a) The multiplicity is given by

$$\Omega(N_{\uparrow}) = \frac{N!}{N_{\uparrow}!(N - N_{\uparrow})!}$$

At $N_{\uparrow} = \frac{N}{2}$ then

$$\Omega(N_{\uparrow}) = \frac{N!}{\left(\frac{N}{2}\right)!\left(N - \frac{N}{2}\right)!} = \frac{N!}{\left(\frac{N}{2}\right)!\left(\frac{N}{2}\right)!}$$

Now we use Stirling's approximation

$$N! \approx N^N e^{-N} \sqrt{2\pi N}$$

Then

$$\Omega(N_{\uparrow}) \approx \frac{N^N e^{-N} \sqrt{2\pi N}}{\left(\left(\frac{N}{2}\right)^{\frac{N}{2}} e^{-\frac{N}{2}} \sqrt{\frac{2\pi N}{2}}\right)^2} = \frac{N^N e^{-N} \sqrt{2\pi N}}{\left(\frac{N}{2}\right)^N e^{-N} \pi N} = \frac{N^N}{N^N} \cdot 2^N \sqrt{\frac{2}{\pi N}} = 2^N \sqrt{\frac{2}{\pi N}}$$

This is the height of the peak

b) Formula for the multiplicity function

We start from

$$\Omega(N_{\uparrow}) = \frac{N!}{N_{\uparrow}!(N - N_{\uparrow})!}$$

We define

$$x = N_{\uparrow} - \frac{N}{2} \Leftrightarrow N_{\uparrow} = x + \frac{N}{2}$$

Therefore we substitute to obtain

$$\Omega(x) = \frac{N!}{\left(x + \frac{N}{2}\right)! \left(N - \left(x + \frac{N}{2}\right)\right)!} = \frac{N!}{\left(x + \frac{N}{2}\right)! \left(\frac{N}{2} - x\right)!}$$

Using Stirling's approximation

$$\begin{aligned} \Omega(x) &= \frac{N^N e^{-N} \sqrt{2\pi N}}{\left(x + \frac{N}{2}\right)^{x+\frac{N}{2}} e^{-(x+\frac{N}{2})} \sqrt{2\pi \left(x + \frac{N}{2}\right)} \left(\frac{N}{2} - x\right)^{\frac{N}{2}-x} e^{-(\frac{N}{2}-x)} \sqrt{2\pi \left(\frac{N}{2} - x\right)}} \\ &= \frac{N^N e^{-N}}{e^{-(x+\frac{N}{2})-(\frac{N}{2}-x)} \left(x + \frac{N}{2}\right)^{x+\frac{N}{2}} \left(\frac{N}{2} - x\right)^{\frac{N}{2}-x}} \sqrt{\frac{2\pi N}{2\pi \left(x + \frac{N}{2}\right) 2\pi \left(\frac{N}{2} - x\right)}} \\ &= \frac{N^N}{\left(x + \frac{N}{2}\right)^{x+\frac{N}{2}} \left(\frac{N}{2} - x\right)^{\frac{N}{2}-x}} \sqrt{\frac{N}{2\pi \left(\frac{N}{2} + x\right) \left(\frac{N}{2} - x\right)}} \\ &= \frac{N^N}{\left(x + \frac{N}{2}\right)^{\frac{N}{2}} \left(\frac{N}{2} - x\right)^{\frac{N}{2}} \left(x + \frac{N}{2}\right)^x \left(\frac{N}{2} - x\right)^{-x}} \sqrt{\frac{N}{2\pi \left(\frac{N}{2} + x\right) \left(\frac{N}{2} - x\right)}} \\ &= \frac{N^N}{\left[\left(x + \frac{N}{2}\right) \left(\frac{N}{2} - x\right)\right]^{\frac{N}{2}} \left(x + \frac{N}{2}\right)^x \left(\frac{N}{2} - x\right)^{-x}} \sqrt{\frac{N}{2\pi \left(\frac{N}{2} + x\right) \left(\frac{N}{2} - x\right)}} \end{aligned}$$

Now we do something analogous to equation 2.26 and define the function

$$\begin{aligned} f(x) &= \left[\left(x + \frac{N}{2}\right) \left(\frac{N}{2} - x\right)\right]^{\frac{N}{2}} \left(x + \frac{N}{2}\right)^x \left(\frac{N}{2} - x\right)^{-x} \\ &= \left[\left(\frac{N}{2}\right)^2 - x^2\right]^{\frac{N}{2}} \left(x + \frac{N}{2}\right)^x \left(\frac{N}{2} - x\right)^{-x} \end{aligned}$$

We take the logarithm in hopes of simplifying

$$\begin{aligned} \ln(f) &= \ln\left(\left[\left(\frac{N}{2}\right)^2 - x^2\right]^{\frac{N}{2}} \left(x + \frac{N}{2}\right)^x \left(\frac{N}{2} - x\right)^{-x}\right) \\ &= \ln\left(\left[\left(\frac{N}{2}\right)^2 - x^2\right]^{\frac{N}{2}}\right) + \ln\left(\left(x + \frac{N}{2}\right)^x\right) + \ln\left(\left(\frac{N}{2} - x\right)^{-x}\right) \\ &= \frac{N}{2} \ln\left(\frac{N^2}{4} - x^2\right) + x \ln\left(\frac{N}{2} + x\right) - x \ln\left(\frac{N}{2} - x\right) \end{aligned}$$

We can use the approximation $\ln(1+x) \approx x$ for $|x| \ll 1$ this gives

$$\ln\left(\frac{N^2}{4} - x^2\right) = \ln\left(\frac{N^2}{4} \left(1 - \frac{4x^2}{N^2}\right)\right) = \ln\left(\frac{N^2}{4}\right) + \ln\left(1 - \frac{4x^2}{N^2}\right) \approx \ln\left(\frac{N^2}{4}\right) - \frac{4x^2}{N^2}$$

$$\ln\left(\frac{N}{2} \pm x\right) = \ln\left(\frac{N}{2}\left(1 \pm \frac{2x}{N}\right)\right) = \ln\left(\frac{N}{2}\right) + \ln\left(1 \pm \frac{2x}{N}\right) \approx \ln\left(\frac{N}{2}\right) \pm \frac{2x}{N}$$

These approximations are valid only in the vicinity of the peak $N_1 = \frac{N}{2} + x$ and for a very large paramagnet. Using the approximations we find

$$\begin{aligned}\ln(f) &\approx \frac{N}{2} \left(\ln\left(\frac{N^2}{4}\right) - \frac{4x^2}{N^2} \right) + x \left(\ln\left(\frac{N}{2}\right) + \frac{2x}{N} \right) - x \left(\ln\left(\frac{N}{2}\right) - \frac{2x}{N} \right) \\ &= \frac{N}{2} \ln\left(\frac{N^2}{4}\right) - \frac{2x^2}{N} + x \ln\left(\frac{N}{2}\right) + \frac{2x^2}{N} - x \ln\left(\frac{N}{2}\right) + \frac{2x^2}{N} \\ &= \frac{N}{2} \ln\left(\frac{N^2}{4}\right) + \frac{2x^2}{N} \\ &= N \ln\left(\frac{N}{2}\right) + \frac{2x^2}{N}\end{aligned}$$

Therefore we have found that

$$f \approx e^{N \ln\left(\frac{N}{2}\right) + \frac{2x^2}{N}} = e^{N \ln\left(\frac{N}{2}\right)} e^{\frac{2x^2}{N}} = \left(\frac{N}{2}\right)^N e^{\frac{2x^2}{N}}$$

We can therefore simplify

$$\begin{aligned}\Omega(x) &= \frac{N^N}{\left(\frac{N}{2}\right)^N e^{\frac{2x^2}{N}}} \sqrt{\frac{N}{2\pi\left(\frac{N}{2} + x\right)\left(\frac{N}{2} - x\right)}} \\ &= 2^N e^{-\frac{2x^2}{N}} \sqrt{\frac{N}{2\pi\left(\frac{N}{2} + x\right)\left(\frac{N}{2} - x\right)}} \\ &= 2^N e^{-\frac{2x^2}{N}} \cdot \sqrt{\frac{N}{2\pi\left(\frac{N^2}{4} - x^2\right)}} \\ &= 2^N e^{-\frac{2x^2}{N}} \cdot \sqrt{\frac{N}{2\pi\frac{N^2}{4}\left(1 - \frac{4x^2}{N^2}\right)}}\end{aligned}$$

We can again use an approximation that since $N \gg x$ then $\left(1 - \frac{4x^2}{N^2}\right) \approx 1$ and we get

$$\Omega(x) \approx 2^N e^{-\frac{2x^2}{N}} \sqrt{\frac{N}{2\pi\frac{N^2}{4}}} = 2^N \sqrt{\frac{2}{\pi N}} e^{-\frac{2x^2}{N}}$$

Compared to the result from a) $\Omega(N_1) = 2^N \sqrt{\frac{2}{\pi N}}$ we see that if $x = 0$ we get the same result.

Some other methods may yield other factors of the exponent as well as additional factors. But as long as the math is correct and approximations reasonable they should still produce the correct result at $x = 0$. If you want more details then let me know - David

c) We have the function

$$\Omega(x) = 2^N \sqrt{\frac{2}{\pi N}} e^{-\frac{2x^2}{N}}$$

The maximum is

$$\Omega_{max} = 2^N \sqrt{\frac{2}{\pi N}}$$

The width can be calculated by solving

$$\begin{aligned} \Omega(x) &= 2^N \sqrt{\frac{2}{\pi N}} e^{-\frac{2x^2}{N}} = \frac{2^N \sqrt{\frac{2}{\pi N}}}{e} \\ &\Downarrow \\ e^{-\frac{2x^2}{N}} &= e^{-1} \\ &\Downarrow \\ \frac{2x^2}{N} &= 1 \\ &\Downarrow \\ x &= \sqrt{\frac{N}{2}} \end{aligned}$$

d) The distance from the maximum where the amplitude drops to 1/e of its maximum in the case of $N = 10^6$ is

$$x = \sqrt{\frac{10^6}{2}} \approx 707.1068$$

Therefore the range of most probable values are within

$$500\,000 \pm 707$$

We can also try to calculate the probability. The total number of microstates in the system is 2^N which means that the probability must be

$$\frac{\Omega(x)}{2^N} = \sqrt{\frac{2}{\pi N}} e^{-\frac{2x^2}{N}}$$

The probability of getting 501 000 is

$$\sqrt{\frac{2}{\pi \cdot 10^6}} e^{-\frac{2 \cdot 1000^2}{10^6}} \approx 1.079819 \cdot 10^{-4}$$

The probability of getting 510 000 is

$$\sqrt{\frac{2}{\pi \cdot 10^6}} e^{-\frac{2 \cdot 10000^2}{10^6}} \approx 1.10419 \cdot 10^{-90}$$

Problem B1

We consider two Einstein solids (A and B), each having N oscillators. The total energy of the two solids is q in harmonic oscillator units, and the multiplicity function of the combined system is

$$\Omega(q_A) = \frac{(q_A + N - 1)!}{q_A!(N - 1)!} \cdot \frac{(q - q_A + N - 1)!}{(q - q_A)!(N - 1)!},$$

where q_A is the number of energy units in Einstein solid A.

Question a:

Show that the average value of the number of energy units in solid A is

$$\overline{q_A} = \frac{\sum_{q_A=0}^q q_A \Omega(q_A)}{\sum_{q_A=0}^q \Omega(q_A)} = \frac{q}{2}. \quad (1)$$

Argue for both the first and the second equality sign.

Hint for the second equality sign: Write $q_A = \frac{q}{2} + x$, reexpress the sums in terms of x , and use symmetry arguments.

The first equality sign is simply the definition for mean value. The denominator is the total number of microstates. The numerator is the possible energies times the number of microstates that can have those energies.

The second equality sign can be derived

$$\frac{\sum_{q_A=0}^q q_A \Omega(q_A)}{\sum_{q_A=0}^q \Omega(q_A)} = \frac{\sum_{x=-\frac{q}{2}}^{\frac{q}{2}} \left(\frac{q}{2} + x\right) \Omega\left(\frac{q}{2} + x\right)}{\sum_{x=-\frac{q}{2}}^{\frac{q}{2}} \Omega\left(\frac{q}{2} + x\right)} = \frac{\sum_{x=-\frac{q}{2}}^{\frac{q}{2}} \frac{q}{2} \Omega\left(\frac{q}{2} + x\right) + x \cdot \Omega\left(\frac{q}{2} + x\right)}{\sum_{x=-\frac{q}{2}}^{\frac{q}{2}} \Omega\left(\frac{q}{2} + x\right)}$$

Both solids have the same number of oscillators which means that the $\Omega_A(k) = \Omega_B(k)$.

Furthermore we know that Ω_A and Ω_B must be symmetric around $\frac{q}{2}$ because $\Omega_A(q_A) = \Omega_B(q - q_A)$. Therefore $\Omega_{tot} = \Omega_A \Omega_B$ must also be symmetric around $\frac{q}{2}$

Therefore all the terms from: $x \cdot \Omega\left(\frac{q}{2} + x\right)$ in the sum will cancel out with each other because we sum from $x = -\frac{q}{2}$ to $x = \frac{q}{2}$. Thus:

$$\frac{\sum_{x=-\frac{q}{2}}^{\frac{q}{2}} \frac{q}{2} \Omega\left(\frac{q}{2} + x\right) + x \cdot \Omega\left(\frac{q}{2} + x\right)}{\sum_{x=-\frac{q}{2}}^{\frac{q}{2}} \Omega\left(\frac{q}{2} + x\right)} = \frac{q \sum_{x=-\frac{q}{2}}^{\frac{q}{2}} \Omega\left(\frac{q}{2} + x\right)}{2 \sum_{x=-\frac{q}{2}}^{\frac{q}{2}} \Omega\left(\frac{q}{2} + x\right)} = \frac{q}{2}$$

Question b:

Show that in this approximation

$$\overline{q_A} = \frac{\int_{-\infty}^{\infty} \left(\frac{q}{2} + x\right) \Omega dx}{\int_{-\infty}^{\infty} \Omega dx} = \frac{q}{2}. \quad (2)$$

Argue for both the first and the second equality sign.

The first equality sign is a result of letting q becoming very large in the following expression

$$\frac{\sum_{x=-\frac{q}{2}}^{\frac{q}{2}} \left(\frac{q}{2} + x\right) \Omega \left(\frac{q}{2} + x\right)}{\sum_{x=-\frac{q}{2}}^{\frac{q}{2}} \Omega \left(\frac{q}{2} + x\right)}$$

The steps in the sum becomes small enough that the sum becomes an integral and it would also be reasonable to integrate to infinity because the multiplicity goes to zero at infinity

Equation 2.27 is given by

$$\Omega \approx \left(\frac{e}{N}\right)^{2N} e^{N \ln\left(\frac{q^2}{4}\right)} e^{-\frac{4Nx^2}{q^2}} = \Omega_{max} e^{-\frac{4Nx^2}{q^2}}$$

Thus:

$$\begin{aligned} \frac{\int_{-\infty}^{\infty} \left(\frac{q}{2} + x\right) \Omega dx}{\int_{-\infty}^{\infty} \Omega dx} &= \frac{\int_{-\infty}^{\infty} \left(\frac{q}{2} + x\right) \Omega_{max} e^{-\frac{4Nx^2}{q^2}} dx}{\int_{-\infty}^{\infty} \Omega_{max} e^{-\frac{4Nx^2}{q^2}} dx} = \frac{\int_{-\infty}^{\infty} \left(\frac{q}{2} + x\right) e^{-\frac{4Nx^2}{q^2}} dx}{\int_{-\infty}^{\infty} e^{-\frac{4Nx^2}{q^2}} dx} \\ &= \frac{\int_{-\infty}^{\infty} \frac{q}{2} e^{-\frac{4Nx^2}{q^2}} + x e^{-\frac{4Nx^2}{q^2}} dx}{\int_{-\infty}^{\infty} e^{-\frac{4Nx^2}{q^2}} dx} \end{aligned}$$

It can be seen that $x e^{-\frac{4Nx^2}{q^2}}$ is an uneven function which means that the integral of this term will be zero. Thus

$$\frac{\int_{-\infty}^{\infty} \frac{q}{2} e^{-\frac{4Nx^2}{q^2}} + x e^{-\frac{4Nx^2}{q^2}} dx}{\int_{-\infty}^{\infty} e^{-\frac{4Nx^2}{q^2}} dx} = \frac{\int_{-\infty}^{\infty} \frac{q}{2} e^{-\frac{4Nx^2}{q^2}} dx}{\int_{-\infty}^{\infty} e^{-\frac{4Nx^2}{q^2}} dx} = \frac{q}{2}$$

Question c:

A useful measure for the fluctuations away from the mean value is the standard deviation σ given by

$$\sigma^2 \equiv \overline{(q_A - \overline{q_A})^2} = \overline{\left(q_A - \frac{q}{2}\right)^2}.$$

Show that

$$\sigma = \frac{q}{2^{3/2}\sqrt{N}}. \quad (3)$$

The deviation away from the mean value must be given by the Gaussian function of the multiplicity because it describes the distribution of q_A . Generally, it applies for Gaussian functions that

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Our multiplicity is also a Gaussian function

$$\Omega \approx \left(\frac{e}{N}\right)^{2N} e^{N \ln\left(\frac{q^2}{4}\right)} e^{-\frac{4Nx^2}{q^2}}$$

Therefore it can be read directly from the function that

$$\frac{1}{2\sigma^2} = \frac{4N}{q^2}$$

$$\Updownarrow$$

$$\sigma^2 = \frac{q^2}{8N}$$

$$\Updownarrow$$

$$\sigma = \frac{q}{2^{3/2}\sqrt{N}}$$

Question d:

Comment on the size of the standard deviation σ of q_A compared to $\overline{q_A}$.

The standard deviation is much smaller than the mean value when N is large while the mean value stays constant at $\frac{q}{2}$

S3

Problem 2.17. Use the methods of this section to derive a formula, similar to equation 2.21, for the multiplicity of an Einstein solid in the “low-temperature” limit, $q \ll N$.

We now let $q \ll N$ where we start from multiplicity of an Einstein solid

$$\Omega(N, q) = \frac{(q + N - 1)!}{q! (N - 1)!}$$

Let us assume that we have a large Einstein solid such that $N \gg 1$ then we can write the same assumption as in the book (Eq 2.17)

$$\Omega(N, q) \approx \frac{(q + N)!}{q! N!}$$

Notice that in the book the derivation from equation 2.17 to 2.21 they found that

The last term becomes negligible compared to the others in the limit $q \gg N$. Exponentiating the first two terms gives

$$\Omega(N, q) \approx e^{N \ln(q/N)} e^N = \left(\frac{eq}{N}\right)^N \quad (\text{when } q \gg N). \quad (2.21)$$

But notice from the equation that we start from $\Omega(N, q) = \Omega(q, N) = \frac{(q+N)!}{q! N!}$. What I am trying to say is that the formula stays the same if we switch between q and N . Therefore, if we in 2.21 write N wherever q is written and similarly write q wherever N is written then we get

$$\Omega(N, q) = \left(\frac{eN}{q}\right)^q, \quad q \ll N$$

No calculations needed :)

Problem 2.30. Consider again the system of two large, identical Einstein solids treated in Problem 2.22.

- (a) For the case $N = 10^{23}$, compute the entropy of this system (in terms of Boltzmann's constant), assuming that *all* of the microstates are allowed. (This is the system's entropy over long time scales.)
- (b) Compute the entropy again, assuming that the system is in its most likely macrostate. (This is the system's entropy over short time scales, except when there is a large and unlikely fluctuation away from the most likely macrostate.)
- (c) Is the issue of time scales really relevant to the entropy of this system?
- (d) Suppose that, at a moment when the system is near its most likely macrostate, you suddenly insert a partition between the solids so that they can no longer exchange energy. Now, even over long time scales, the entropy is given by your answer to part (b). Since this number is less than your answer to part (a), you have, in a sense, caused a violation of the second law of thermodynamics. Is this violation significant? Should we lose any sleep over it?

a) From 2.22 we found that

$$\Omega_{tot} = \frac{2^{4N}}{\sqrt{8\pi N}}$$

The entropy is given by

$$\frac{S}{k} = \ln\left(\frac{2^{4N}}{\sqrt{8\pi N}}\right) = 4N \ln(2) - \frac{1}{2} \ln(8\pi N)$$

$$\frac{S}{k} = 4 \cdot 10^{23} \cdot \ln(2) - \frac{1}{2} \ln(8\pi \cdot 10^{23}) = 2.77258872223978123766864 \cdot 10^{23}$$

$$S = 1.3806 \cdot 10^{-23} \frac{J}{K} \cdot 2.77 \cdot 10^{23} = 3.82 \frac{J}{K}$$

b) From 2.22 we found that

$$\Omega_{tot} = \left(\frac{2^{2N}}{\sqrt{4\pi N}}\right)^2 = \frac{2^{4N}}{4\pi N}$$

The entropy is given by

$$\frac{S}{k} = \ln\left(\frac{2^{4N}}{4\pi N}\right) = 4N \ln(2) - \ln(4\pi N)$$

$$\frac{S}{k} = 4 \cdot 10^{23} \cdot \ln(2) - \ln(4\pi \cdot 10^{23}) = 2.77258872223978123766837 \cdot 10^{23}$$

$$S = 1.3806 \cdot 10^{-23} \frac{J}{K} \cdot 2.77 \cdot 10^{23} = 3.82 \frac{J}{K}$$

c) The answer is approximately the same!!! So clearly the timescale does not matter!

- d) Again the answer is nearly the same. Just count how many decimal places we have to go before we see any numerical difference. The violation is therefore not significant

Problem 2.34. Show that during the quasistatic isothermal expansion of a monatomic ideal gas, the change in entropy is related to the heat input Q by the simple formula

$$\Delta S = \frac{Q}{T}.$$

In the following chapter I'll prove that this formula is valid for *any* quasistatic process. Show, however, that it is *not* valid for the free expansion process described above.

From the book we have this equation and note what is written about it

$$\Delta S = Nk \ln \frac{V_f}{V_i} \quad (U, N \text{ fixed}). \quad (2.51)$$

This formula applies, for instance, to the quasistatic isothermal expansion considered in Section 1.5, where the gas pushes on a piston, doing mechanical work, while we simultaneously supply heat from outside to keep the gas at constant temperature. In this case we can think of the entropy increase as being caused by the heat

The isothermal expansion referred to in section 1.5 is

$$\begin{aligned} W &= - \int_{V_i}^{V_f} P dV = -NkT \int_{V_i}^{V_f} \frac{1}{V} dV \\ &= -NkT (\ln V_f - \ln V_i) = NkT \ln \frac{V_i}{V_f}. \end{aligned} \quad (1.30)$$

We see that

$$W = NkT \ln \left(\frac{V_i}{V_f} \right) = -NkT \ln \left(\frac{V_f}{V_i} \right) = -T\Delta S$$

From the first law of thermodynamics

$$\Delta U = Q + W$$

Since we have an isothermal process then $\Delta U = 0$ which comes from the equipartition theorem $U = Nf \frac{1}{2} kT$. Therefore

$$Q + W = Q - T\Delta S = 0$$

$$\Updownarrow$$

$$\Delta S = \frac{Q}{T}$$

Problem 3.5. Starting with the result of Problem 2.17, find a formula for the temperature of an Einstein solid in the limit $q \ll N$. Solve for the energy as a function of temperature to obtain $U = N\epsilon e^{-\epsilon/kT}$ (where ϵ is the size of an energy unit).

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The answer to 2.17 is

$$\Omega(N, q) = \left(\frac{eN}{q}\right)^q$$

We can calculate the entropy with the formula

$$\begin{aligned} S &= k \ln(\Omega) \\ &= qk \ln\left(\frac{eN}{q}\right) \end{aligned}$$

It applies that $q = \frac{U}{\hbar\omega}$. This is just Plancks relation

$$S = \frac{U}{\hbar\omega} k \ln\left(\frac{eN\hbar\omega}{U}\right) = \frac{Uk}{\hbar\omega} \ln(eN\hbar\omega) - \frac{Uk}{\hbar\omega} \ln(U)$$

We can calculate the temperature by taking the derivative

$$\left(\frac{\partial S}{\partial U}\right)_{V,N} = \frac{1}{T}$$

We can now take the derivative of our expression of S

$$\begin{aligned} \left(\frac{\partial S}{\partial U}\right)_{V,N} &= \frac{k}{\hbar\omega} \ln(eN\hbar\omega) - \frac{k}{\hbar\omega} \ln(U) - \frac{k}{\hbar\omega} \\ &= \frac{k}{\hbar\omega} \ln(N\hbar\omega) - \frac{k}{\hbar\omega} \ln(U) - \frac{k}{\hbar\omega} + \frac{k}{\hbar\omega} \ln(e) \\ &= \frac{k}{\hbar\omega} \ln\left(\frac{N\hbar\omega}{U}\right) \end{aligned}$$

We not have the relation where we can isolate the energy

$$\begin{aligned} \frac{k}{\hbar\omega} \ln\left(\frac{N\hbar\omega}{U}\right) &= \frac{1}{T} \\ \Downarrow \\ \ln\left(\frac{N\hbar\omega}{U}\right) &= \frac{\hbar\omega}{kT} \\ \Downarrow \\ \frac{N\hbar\omega}{U} &= e^{\frac{\hbar\omega}{kT}} \\ \Downarrow \\ U &= N\hbar\omega e^{-\frac{\hbar\omega}{kT}} \end{aligned}$$

In my case it appears I have defined $\epsilon = \hbar\omega$. But this is what ϵ physically is. You will also see this later on

Problem 3.8. Starting with the result of Problem 3.5, calculate the heat capacity of an Einstein solid in the low-temperature limit. Sketch the predicted heat capacity as a function of temperature. (Note: Measurements of heat capacities of actual solids at low temperatures do not confirm the prediction that you will make in this problem. A more accurate model of solids at low temperatures is presented in Section 7.5.)

The heat capacity is defined as

$$C_V = \left(\frac{\partial U}{\partial T} \right)_{N,V}$$

We had already established our expression

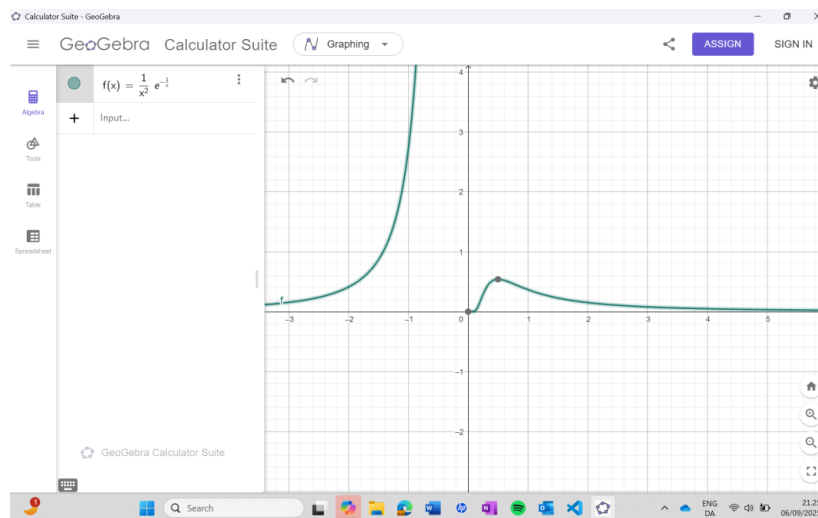
$$U = N\hbar\omega e^{-\frac{\hbar\omega}{kT}}$$

Let us take the derivative with respect to the temperature using the chain rule

$$\begin{aligned} C_V &= N\hbar\omega e^{-\frac{\hbar\omega}{kT}} \frac{\partial}{\partial T} \left(-\frac{\hbar\omega}{kT} \right) \\ &= N\hbar\omega e^{-\frac{\hbar\omega}{kT}} \frac{\hbar\omega}{kT^2} \\ &= \frac{N(\hbar\omega)^2}{kT^2} e^{-\frac{\hbar\omega}{kT}} \end{aligned}$$

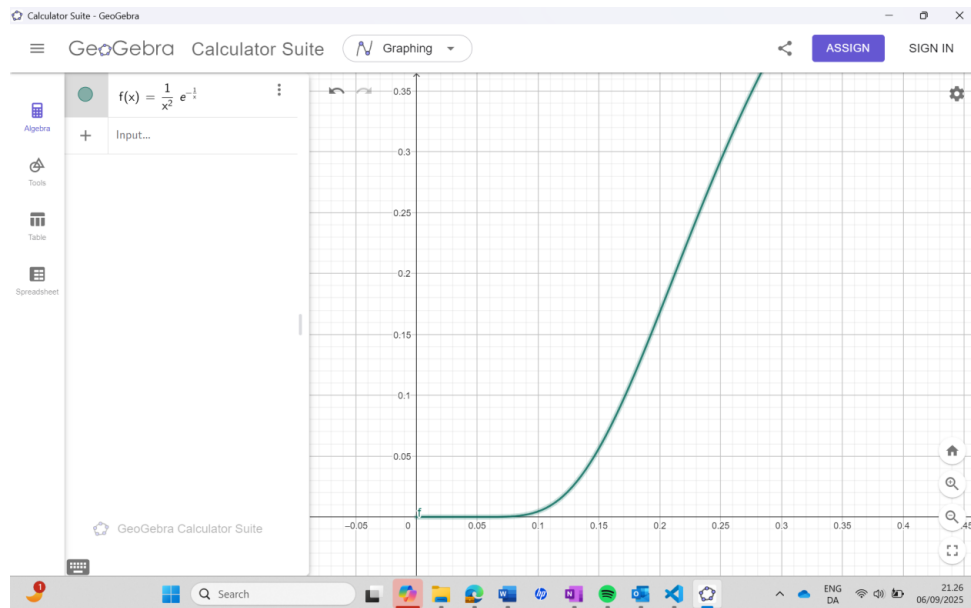
To sketch the function we essentially just plot

$$f(x) = \frac{1}{x^2} e^{-\frac{1}{x}}$$



Well this formula is technically only valid when x is positive and very small due to the assumptions we made on the way so this is a better representation

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Problem 3.23. Show that the entropy of a two-state paramagnet, expressed as a function of temperature, is $S = Nk[\ln(2 \cosh x) - x \tanh x]$, where $x = \mu B/kT$. Check that this formula has the expected behavior as $T \rightarrow 0$ and $T \rightarrow \infty$.

I will use the same steps as outlined in the book

We know that the multiplicity is given by

$$\Omega = \frac{N!}{N_{\uparrow}!(N - N_{\uparrow})!}$$

We can calculate the entropy by the formula

$$\begin{aligned} \frac{S}{k} &= \ln(\Omega) \\ &= \ln\left(\frac{N!}{N_{\uparrow}!(N - N_{\uparrow})!}\right) \\ &= \ln(N!) - \ln(N_{\uparrow}!) - \ln((N - N_{\uparrow})!) \end{aligned}$$

Let us now use Stirlings formula

$$\ln(N!) = N \ln(N) - N$$

This simplifies our expression to

$$\begin{aligned} \frac{S}{k} &\approx N \ln(N) - N - N_{\uparrow} \ln(N_{\uparrow}) + N_{\uparrow} - (N - N_{\uparrow}) \ln(N - N_{\uparrow}) + (N - N_{\uparrow}) \\ &= N \ln(N) - N_{\uparrow} \ln(N_{\uparrow}) - (N - N_{\uparrow}) \ln(N - N_{\uparrow}) \end{aligned}$$

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We can calculate the temperature to get some expression for the temperature as a function of energy

$$\frac{1}{T} = \left(\frac{\partial S}{\partial U} \right)_{N,B} = \frac{\partial S}{\partial N_{\uparrow}} \frac{\partial N_{\uparrow}}{\partial U}$$

We essentially have to calculate two derivatives so let us divide this task into two parts. First let us focus on calculating $\frac{\partial N_{\uparrow}}{\partial U}$. We know that our energy is given by

$$U = \mu B(N_{\downarrow} - N_{\uparrow}) = -\mu B(2N_{\uparrow} - N)$$

$\Downarrow$

$$N_{\uparrow} = -\frac{U}{2\mu B} + \frac{N}{2}$$

$\Downarrow$

$$\frac{\partial N_{\uparrow}}{\partial U} = -\frac{1}{2\mu B}$$

Nice. One down. Let us try to calculate the other $\frac{\partial S}{\partial N_{\uparrow}}$

$$\frac{S}{k} \approx N \ln(N) - N_{\uparrow} \ln(N_{\uparrow}) - (N - N_{\uparrow}) \ln(N - N_{\uparrow})$$

$\Downarrow$

$$\frac{1}{k} \frac{\partial S}{\partial N_{\uparrow}} = -\ln(N_{\uparrow}) - 1 + \ln(N - N_{\uparrow}) + 1$$

$\Downarrow$

$$\frac{\partial S}{\partial N_{\uparrow}} = k(-\ln(N_{\uparrow}) + \ln(N - N_{\uparrow}))$$

We can now combine our two results

$$\begin{aligned} \frac{1}{T} &= \frac{\partial S}{\partial N_{\uparrow}} \frac{\partial N_{\uparrow}}{\partial U} \\ &= -\frac{k}{2\mu B} (-\ln(N_{\uparrow}) + \ln(N - N_{\uparrow})) \\ &= -\frac{k}{2\mu B} \left(\ln \left(\frac{N - N_{\uparrow}}{N_{\uparrow}} \right) \right) \\ &= \frac{k}{2\mu B} \left(\ln \left(\frac{N_{\uparrow}}{N - N_{\uparrow}} \right) \right) \end{aligned}$$

Now we can insert

$$N_{\uparrow} = -\frac{U}{2\mu B} + \frac{N}{2}$$

We get

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$$\begin{aligned}
\frac{1}{T} &= \frac{k}{2\mu B} \left(\ln \left(\frac{\frac{N}{2} - \frac{U}{2\mu B}}{N - \frac{N}{2} + \frac{U}{2\mu B}} \right) \right) \\
&= \frac{k}{2\mu B} \left(\ln \left(\frac{N - \frac{U}{\mu B}}{N + \frac{U}{\mu B}} \right) \right) \\
&\Downarrow \\
\ln \left(\frac{N - \frac{U}{\mu B}}{N + \frac{U}{\mu B}} \right) &= \frac{2\mu B}{kT} \\
&\Downarrow \\
N - \frac{U}{\mu B} &= \left(N + \frac{U}{\mu B} \right) e^{\frac{2\mu B}{kT}} \\
&\Downarrow \\
N - N e^{\frac{2\mu B}{kT}} &= \frac{U}{\mu B} e^{\frac{2\mu B}{kT}} + \frac{U}{\mu B} \\
&\Downarrow \\
N \left(1 - e^{\frac{2\mu B}{kT}} \right) &= \frac{U}{\mu B} \left(1 + e^{\frac{2\mu B}{kT}} \right) \\
&\Downarrow \\
U &= N\mu B \frac{1 - e^{\frac{2\mu B}{kT}}}{1 + e^{\frac{2\mu B}{kT}}} \\
&= -N\mu B \frac{e^{\frac{2\mu B}{kT}} - 1}{e^{\frac{2\mu B}{kT}} + 1} \\
&= -N\mu B \frac{e^{\frac{\mu B}{kT}} - e^{-\frac{\mu B}{kT}}}{e^{\frac{\mu B}{kT}} + e^{-\frac{\mu B}{kT}}} \\
&= -N\mu B \tanh \left(\frac{\mu B}{kT} \right)
\end{aligned}$$

Okay. Let us take a moment to breathe and think about what we have. We still want to compute

$$\frac{S}{k} \approx N \ln(N) - N_{\uparrow} \ln(N_{\uparrow}) - (N - N_{\uparrow}) \ln(N - N_{\uparrow})$$

We know we can rewrite $N_{\uparrow}$ using this

$$N_{\uparrow} = -\frac{U}{2\mu B} + \frac{N}{2}$$

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And now we have managed to connect our energy with temperature with this

$$U = -N\mu B \tanh\left(\frac{\mu B}{kT}\right)$$

So logically we want to relate $N_{\uparrow}$ with temperature by inserting U

$$\begin{aligned} N_{\uparrow} &= -\frac{\left(-N\mu B \tanh\left(\frac{\mu B}{kT}\right)\right)}{2\mu B} + \frac{N}{2} \\ &= \frac{N}{2} \tanh\left(\frac{\mu B}{kT}\right) + \frac{N}{2} \end{aligned}$$

Let us use our hint and convert it into exponentials

$$N_{\uparrow} = \frac{N}{2} \left(\tanh\left(\frac{\mu B}{kT}\right) + 1 \right) = \frac{N}{2} \left(\frac{e^{\frac{\mu B}{kT}} - e^{-\frac{\mu B}{kT}}}{e^{\frac{\mu B}{kT}} + e^{-\frac{\mu B}{kT}}} + 1 \right) = \frac{N e^{\frac{\mu B}{kT}}}{e^{\frac{\mu B}{kT}} + e^{-\frac{\mu B}{kT}}}$$

We might as well make life easier for us and calculate $N - N_{\uparrow}$

$$N - N_{\uparrow} = N - \frac{N e^{\frac{\mu B}{kT}}}{e^{\frac{\mu B}{kT}} + e^{-\frac{\mu B}{kT}}} = N \left(1 - \frac{e^{\frac{\mu B}{kT}}}{e^{\frac{\mu B}{kT}} + e^{-\frac{\mu B}{kT}}} \right) = \frac{N e^{-\frac{\mu B}{kT}}}{e^{\frac{\mu B}{kT}} + e^{-\frac{\mu B}{kT}}}$$

Okay let us consolidate

$$\begin{aligned} N_{\uparrow} &= \frac{N e^{\frac{\mu B}{kT}}}{e^{\frac{\mu B}{kT}} + e^{-\frac{\mu B}{kT}}} \\ N - N_{\uparrow} &= \frac{N e^{-\frac{\mu B}{kT}}}{e^{\frac{\mu B}{kT}} + e^{-\frac{\mu B}{kT}}} \end{aligned}$$

Let me define $x \equiv \frac{\mu B}{kT}$ so I don't have to write it all the time

$$\begin{aligned} N_{\uparrow} &= \frac{N e^x}{e^x + e^{-x}} \\ N - N_{\uparrow} &= \frac{N e^{-x}}{e^x + e^{-x}} \end{aligned}$$

We can insert these into our entropy and see what we get

$$\begin{aligned} \frac{S}{k} &\approx N \ln(N) - N_{\uparrow} \ln(N_{\uparrow}) - (N - N_{\uparrow}) \ln(N - N_{\uparrow}) \\ &= N \ln(N) - \frac{N e^x}{e^x + e^{-x}} \ln\left(\frac{N e^x}{e^x + e^{-x}}\right) - \left(\frac{N e^{-x}}{e^x + e^{-x}}\right) \ln\left(\frac{N e^{-x}}{e^x + e^{-x}}\right) \\ &\quad \Updownarrow \\ \frac{S}{Nk} &= \ln(N) - \frac{e^x}{e^x + e^{-x}} \ln\left(\frac{N e^x}{e^x + e^{-x}}\right) - \left(\frac{e^{-x}}{e^x + e^{-x}}\right) \ln\left(\frac{N e^{-x}}{e^x + e^{-x}}\right) \end{aligned}$$

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$$= \ln(N) - \frac{e^x}{e^x + e^{-x}} \ln(Ne^x) + \frac{e^x}{e^x + e^{-x}} \ln(e^x + e^{-x}) - \frac{e^{-x}}{e^x + e^{-x}} \ln(Ne^{-x}) \\ + \frac{e^{-x}}{e^x + e^{-x}} \ln(e^x + e^{-x})$$

We can combine what is marked in red

$$\frac{S}{Nk} = \ln(N) - \frac{e^x}{e^x + e^{-x}} \ln(Ne^x) - \frac{e^{-x}}{e^x + e^{-x}} \ln(Ne^{-x}) + \left(\frac{e^x + e^{-x}}{e^x + e^{-x}} \right) \ln(e^x + e^{-x}) \\ = \ln(N) - \frac{e^x}{e^x + e^{-x}} (\ln(N) + \ln(e^x)) - \frac{e^{-x}}{e^x + e^{-x}} (\ln(N) + \ln(e^{-x})) - \ln(e^x + e^{-x}) \\ = \ln(N) \left(1 - \frac{e^x + e^{-x}}{e^x + e^{-x}} \right) - \frac{e^x \ln(e^x)}{e^x + e^{-x}} - \frac{e^{-x} \ln(e^{-x})}{e^x + e^{-x}} + \left(\frac{e^x + e^{-x}}{e^x + e^{-x}} \right) \ln(e^x + e^{-x})$$

Okay now notice that the fractions marked green is actually 1 and the logarithms cancels in those marked red so

$$\frac{S}{Nk} = -\frac{xe^x}{e^x + e^{-x}} + \frac{xe^{-x}}{e^x + e^{-x}} + \ln(e^x + e^{-x}) \\ = -x \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right) + \ln(e^x + e^{-x})$$

Now introduce the hyperbolic functions

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad \cosh(x) = \frac{e^x + e^{-x}}{2}$$

This gives

$$\frac{S}{Nk} = \ln(2 \cosh(x)) - x \tanh(x)$$

↕

$$S = Nk[\ln(2 \cosh(x)) - x \tanh(x)], \quad x = \frac{\mu B}{kT}$$

We now calculate the limits. If $T \rightarrow 0$ then $x \rightarrow \infty$:

$$\lim_{T \rightarrow 0} S = \lim_{x \rightarrow \infty} S = \lim_{x \rightarrow \infty} Nk \left[\ln(e^x + e^{-x}) - x \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right) \right]$$

Nobody got time to calculate these limits so we do the same as we did in our first week. Things that grow fast is zero if there is something that grows even faster. So our limit is the same as

$$= \lim_{x \rightarrow \infty} Nk \left[\ln(e^x) - x \left(\frac{e^x}{e^x} \right) \right] \\ = \lim_{x \rightarrow \infty} Nk[x - x] = 0$$

This makes sense because as $T=0$ we are at ground state which only has $\Omega = 1$ and therefore

$$S = k \ln(1) = 0$$

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We calculate the other limit

$$\begin{aligned}\lim_{T \rightarrow \infty} S &= \lim_{x \rightarrow 0} S = \lim_{x \rightarrow 0} Nk \left[\ln(e^x + e^{-x}) - x \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right) \right] \\ &= \lim_{x \rightarrow 0} Nk \left[\ln(1 + 1) - 0 \cdot \left(\frac{1 - 1}{1 + 1} \right) \right] \\ &= Nk \ln(2)\end{aligned}$$

At infinite temperature we have maximum entropy since

$$\left(\frac{\partial S}{\partial U} \right)_{V,N} = \lim_{T \rightarrow 0} \frac{1}{T} = 0$$

This is the point where half the states are up and half is down. In last TØ in problem 2.24 we calculated the multiplicity to be

$$\Omega = 2^N \sqrt{\frac{2}{\pi N}}$$

This gives

$$S = k \ln(\Omega) = k \ln \left(2^N \sqrt{\frac{2}{\pi N}} \right) = k \ln(2^N) + k \ln \left(\sqrt{\frac{2}{\pi N}} \right) = Nk \ln(2) + k \ln \left(\sqrt{\frac{2}{\pi N}} \right)$$

The second term is very small for large N, så

$$S \approx Nk \ln(2)$$

Nice both limits make sense.

Note: The math itself is not difficult compared to getting the idea. My best advice here is to take some starting point and try to calculate from there. In this case it was the multiplicity which naturally led us to $S = k \ln(\Omega)$. Once you are stuck you have to start asking yourself two questions. 1. what you are missing in your expression. 2. what should not be in your expression, which in our case is $N_{\uparrow}$. And then you start looking for formulas in the book. Question 1 makes us look for something related to temperature, which naturally leads us to $\frac{\partial S}{\partial U} = \frac{1}{T}$. Question 2 makes us look for something related to $N_{\uparrow}$ which naturally leads us to the expression of $N_{\uparrow}$ with U. And then it is matter of repeating these questions over and over again every time you are stuck until they converge to a final solution. If you can master this skill then you are set for the rest of the course. – David

Problem 3.25. In Problem 2.18 you showed that the multiplicity of an Einstein solid containing N oscillators and q energy units is approximately

$$\Omega(N, q) \approx \left(\frac{q+N}{q} \right)^q \left(\frac{q+N}{N} \right)^N.$$

- Starting with this formula, find an expression for the entropy of an Einstein solid as a function of N and q . Explain why the factors omitted from the formula have no effect on the entropy, when N and q are large.
- Use the result of part (a) to calculate the temperature of an Einstein solid as a function of its energy. (The energy is $U = q\epsilon$, where ϵ is a constant.) Be sure to simplify your result as much as possible.
- Invert the relation you found in part (b) to find the energy as a function of temperature, then differentiate to find a formula for the heat capacity.
- Show that, in the limit $T \rightarrow \infty$, the heat capacity is $C = Nk$. (Hint: When x is very small, $e^x \approx 1 + x$.) Is this the result you would expect? Explain.
- Make a graph (possibly using a computer) of the result of part (c). To avoid awkward numerical factors, plot C/Nk vs. the dimensionless variable $t = kT/\epsilon$, for t in the range from 0 to about 2. Discuss your prediction for the heat capacity at low temperature, comparing to the data for lead, aluminum, and diamond shown in Figure 1.14. Estimate the value of ϵ , in electron-volts, for each of those real solids.

a) We calculate the entropy

$$\begin{aligned} \frac{S}{k} &= \ln(\Omega) = \ln \left(\left(\frac{q+N}{q} \right)^q \left(\frac{q+N}{N} \right)^N \right) \\ &= q \ln \left(\frac{q+N}{q} \right) + N \ln \left(\frac{q+N}{N} \right) \\ &= q \ln(q+N) - q \ln(q) + N \ln(q+N) - N \ln(N) \\ &= (N+q) \ln(q+N) - q \ln(q) - N \ln(N) \end{aligned}$$

Problem 2.18. Use Stirling's approximation to show that the multiplicity of an Einstein solid, for any large values of N and q , is approximately

$$\Omega(N, q) \approx \frac{\left(\frac{q+N}{q} \right)^q \left(\frac{q+N}{N} \right)^N}{\sqrt{2\pi q(q+N)/N}}.$$

We essentially just omit a factor of $\frac{1}{2} \ln \left(\frac{2\pi q(q+N)}{N} \right)$. This factor is not multiplied with q or N , which means that this factor is very small compared to all the others

b) If we have to calculate the temperature then we have to invoke the formula

$$\left(\frac{\partial S}{\partial U} \right)_{V,N} = \frac{1}{T}$$

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We can use that $q = \frac{U}{\epsilon}$ and find an expression for entropy

$$\begin{aligned}\frac{S}{k} &= (N + q) \ln(q + N) - q \ln(q) - N \ln(N) \\ &= \left(N + \frac{U}{\epsilon}\right) \ln\left(\frac{U}{\epsilon} + N\right) - \frac{U}{\epsilon} \ln\left(\frac{U}{\epsilon}\right) - N \ln(N) \\ &\quad \Downarrow \\ S &= k \left[\left(N + \frac{U}{\epsilon}\right) \ln\left(\frac{U}{\epsilon} + N\right) - \frac{U}{\epsilon} \ln\left(\frac{U}{\epsilon}\right) - N \ln(N) \right]\end{aligned}$$

I can now calculate the derivative

$$\begin{aligned}\left(\frac{\partial S}{\partial U}\right)_{V,N} &= k \left[\frac{1}{\epsilon} \ln\left(\frac{U}{\epsilon} + N\right) + \frac{\left(N + \frac{U}{\epsilon}\right)}{\left(N + \frac{U}{\epsilon}\right)} - \frac{1}{\epsilon} \ln\left(\frac{U}{\epsilon}\right) - \frac{\frac{U}{\epsilon}}{\frac{U}{\epsilon}} \right] \\ &= k \left[\frac{1}{\epsilon} \ln\left(\frac{U}{\epsilon} + N\right) - \frac{1}{\epsilon} \ln\left(\frac{U}{\epsilon}\right) \right] \\ &= \frac{k}{\epsilon} \ln\left(\frac{U}{\epsilon} + N\right) - \frac{k}{\epsilon} \ln\left(\frac{U}{\epsilon}\right) \\ &= \frac{k}{\epsilon} \ln\left(\frac{\frac{U}{\epsilon} + N}{\frac{U}{\epsilon}}\right) \\ &= \frac{k}{\epsilon} \ln\left(1 + \frac{N\epsilon}{U}\right) = \frac{1}{T} \\ &\quad \Downarrow \\ T &= \frac{\epsilon}{k} \frac{1}{\ln\left(1 + \frac{N\epsilon}{U}\right)}\end{aligned}$$

c) First we invert it

$$\begin{aligned}\ln\left(1 + \frac{N\epsilon}{U}\right) &= \frac{\epsilon}{kT} \\ &\quad \Downarrow \\ 1 + \frac{N\epsilon}{U} &= e^{\frac{\epsilon}{kT}} \\ &\quad \Downarrow \\ \frac{N\epsilon}{U} &= e^{\frac{\epsilon}{kT}} - 1 \\ &\quad \Downarrow \\ U &= \frac{N\epsilon}{e^{\frac{\epsilon}{kT}} - 1}\end{aligned}$$

The heat capacity can now be found by differentiating

$$\begin{aligned}
 C_V &= \left(\frac{\partial U}{\partial T} \right)_{V,N} = N\epsilon \frac{\partial}{\partial T} \left(\frac{1}{e^{\frac{\epsilon}{kT}} - 1} \right) \\
 &= - \frac{N\epsilon}{\left(e^{\frac{\epsilon}{kT}} - 1 \right)^2} \frac{\partial}{\partial T} \left(e^{\frac{\epsilon}{kT}} - 1 \right) \\
 &= - \frac{N\epsilon e^{\frac{\epsilon}{kT}}}{\left(e^{\frac{\epsilon}{kT}} - 1 \right)^2} \frac{\partial}{\partial T} \left(\frac{\epsilon}{kT} \right) \\
 &= - \frac{N\epsilon^2 e^{\frac{\epsilon}{kT}}}{k \left(e^{\frac{\epsilon}{kT}} - 1 \right)^2} \frac{\partial}{\partial T} \left(\frac{1}{T} \right) \\
 &= \frac{N\epsilon^2 e^{\frac{\epsilon}{kT}}}{kT^2 \left(e^{\frac{\epsilon}{kT}} - 1 \right)^2}
 \end{aligned}$$

d) We now calculate the limit of the heat capacity when $T \rightarrow \infty$

We use the approximation that $e^{\frac{\epsilon}{kT}} \approx 1 + \frac{\epsilon}{kT}$ in the denominator

$$\begin{aligned}
 \lim_{T \rightarrow \infty} C_V &= \lim_{T \rightarrow \infty} \frac{N\epsilon^2 e^{\frac{\epsilon}{kT}}}{kT^2 \left(e^{\frac{\epsilon}{kT}} - 1 \right)^2} \\
 &= \lim_{T \rightarrow \infty} \frac{N\epsilon^2}{kT^2 \left(1 + \frac{\epsilon}{kT} - 1 \right)^2} \\
 &= \lim_{T \rightarrow \infty} \frac{N\epsilon^2}{kT^2 \left(\frac{\epsilon}{kT} \right)^2} \\
 &= Nk
 \end{aligned}$$

From the equipartition theorem

$$U_{\text{thermal}} = Nf \cdot \frac{1}{2} kT$$

Then the heat capacity is

$$C_V = \frac{f}{2} Nk$$

Each oscillator has two degrees of freedom. One for potential energy and one for kinetic energy which gives

$$C_V = Nk$$

A diatomic molecule can also *vibrate*, as if the two atoms were held together by a spring. This vibration should count as *two* degrees of freedom, one for the vibrational kinetic energy and one for the potential energy. (You may recall from classical mechanics that the average kinetic and potential energies of a simple harmonic oscillator are equal—a result that is consistent with the equipartition theorem.) More complicated molecules can vibrate in a variety of ways: stretching, flexing, twisting. Each “mode” of vibration counts as two degrees of freedom.

e) We do plotting after rewriting the expression

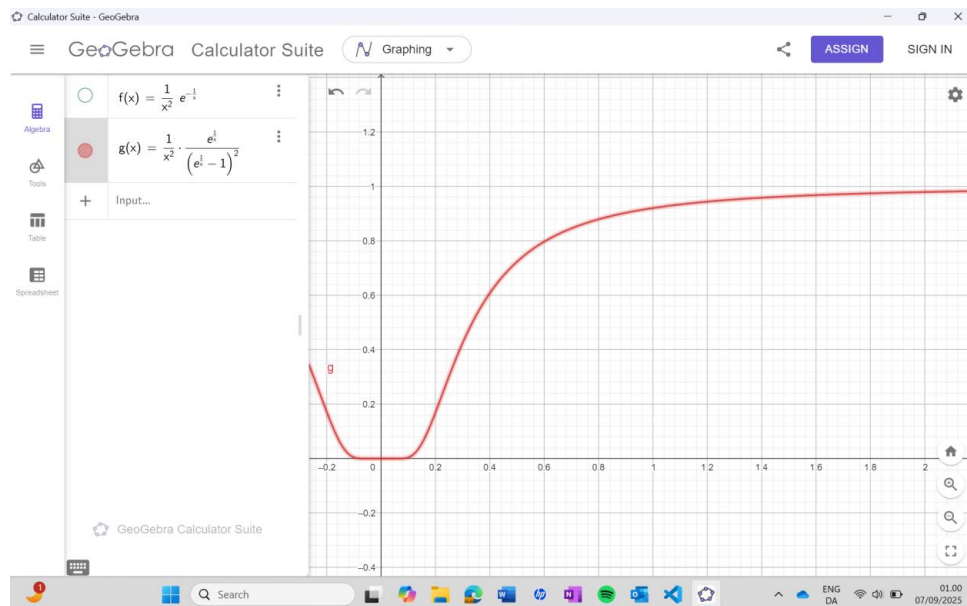
$$C_V = \frac{N\epsilon^2 e^{\frac{\epsilon}{kT}}}{kT^2 \left(e^{\frac{\epsilon}{kT}} - 1\right)^2}$$

$$\Downarrow$$

$$\frac{C}{Nk} = \left(\frac{\epsilon}{kT}\right)^2 \frac{e^{\frac{\epsilon}{kT}}}{\left(e^{\frac{\epsilon}{kT}} - 1\right)^2}$$

We define $t = \frac{kT}{\epsilon}$ such that we have to plot

$$f(t) = \frac{1}{t^2} \frac{e^{\frac{1}{t}}}{\left(e^{\frac{1}{t}} - 1\right)^2}$$



Ja jeg har seriøst brugt 3 timer på de sidste 2 opgaver kl er nu 1 om natten :(

Problem C1**Question a:**

Find the number of ways $\Omega(n_A, n_C)$ one can break n_A AT-pairs and n_C CG-pairs. Show that for $n_i \gg 1$, $N_i \gg 1$, and $N_i - n_i \gg 1$ ($i = A, C$), the entropy S of the helix is

$$\frac{S}{k} = N_A \ln(N_A) - n_A \ln(n_A) - (N_A - n_A) \ln(N_A - n_A) + N_C \ln(N_C) - n_C \ln(n_C) - (N_C - n_C) \ln(N_C - n_C) \quad (2)$$

The number of ways that AT pairs can be broken are given by the binomial distribution

$$\Omega(n_A) = \binom{N_A}{n_A} = \frac{N_A!}{n_A! \cdot (N_A - n_A)!}$$

Similarly for CG pairs

$$\Omega(n_C) = \binom{N_C}{n_C} = \frac{N_C!}{n_C! \cdot (N_C - n_C)!}$$

The total multiplicity is

$$\Omega(n_A, n_C) = \Omega(n_A) \Omega(n_C) = \frac{N_A!}{n_A! \cdot (N_A - n_A)!} \cdot \frac{N_C!}{n_C! \cdot (N_C - n_C)!}$$

The entropy is given by

$$\begin{aligned} \frac{S}{k} &= \ln(\Omega) = \ln\left(\frac{N_A!}{n_A! \cdot (N_A - n_A)!} \cdot \frac{N_C!}{n_C! \cdot (N_C - n_C)!}\right) \\ &= \ln(N_A!) - \ln(n_A!) - \ln((N_A - n_A)!) + \ln(N_C!) - \ln(n_C!) - \ln((N_C - n_C)!) \end{aligned}$$

Now using Stirling's approximation

$$\begin{aligned} \frac{S}{k} &\approx N_A \ln(N_A) - N_A + n_A \ln(n_A) - n_A + (N_A - n_A) \ln(N_A - n_A) + (N_A - n_A) + N_C \ln(N_C) - N_C \\ &\quad - n_C \ln(n_C) + n_C - (N_C - n_C) \ln(N_C - n_C) + (N_C - n_C) \\ &= N_A \ln(N_A) - n_A \ln(n_A) - (N_A - n_A) \ln(N_A - n_A) + N_C \ln(N_C) - n_C \ln(n_C) \\ &\quad - (N_C - n_C) \ln(N_C - n_C) \end{aligned}$$

Question b:

Use Eq. (1) and Eq. (2) to show that in equilibrium

$$\frac{N_A - n_A^{\text{eq}}}{n_A^{\text{eq}}} = \left(\frac{N_C - n_C^{\text{eq}}}{n_C^{\text{eq}}} \right)^\gamma$$

and give a formula for γ . Here, n_i^{eq} are the equilibrium values of n_i .

$$E = n_A E_A + n_C E_C$$

$$\Downarrow$$

$$n_C = \frac{E - n_A E_A}{E_C}$$

At equilibrium then the entropy should be at its maximum

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$$\begin{aligned} \left(\frac{\partial S}{\partial n_A}\right) &= \left(\frac{\partial S_A}{\partial n_A}\right) + \left(\frac{\partial S_C}{\partial n_A}\right) = \left(\frac{\partial S_A}{\partial n_A}\right) + \left(\frac{\partial n_C}{\partial n_A}\right) \left(\frac{\partial S_C}{\partial n_C}\right) = \left(\frac{\partial S_A}{\partial n_A}\right) - \frac{E_A}{E_C} \left(\frac{\partial S_C}{\partial n_C}\right) = 0 \\ &\Downarrow \\ \left(\frac{\partial S_A}{\partial n_A}\right) &= \frac{E_A}{E_C} \left(\frac{\partial S_C}{\partial n_C}\right) \end{aligned}$$

If we look at our entropy expression

$$\begin{aligned} \frac{S}{k} &= \frac{S_A + S_C}{k} = N_A \ln(N_A) - n_A \ln(n_A) - (N_A - n_A) \ln(N_A - n_A) + N_C \ln(N_C) - n_C \ln(n_C) \\ &\quad - (N_C - n_C) \ln(N_C - n_C) \end{aligned}$$

Notice that

$$\begin{aligned} \frac{S_A}{k} &= N_A \ln(N_A) - n_A \ln(n_A) - (N_A - n_A) \ln(N_A - n_A) \\ \frac{S_C}{k} &= N_C \ln(N_C) - n_C \ln(n_C) - (N_C - n_C) \ln(N_C - n_C) \end{aligned}$$

The we find that

$$\begin{aligned} \frac{1}{k} \frac{\partial S_{A,C}}{\partial n_{A,C}} &= -\ln(n_{A,C}) - \frac{n_{A,C}}{n_{A,C}} - \ln(N_{A,C} - n_{A,C}) \cdot (-1) - \frac{(N_{A,C} - n_{A,C})}{(N_{A,C} - n_{A,C})} \cdot (-1) \\ &= \ln(N_{A,C} - n_{A,C}) - \ln(n_{A,C}) \\ &= \ln\left(\frac{N_{A,C} - n_{A,C}}{n_{A,C}}\right) \end{aligned}$$

Therefore we obtain

$$\begin{aligned} \ln\left(\frac{N_A - n_A}{n_A}\right) &= \frac{E_A}{E_C} \ln\left(\frac{N_C - n_C}{n_C}\right) \\ &\Downarrow \\ \frac{N_A - n_A}{n_A} &= \left(\frac{N_C - n_C}{n_C}\right)^{\frac{E_A}{E_C}} \end{aligned}$$

Thus

$$\gamma = \frac{E_A}{E_C}$$

Question c:

Consider now the special case $N_A = N_C$ and $E_A = E_C$. Calculate $S(E)$ and then the energy $E(T)$ as a function of temperature T . Interpret the limits $\lim_{T \rightarrow 0} E(T)$ and $\lim_{T \rightarrow \infty} E(T)$ physically.

At equilibrium if $N_A = N_C$ and $E_A = E_C$ then

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$$\frac{N_A - n_A}{n_A} = \left(\frac{N_C - n_C}{n_C} \right)^{\frac{E_A}{E_C}}$$

$\Downarrow$

$$n_A = n_C$$

Thus the energy can be described as

$$E = n_A E_A + n_C E_C = 2n_A E_A$$

$\Downarrow$

$$n_A = \frac{E}{2E_A}$$

Therefore the entropy can be reduced to

$$\begin{aligned} \frac{S}{k} &= N_A \ln(N_A) - n_A \ln(n_A) - (N_A - n_A) \ln(N_A - n_A) + N_C \ln(N_C) - n_C \ln(n_C) \\ &\quad - (N_C - n_C) \ln(N_C - n_C) \end{aligned}$$

$\Downarrow$

$$\begin{aligned} \frac{S}{2k} &= N_A \ln(N_A) - n_A \ln(n_A) - (N_A - n_A) \ln(N_A - n_A) \\ &= N_A \ln(N_A) - \frac{E}{2E_A} \ln\left(\frac{E}{2E_A}\right) - \left(N_A - \frac{E}{2E_A}\right) \ln\left(N_A - \frac{E}{2E_A}\right) \\ &= N_A \ln(N_A) - N_A \ln\left(N_A - \frac{E}{2E_A}\right) + \frac{E}{2E_A} \ln\left(N_A - \frac{E}{2E_A}\right) - \frac{E}{2E_A} \ln\left(\frac{E}{2E_A}\right) \\ &= N_A \ln\left(\frac{N_A}{N_A - \frac{E}{2E_A}}\right) + \frac{E}{2E_A} \ln\left(\frac{N_A - \frac{E}{2E_A}}{\frac{E}{2E_A}}\right) \end{aligned}$$

This cannot be reduced anymore

$$\left(\frac{\partial S}{\partial E} \right)_{V,N} = \frac{1}{T}$$

Let $x = \frac{E}{2E_A}$ then

$$\frac{S}{2k} = N_A \ln\left(\frac{N_A}{N_A - x}\right) + x \ln\left(\frac{N_A - x}{x}\right)$$

Then

$$\left(\frac{\partial S}{\partial E} \right) = \left(\frac{\partial S}{\partial x} \right) \left(\frac{\partial x}{\partial E} \right) = \left(\frac{\partial S}{\partial x} \right) \cdot \frac{1}{2E_A} = \frac{1}{T}$$

$\Downarrow$

$$\left(\frac{\partial S}{\partial x} \right) = \frac{2E_A}{T}$$

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We can split the derivative into two parts. The first derivative

$$\left(N_A \ln\left(\frac{N_A}{N_A - x}\right)\right)' = N_A \cdot \left(\frac{N_A - x}{N_A}\right)' \cdot \left(\frac{N_A}{N_A - x}\right)' = (N_A - x) \cdot \left(-\frac{N_A}{(N_A - x)^2}\right) \cdot (-1) = \frac{N_A}{N_A - x}$$

The second derivative

$$\begin{aligned}\left(x \ln\left(\frac{N_A - x}{x}\right)\right)' &= \ln\left(\frac{N_A - x}{x}\right) + x \cdot \left(\ln\left(\frac{N_A - x}{x}\right)\right)' \\ &= \ln\left(\frac{N_A - x}{x}\right) + x \cdot \frac{x}{N_A - x} \cdot \left(\frac{N_A - x}{x}\right)' \\ &= \ln\left(\frac{N_A - x}{x}\right) + x \cdot \frac{x}{N_A - x} \cdot \left(-\frac{N_A - x}{x^2} - \frac{1}{x}\right) \\ &= \ln\left(\frac{N_A - x}{x}\right) + x \cdot \frac{x}{N_A - x} \cdot \left(\frac{x - N_A - x}{x^2}\right) \\ &= \ln\left(\frac{N_A - x}{x}\right) - \frac{N_A}{N_A - x}\end{aligned}$$

Using these results we get:

$$\left(\frac{\partial S}{\partial x}\right) = 2k \left(\frac{N_A}{N_A - x} + \ln\left(\frac{N_A - x}{x}\right) - \frac{N_A}{N_A - x}\right) = \frac{2E_A}{T}$$

$\Downarrow$

$$\ln\left(\frac{N_A - x}{x}\right) = \frac{E_A}{kT}$$

$\Downarrow$

$$\frac{N_A - x}{x} = e^{\frac{E_A}{kT}}$$

$\Downarrow$

$$N_A = x + x e^{\frac{E_A}{kT}}$$

$\Downarrow$

$$x = \frac{N_A}{1 + e^{\frac{E_A}{kT}}} = \frac{E}{2E_A}$$

$\Downarrow$

$$E(T) = \frac{2E_A N_A}{1 + e^{\frac{E_A}{kT}}}$$

If $T \rightarrow 0$ then $e^{\frac{E_A}{kT}} \rightarrow \infty$ and $E \rightarrow 0$ which is to be expected. There will not be any energy to break any bonds if the temperature is zero

If $T \rightarrow \infty$ then $e^{\frac{E_A}{kT}} \rightarrow 1$ and $E \rightarrow E_A N_A$ which means that there is enough energy to break half of the bonds. This makes sense because at infinite temperature we reach maximum entropy

Problem 3.36. Consider an Einstein solid for which both N and q are much greater than 1. Think of each oscillator as a separate “particle.”

(a) Show that the chemical potential is

$$\mu = -kT \ln\left(\frac{N+q}{N}\right).$$

(b) Discuss this result in the limits $N \gg q$ and $N \ll q$, concentrating on the question of how much S increases when another particle carrying no energy is added to the system. Does the formula make intuitive sense?

For any large Einstein solid we know that

$$\Omega(N, q) \approx \left(\frac{q+N}{q}\right)^q \left(\frac{q+N}{N}\right)^N \left(\frac{2\pi q(q+N)}{N}\right)^{-\frac{1}{2}}, \quad N, q \gg 1$$

Ignore the last term because it is much smaller compared to the first two terms

$$\Omega(N, q) \approx \left(\frac{q+N}{q}\right)^q \left(\frac{q+N}{N}\right)^N, \quad N, q \gg 1$$

The chemical potential is given by

$$\left(\frac{\partial S}{\partial N_i}\right)_{U, V, N_j} = -\frac{\mu}{T}$$

We need to calculate the entropy according to this formula so

$$\begin{aligned} \frac{S}{k} &= \ln(\Omega) = \ln\left(\left(\frac{q+N}{q}\right)^q \left(\frac{q+N}{N}\right)^N\right) \\ &= q \ln\left(\frac{q+N}{q}\right) + N \ln\left(\frac{q+N}{N}\right) \end{aligned}$$

Since each oscillator has to be considered a particle then N in Ω is the same as N in the partial derivative and we can therefore calculate the partial derivative

$$\begin{aligned} \frac{1}{k} \frac{\partial S}{\partial N} &= q \frac{\partial}{\partial N} \left[\ln\left(\frac{q+N}{q}\right) \right] + \ln\left(\frac{q+N}{N}\right) + N \frac{\partial}{\partial N} \left[\ln\left(\frac{q+N}{N}\right) \right] \\ &= \frac{q}{q+N} \frac{\partial}{\partial N} \left[\frac{q+N}{q} \right] + \ln\left(\frac{q+N}{N}\right) + \frac{N}{q+N} \frac{\partial}{\partial N} \left[\frac{q+N}{N} \right] \\ &= \frac{q^2}{q+N} \cdot \frac{\partial}{\partial N} \left[1 + \frac{N}{q} \right] + \ln\left(\frac{q+N}{N}\right) + \frac{N^2}{q+N} \frac{\partial}{\partial N} \left[\frac{q}{N} + 1 \right] \end{aligned}$$

$$\begin{aligned}
&= \frac{q^2}{q+N} \cdot \frac{1}{q} + \ln\left(\frac{q+N}{N}\right) - \frac{N^2}{q+N} \cdot \frac{q}{N^2} \\
&= \ln\left(\frac{q+N}{N}\right)
\end{aligned}$$

Thus we get that

$$\begin{aligned}
\left(\frac{\partial S}{\partial N}\right)_{U,V} &= k \ln\left(\frac{q+N}{N}\right) = -\frac{\mu}{T} \\
&\Downarrow \\
\mu &= -kT \ln\left(\frac{q+N}{N}\right) = -kT \ln\left(1 + \frac{q}{N}\right)
\end{aligned}$$

If $N \gg q$ then we can use the approximation $\ln(1+x) \approx x$, $|x| \ll 1$

$$\mu = -\frac{kTq}{N}$$

If $N \ll q$ then we can approximate

$$\mu \approx -kT \ln\left(\frac{q}{N}\right)$$

Well does this make sense? Let us try to justify. If we look at the formula $\left(\frac{\partial S}{\partial N_i}\right)_{U,V,N_j} = -\frac{\mu}{T}$ then the chemical potential seems to be a measure of how much the entropy will increase for each particle added. So let us try to look at this quantity instead

$$\begin{aligned}
\left(\frac{\partial S}{\partial N}\right)_{U,V} &= \frac{kq}{N} \approx 0, \quad N \gg q \\
\left(\frac{\partial S}{\partial N}\right)_{U,V} &= k \ln\left(\frac{q}{N}\right) \gg 0, \quad q \gg N
\end{aligned}$$

If we have many more oscillators than energy quanta ($N \gg q$) then it is limited how many energy quanta you can distribute to each oscillator. In the extreme case where $q = 1$ you would only be able to increase the multiplicity by 1 for each oscillator which is practically no increase in entropy and hence why $\left(\frac{\partial S}{\partial N}\right)_{U,V} \approx 0$

If we have many more energy quanta than oscillators ($q \gg N$) then each oscillator will be able to distribute the energy quanta in many more ways. Hence, $\left(\frac{\partial S}{\partial N}\right)_{U,V} = k \ln\left(\frac{q}{N}\right) \gg 0$

Problem 6.12. Cold interstellar molecular clouds often contain the molecule cyanogen (CN), whose first rotational excited states have an energy of 4.7×10^{-4} eV (above the ground state). There are actually three such excited states, all with the same energy. In 1941, studies of the absorption spectrum of starlight that passes through these molecular clouds showed that for every ten CN molecules that are in the ground state, approximately three others are in the three first excited states (that is, an average of one in each of these states). To account for this data, astronomers suggested that the molecules might be in thermal equilibrium with some “reservoir” with a well-defined temperature. What is that temperature?*

First equation that comes in mind for me is

$$\frac{P(s_2)}{P(s_1)} = e^{-\frac{E(s_2) - E(s_1)}{kT}}$$

Remember we also need to be careful because the first rotational excited is triply degenerate (3 states at the same energy). We can describe the probability as

$$P(s) = \frac{1}{Z} e^{-\beta E(s)}$$

Instead of the probability of each state we can describe the probability of each energy level as

$$P(E) = \frac{d(E)}{Z} e^{-\beta E}$$

Where $d(E)$ is the degeneracy of each energy level. Now we have a proper equation that we can use

$$\frac{P(E_2)}{P(E_1)} = \frac{d(E_2)}{d(E_1)} e^{-\frac{E_2 - E_1}{kT}}$$

Since we have 10 molecules in ground state and 3 molecules in the first excited state then the probability is 3/10 but the degeneracy is 3/1 so

$$3e^{-\frac{E_2 - E_1}{kT}} = \frac{3}{10}$$

$$\Updownarrow$$

$$-\frac{E_2 - E_1}{kT} = \ln\left(\frac{1}{10}\right)$$

$$\Updownarrow$$

$$T = -\frac{E_2 - E_1}{k \ln\left(\frac{1}{10}\right)} = \frac{E_2 - E_1}{k \ln(10)} = \frac{4.7 \cdot 10^{-4} \text{ eV}}{1.3806488 \cdot 10^{-23} \frac{\text{J}}{\text{K}} \cdot \ln(10)} = 2.3687 \text{ K}$$

Problem 6.13. At *very* high temperatures (as in the very early universe), the proton and the neutron can be thought of as two different states of the same particle, called the “nucleon.” (The reactions that convert a proton to a neutron or vice versa require the absorption of an electron or a positron or a neutrino, but all of these particles tend to be very abundant at sufficiently high temperatures.) Since the neutron’s mass is higher than the proton’s by 2.3×10^{-30} kg, its energy is higher by this amount times c^2 . Suppose, then, that at some very early time, the nucleons were in thermal equilibrium with the rest of the universe at 10^{11} K. What fraction of the nucleons at that time were protons, and what fraction were neutrons?

Einstein’s infamous formula!

$$E = mc^2$$

We have that the energy difference between a proton and neutron is

$$E_N - E_P = (m_N - m_P)c^2$$

We also know that

$$\frac{P(s_2)}{P(s_1)} = e^{-\frac{E(s_2) - E(s_1)}{kT}}$$

↓

$$\frac{P_N}{P_P} = e^{-\frac{(m_N - m_P)c^2}{kT}} = e^{-\frac{2.3 \cdot 10^{-30} \text{ kg} \left(2.9979 \cdot 10^8 \frac{\text{m}}{\text{s}}\right)^2}{1.3806 \cdot 10^{-23} \frac{\text{J}}{\text{K}} \cdot 10^{11} \text{ K}}} = 0.8609$$

The fraction of protons out of all nucleons is

$$\frac{P_P}{P_N + P_P} = \frac{P_P}{0.8609P_P + P_P} = \frac{1}{0.8609 + 1} = 0.53737$$

The fraction of neutrons out of all nucleons is

$$\frac{P_N}{P_N + P_P} = 1 - 0.53737 = 0.46263$$

Problem 6.16. Prove that, for any system in equilibrium with a reservoir at temperature T , the average value of the energy is

$$\bar{E} = -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{\partial}{\partial \beta} \ln Z,$$

where $\beta = 1/kT$. These formulas can be extremely useful when you have an explicit formula for the partition function.

Start with the definition of the mean value which is the sum of the energy of state times the probability of that given state

$$\bar{E} = \sum_s E(s)P(s) = \frac{1}{Z} \sum_s E(s)e^{-\beta E(s)} = -\frac{1}{Z} \sum_s \frac{\partial}{\partial \beta} e^{-\beta E(s)} = -\frac{1}{Z} \frac{\partial Z}{\partial \beta}$$

To prove the last equation we can use the chain rule

$$\frac{\partial \ln(Z)}{\partial \beta} = \frac{\partial Z}{\partial Z} \frac{\partial \ln(Z)}{\partial \beta} = \frac{\partial \ln(Z)}{\partial Z} \frac{\partial Z}{\partial \beta} = \frac{1}{Z} \frac{\partial Z}{\partial \beta}$$

Problem 6.20. This problem concerns a collection of N identical harmonic oscillators (perhaps an Einstein solid or the internal vibrations of gas molecules) at temperature T . As in Section 2.2, the allowed energies of each oscillator are $0, hf, 2hf$, and so on.

(a) Prove by long division that

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

For what values of x does this series have a finite sum?

- (b) Evaluate the partition function for a single harmonic oscillator. Use the result of part (a) to simplify your answer as much as possible.
- (c) Use formula 6.25 to find an expression for the average energy of a single oscillator at temperature T . Simplify your answer as much as possible.
- (d) What is the total energy of the system of N oscillators at temperature T ? Your result should agree with what you found in Problem 3.25.
- (e) If you haven't already done so in Problem 3.25, compute the heat capacity of this system and check that it has the expected limits as $T \rightarrow 0$ and $T \rightarrow \infty$.

a) Remember how to do long division? Because I sure don't. Had to look it up on the internet.

The image shows a handwritten long division on a piece of graph paper. The divisor is $1-x$ and the dividend is $1+x+x^2+x^3+\dots$. The division proceeds as follows:

$$\begin{array}{r} 1+x+x^2+x^3+\dots \\ 1-x \overline{) 1+x+x^2+x^3+\dots} \\ \underline{1-x} \\ x \\ \underline{x-x^2} \\ x^2 \\ \underline{x^2-x^3} \\ x^3 \\ \underline{x^3-x^4} \\ x^4 \\ \vdots \end{array}$$

A black pen with the brand name "isfit" is lying diagonally across the bottom right of the graph paper.

b) The partition function is defined as

$$Z = \sum_s e^{-\beta E(s)}$$

From quantum mechanics the energy levels in a harmonic oscillator is

$$E_n = \left(n + \frac{1}{2}\right) \hbar \omega$$

In the context of statistical mechanics we always define the zero-point energy as the ground state. Ask me if you don't understand why.

$$E_n = n \hbar \omega$$

Therefore

$$Z = \sum_{n=0} e^{-\beta n \hbar \omega} = \sum_{n=0} (e^{-\beta \hbar \omega})^n$$

Our infinite series from a) says that

$$\sum_{n=0} x^n = \frac{1}{1-x}$$

Therefore we have that

$$Z = \frac{1}{1 - e^{-\beta \hbar \omega}} = \frac{1}{1 - e^{-\beta \hbar f}}$$

c) The average energy can found from the formula

$$\begin{aligned} \overline{E} &= -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = \frac{1}{Z} \frac{1}{(1 - e^{-\beta \hbar f})^2} \frac{\partial}{\partial \beta} (1 - e^{-\beta \hbar f}) \\ &= -\frac{1}{Z} \frac{1}{(1 - e^{-\beta \hbar f})^2} \frac{\partial}{\partial \beta} (e^{-\beta \hbar f}) \\ &= -\frac{1}{Z} \frac{1}{(1 - e^{-\beta \hbar f})^2} e^{-\beta \hbar f} \frac{\partial}{\partial \beta} (-\beta \hbar f) \\ &= \frac{1}{Z} \frac{1}{(1 - e^{-\beta \hbar f})^2} e^{-\beta \hbar f} \hbar f \end{aligned}$$

Let's make it a little prettier

$$\overline{E} = \frac{\hbar f e^{-\beta \hbar f}}{1 - e^{-\beta \hbar f}} = \frac{\hbar f}{\frac{1 - e^{-\beta \hbar f}}{e^{-\beta \hbar f}}} = \frac{\hbar f}{e^{\beta \hbar f} - 1}$$

d) From 3.25 we found that the energy is

$$U = \frac{N\epsilon}{e^{\frac{\epsilon}{kT}} - 1}$$

Since average values are additive then

$$\overline{U} = N\overline{E} = \frac{Nhf}{e^{\beta hf} - 1}$$

Let $\epsilon = hf$ and we have agreement with our results from problem 3.25

e) We already did this in previous TØ

Problem D1

An alternative definition of the entropy is

$$S = -k \sum_r p_r \ln(p_r), \quad (1)$$

where p_r is the probability of the system to be in microstate r and the sum is over all microstates accessible to the system. We take Eq. (1) to hold for any probability distribution p_r with $\sum_r p_r = 1$ and $p_r \geq 0$.

Question a:

For an isolated system in equilibrium, $p_r = p_0 \equiv 1/\Omega$ for all accessible states r . Show that in this case, Eq. (1) reduces to our basic definition given by Schroeder Eq. (2.45), i.e. $S = S_0 \equiv k \ln(\Omega)$.

Let us insert and see what happens

$$S = -k \sum_r \frac{1}{\Omega} \ln\left(\frac{1}{\Omega}\right)$$

Since we sum over all accessible states then we have Ω terms such that

$$S = k \frac{\Omega}{\Omega} \ln(\Omega) = k \ln(\Omega)$$

Question b:

Take a system with another probability distribution p_r and entropy S . Show that

$$S_0 - S = k \sum_r p_0 \frac{p_r}{p_0} \ln\left(\frac{p_r}{p_0}\right). \quad (2)$$

From the new definition

$$S = -k \sum_r p_r \ln(p_r)$$

$$S_0 = -k \sum_r p_0 \ln(p_0)$$

Note on S_0 . It may be weird that we are summing over r but the concept is the same. We are summing over all microstates but the probability distribution but p_r is different for each state while $p_0 = \frac{1}{\Omega}$ is constant for each state

$$S_0 - S = k \sum_r p_r \ln(p_r) - k \sum_r p_0 \ln(p_0)$$

$$= k \sum_r \frac{p_0 p_r}{p_0} \ln(p_r) - k \sum_r p_0 \ln(p_0)$$

Notice that for the second term there is no dependence on r so we can do the following manipulation

$$k \sum_r p_0 \ln(p_0) = k \ln(p_0) \sum_r p_0$$

Since the total probability of all microstates must equal 1 then

$$\sum_r p_0 = \sum_r p_r = \sum_r \frac{p_0 p_r}{p_0} = 1$$

Therefore we can write the second term as

$$k \ln(p_0) \sum_r p_0 = k \ln(p_0) \sum_r \frac{p_0 p_r}{p_0} = k \sum_r \frac{p_0 p_r}{p_0} \ln(p_0)$$

We can now continue where we left off

$$\begin{aligned} S_0 - S &= k \sum_r \frac{p_0 p_r}{p_0} \ln(p_r) - k \sum_r p_0 \ln(p_0) \\ &= k \sum_r \frac{p_0 p_r}{p_0} \ln(p_r) - k \sum_r \frac{p_0 p_r}{p_0} \ln(p_0) \\ &= k \sum_r \frac{p_0 p_r}{p_0} \ln(p_r) - \frac{p_0 p_r}{p_0} \ln(p_0) \\ &= k \sum_r \frac{p_0 p_r}{p_0} \ln\left(\frac{p_r}{p_0}\right) \end{aligned}$$

Question c:

Show that we can write

$$S_0 - S = k p_0 \sum_r f(p_r/p_0) \quad (3)$$

with $f(x) = x \ln(x) - x + 1$.

Let's insert and check

$$\begin{aligned} S_0 - S &= k p_0 \sum_r \frac{p_r}{p_0} \ln\left(\frac{p_r}{p_0}\right) - \frac{p_r}{p_0} + 1 \\ &= k \sum_r p_r \ln\left(\frac{p_r}{p_0}\right) - p_r + p_0 \end{aligned}$$

$$= k \sum_r p_0 \frac{p_r}{p_0} \ln \left(\frac{p_r}{p_0} \right) - k \sum_r p_r + k \sum_r p_0$$

Using the fact that

$$\sum_r p_r = \sum_r p_0 = 1$$

Then

$$S_0 - S = k \sum_r p_0 \frac{p_r}{p_0} \ln \left(\frac{p_r}{p_0} \right)$$

Question d:

Show that $f(x) > 0$ for all $x \geq 0$, except for $x = 1$ where $f(x) = 0$. *Hint:* Look at the sign of $f'(x)$.

$$f(x) = x \ln(x) - x + 1$$

$$f'(x) = \frac{x}{x} + \ln(x) - 1 = \ln(x)$$

Notice that for $x > 1$ then $f'(x) > 0$ and the function is always rising

However for $0 < x < 1$ then $f'(x) < 0$ and the function is decreasing. The minimum of the function must therefore be $x = 1$

The value of the function at the minimum is

$$f(1) = \ln(1) - 1 + 1 = 0$$

Therefore the function $f(x)$ must be greater than 0 for all $x > 0$

Finally we should also check the point $x = 0$

$$f(0) = 1$$

Thus we can conclude that $f(x) > 0$ for all $x \geq 0$

Question e:

Prove that $S \leq S_0$ for all probability distributions p_r over microstates, and that the equality sign only holds for $p_r = p_0$.

You have now shown that the entropy is maximum when the probability is constant for all accessible states for an isolated system, i.e. when it is in equilibrium.

We can describe

$$S_0 - S = kp_0 \sum_r f\left(\frac{p_r}{p_0}\right)$$

Since $\frac{p_r}{p_0} > 0$ and since $f(x) > 0$ for all $x \geq 0$ except when $x = 1$ then in all those cases:

$$S_0 - S = kp_0 \sum_r f\left(\frac{p_r}{p_0}\right) \geq 0$$

$$\Updownarrow$$

$$S_0 \geq S$$

However in the special case where $x = 1$ when $p_r = p_0$ then

$$S_0 - S = kp_0 \sum_r f\left(\frac{p_r}{p_0}\right) = kp_0 \sum_r 0 = 0$$

$$\Updownarrow$$

$$S_0 = S$$

Problem D2

Consider a particle in a harmonic oscillator with energy levels nhf , where $n = 0, 1, 2, \dots$, at temperature T .

Question a:

Show that the partition function Z of the particle is

$$Z \simeq \left(\frac{kT}{hf} \right)^\kappa$$

for $kT \gg hf$ and give a value for κ .

The definition of the partition function is

$$Z = \sum_E d(E) e^{-\beta E}$$

The degeneracy of each energy level is 1 so

$$Z = \sum_n e^{-\frac{nhf}{kT}}$$

From the Taylor series it applies that

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

Therefore

$$Z = \frac{1}{1 - e^{-\frac{hf}{kT}}}$$

For $kT \gg hf$ we can use the Taylor approximation $e^x \approx 1 + x$, $|x| \ll 1$ which gives us

$$Z = \frac{1}{1 - \left(1 - \frac{hf}{kT}\right)} = \frac{kT}{hf}$$

Therefore $\kappa = 1$

Consider now a particle, which can sit in two harmonic oscillators. Oscillator A has energy levels nhf , and oscillator B has energy levels $\epsilon_0 + nhf$, where $\epsilon_0 > 0$ and $n = 0, 1, 2, \dots$. The particle can hop freely between the two oscillators, and the available states for the particle are therefore the harmonic oscillator levels in A and B .

Question b:

Give an expression for the partition function for the particle, when the temperature is T . What is the probability P_A that the particle sits in oscillator A ?

The partition function is given by

$$\begin{aligned} Z &= \sum_E e^{-\beta E} = \sum_{n=0} e^{-\beta nhf} + \sum_{n=0} e^{-\beta(\epsilon_0 + nhf)} \\ &= \sum_n e^{-\beta nhf} + e^{-\beta\epsilon_0} \sum_n e^{-\beta nhf} \\ &= (1 + e^{-\beta\epsilon_0}) \sum_n e^{-\beta nhf} \\ &= \frac{1 + e^{-\beta\epsilon_0}}{1 - e^{-\beta hf}} \end{aligned}$$

The probability that the particle sits in oscillator A is

$$\begin{aligned} P(s) &= \frac{1}{Z} e^{-\beta E(s)} \\ &\Downarrow \\ P_A &= \frac{1}{Z} \sum_n e^{-\beta nhf} = \frac{1}{Z} \cdot \frac{1}{1 - e^{-\beta hf}} = \frac{1}{\frac{1 + e^{-\beta\epsilon_0}}{1 - e^{-\beta hf}}} \cdot \frac{1}{1 - e^{-\beta hf}} = \frac{1}{1 + e^{-\beta\epsilon_0}} \end{aligned}$$

This is the answer. We used the same Taylor series on the way to get here $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$

Question c:

Give an expression for the average energy $E(T)$ of the particle. Show that $E(T) \approx a + bT$ when kT is large compared to both hf and ϵ_0 , and give values for a and b .

The average energy is defined as

$$E = -\frac{1}{Z} \frac{\partial Z}{\partial \beta}$$

We find that

$$\begin{aligned}
E &= -\frac{1}{Z} \frac{\partial}{\partial \beta} \left(\frac{1 + e^{-\beta \epsilon_0}}{1 - e^{-\beta hf}} \right) \\
&= -\frac{1}{Z} \left[\frac{1}{1 - e^{-\beta hf}} \cdot \frac{\partial}{\partial \beta} (1 + e^{-\beta \epsilon_0}) + (1 + e^{-\beta \epsilon_0}) \cdot \frac{\partial}{\partial \beta} \left(\frac{1}{1 - e^{-\beta hf}} \right) \right] \\
&= -\frac{1}{Z} \left[\frac{e^{-\beta \epsilon_0}}{1 - e^{-\beta hf}} \cdot \frac{\partial}{\partial \beta} (-\beta \epsilon_0) - \frac{(1 + e^{-\beta \epsilon_0})}{(1 - e^{-\beta hf})^2} \cdot \frac{\partial}{\partial \beta} (1 - e^{-\beta hf}) \right] \\
&= -\frac{1}{Z} \left[\frac{-\epsilon_0 e^{-\beta \epsilon_0}}{1 - e^{-\beta hf}} + \frac{e^{-\beta hf} (1 + e^{-\beta \epsilon_0})}{(1 - e^{-\beta hf})^2} \cdot \frac{\partial}{\partial \beta} (-\beta hf) \right] \\
&= -\frac{1}{Z} \left[\frac{-\epsilon_0 e^{-\beta \epsilon_0}}{1 - e^{-\beta hf}} - \frac{hf e^{-\beta hf} (1 + e^{-\beta \epsilon_0})}{(1 - e^{-\beta hf})^2} \right] \\
&= \frac{1}{Z} \left[\frac{\epsilon_0 e^{-\beta \epsilon_0}}{1 - e^{-\beta hf}} + \frac{hf e^{-\beta hf} (1 + e^{-\beta \epsilon_0})}{(1 - e^{-\beta hf})^2} \right]
\end{aligned}$$

Let us insert Z and see how much we can reduce this expression

$$\begin{aligned}
&= \frac{1}{\frac{1 + e^{-\beta \epsilon_0}}{1 - e^{-\beta hf}}} \left[\frac{\epsilon_0 e^{-\beta \epsilon_0}}{1 - e^{-\beta hf}} + \frac{hf e^{-\beta hf} (1 + e^{-\beta \epsilon_0})}{(1 - e^{-\beta hf})^2} \right] \\
&= \frac{1}{1 + e^{-\beta \epsilon_0}} \left[\epsilon_0 e^{-\beta \epsilon_0} + \frac{hf e^{-\beta hf} (1 + e^{-\beta \epsilon_0})}{1 - e^{-\beta hf}} \right] \\
&= \frac{\epsilon_0 e^{-\beta \epsilon_0}}{1 + e^{-\beta \epsilon_0}} + \frac{hf e^{-\beta hf}}{1 - e^{-\beta hf}} \\
&= \frac{\epsilon_0}{\frac{1 + e^{-\beta \epsilon_0}}{e^{-\beta \epsilon_0}}} + \frac{hf}{\frac{1 - e^{-\beta hf}}{e^{-\beta hf}}} \\
&= \frac{\epsilon_0}{e^{\beta \epsilon_0} + 1} + \frac{hf}{e^{\beta hf} - 1} \\
&= \frac{\frac{\epsilon_0}{e^{\frac{\epsilon_0}{kT}} + 1}} + \frac{\frac{hf}{e^{\frac{hf}{kT}} - 1}}
\end{aligned}$$

Okay this is our final expression. We are now asked to approximate it for $kT \gg hf$ and $kT \gg \epsilon_0$ where we again use the Taylor approximation $e^x \approx 1 + x$, $|x| \ll 1$ thus

$$E = \frac{\epsilon_0}{e^{\frac{\epsilon_0}{kT}} + 1} + \frac{hf}{e^{\frac{hf}{kT}} - 1}$$

$$\approx \frac{\epsilon_0}{1 + \frac{\epsilon_0}{kT} + 1} + \frac{hf}{1 + \frac{hf}{kT} - 1}$$

$$= \frac{\epsilon_0}{2 + \frac{\epsilon_0}{kT}} + kT$$

Lets make another approximation that $2 \gg \frac{\epsilon_0}{kT}$ then

$$E \approx \frac{\epsilon_0}{2} + kT$$

Problem D3

Consider a single particle, which can occupy any of the 6 energy levels shown in figure 1. Three of the energy levels are at the energy ϵ_1 , two are at the energy ϵ_2 , and one is at the energy ϵ_3 . Here, $\epsilon_3 > \epsilon_2 > \epsilon_1$. The system is in thermal equilibrium with a reservoir at temperature T .

Question a:

Give an expression for the partition function Z of the system.

Just take care of the degeneracy of each energy and remember that each energy level is a state, s

$$Z = \sum_s e^{-\beta E(s)} = 3e^{-\beta\epsilon_1} + 2e^{-\beta\epsilon_2} + e^{-\beta\epsilon_3}$$

Question b:

Compute the mean energy $\bar{E}$ of the system.

The formula for mean energy is the derivative of the partition function

$$\begin{aligned}\bar{E} &= -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{1}{Z} [-3\epsilon_1 e^{-\beta\epsilon_1} - 2\epsilon_2 e^{-\beta\epsilon_2} - \epsilon_3 e^{-\beta\epsilon_3}] \\ &= \frac{3\epsilon_1 e^{-\beta\epsilon_1} + 2\epsilon_2 e^{-\beta\epsilon_2} + \epsilon_3 e^{-\beta\epsilon_3}}{3e^{-\beta\epsilon_1} + 2e^{-\beta\epsilon_2} + e^{-\beta\epsilon_3}}\end{aligned}$$

Question c:

Compute the limits $T \rightarrow 0$ and $T \rightarrow \infty$ of $\bar{E}$ and explain both limits physically.

When $T \rightarrow 0$ then $\beta = \frac{1}{kT} \rightarrow \infty$

$$\lim_{T \rightarrow 0} E = \lim_{\beta \rightarrow \infty} E = \lim_{\beta \rightarrow \infty} \left[\frac{3\epsilon_1 e^{-\beta\epsilon_1} + 2\epsilon_2 e^{-\beta\epsilon_2} + \epsilon_3 e^{-\beta\epsilon_3}}{3e^{-\beta\epsilon_1} + 2e^{-\beta\epsilon_2} + e^{-\beta\epsilon_3}} \right]$$

So all function appears to approach zero. But the one that approaches zero slowest is the one with ϵ_1 in the exponent since $\epsilon_1 < \epsilon_2 < \epsilon_3$. Therefore, the exponential function with ϵ_1 is relatively much larger than the other exponents in the limit and we can approximate them away as zero

$$\lim_{T \rightarrow 0} E = \lim_{\beta \rightarrow \infty} \left[\frac{3\epsilon_1 e^{-\beta\epsilon_1} + 2\epsilon_2 e^{-\beta\epsilon_2} + \epsilon_3 e^{-\beta\epsilon_3}}{3e^{-\beta\epsilon_1} + 2e^{-\beta\epsilon_2} + e^{-\beta\epsilon_3}} \right] = \lim_{\beta \rightarrow \infty} \left[\frac{3\epsilon_1 e^{-\beta\epsilon_1}}{3e^{-\beta\epsilon_1}} \right] = \epsilon_1$$

This makes sense since they will all be in ground state

When $T \rightarrow \infty$ then $\beta = \frac{1}{kT} \rightarrow 0$

$$\lim_{T \rightarrow \infty} E = \lim_{\beta \rightarrow 0} E = \lim_{\beta \rightarrow 0} \left[\frac{3\epsilon_1 e^{-\beta\epsilon_1} + 2\epsilon_2 e^{-\beta\epsilon_2} + \epsilon_3 e^{-\beta\epsilon_3}}{3e^{-\beta\epsilon_1} + 2e^{-\beta\epsilon_2} + e^{-\beta\epsilon_3}} \right] = \left[\frac{3\epsilon_1 + 2\epsilon_2 + \epsilon_3}{3 + 2 + 1} \right] = \frac{3\epsilon_1 + 2\epsilon_2 + \epsilon_3}{6}$$

This makes sense since infinite temperature means maximum entropy. So all energy levels will be equally occupied which gives us this average energy

S5

Problem 5.23. By subtracting μN from U , H , F , or G , one can obtain four new thermodynamic potentials. Of the four, the most useful is the **grand free energy** (or **grand potential**),

$$\Phi \equiv U - TS - \mu N.$$

- (a) Derive the thermodynamic identity for Φ , and the related formulas for the partial derivatives of Φ with respect to T , V , and μ .
 - (b) Prove that, for a system in thermal and diffusive equilibrium (with a reservoir that can supply both energy and particles), Φ tends to decrease.
- a) Let us find the thermodynamic identity of Φ by seeing what happens if we make a small change

$$d\Phi = dU - TdS - SdT - \mu dN - Nd\mu$$

We can also use equation 3.58

$$dU = TdS - PdV + \mu dN$$

Then we get

$$\begin{aligned} d\Phi &= TdS - PdV + \mu dN - TdS - SdT - \mu dN - Nd\mu \\ &= -PdV - SdT - Nd\mu \\ &= \left(\frac{\partial\Phi}{\partial V}\right)_{T,\mu} dV + \left(\frac{\partial\Phi}{\partial T}\right)_{V,\mu} dT + \left(\frac{\partial\Phi}{\partial\mu}\right)_{T,V} d\mu \end{aligned}$$

Therefore we can see that

$$\left(\frac{\partial\Phi}{\partial V}\right)_{T,\mu} = -P, \quad \left(\frac{\partial\Phi}{\partial T}\right)_{V,\mu} = -S, \quad \left(\frac{\partial\Phi}{\partial\mu}\right)_{T,V} = -N$$

- b) From equation 3.58

$$dU = TdS - PdV + \mu dN$$

$$\Updownarrow$$

$$dS = \frac{1}{T}(dU + PdV + \mu dN)$$

When we have thermal equilibrium then $\frac{\partial S_A}{\partial U_A} = \frac{\partial S_B}{\partial U_B} = \frac{1}{T}$. So the temperature is the same in both places and constant, which means $dT = 0$.

When we have diffusive equilibrium then $\left(\frac{\partial S_A}{\partial N_A}\right)_{U_A, V_A, N_B} = \left(\frac{\partial S_B}{\partial N_B}\right)_{U_B, V_B, N_A} = -\frac{\mu}{T}$. But since T is constant then μ must also be constant and $d\mu = 0$

Finally we also keep the volume constant so $dV = 0$

Since we want equilibrium then the change in entropy has to be zero. Our total entropy consists of a contribution from our system and for our reservoir must be zero

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$$dS_{tot} = dS_{sys} + dS_R = 0$$

$$\Updownarrow$$

$$dS_R = -dS_{sys} = -\frac{1}{T_{sys}}(dU_{sys} + P_{sys}dV_{sys} + \mu_{sys}dN_{sys})$$

But we just said that $T_{sys} = T_R$ and $\mu_{sys} = \mu_R$ and $dV = 0$ so

$$dS_R = -\frac{1}{T}(dU_{sys} + \mu dN_{sys})$$

Therefore the total entropy is

$$\begin{aligned} dS_{tot} &= dS_{sys} + dS_R = dS_{sys} - \frac{1}{T}(dU_{sys} + \mu dN_{sys}) \\ &= -\frac{1}{T}(dU_{sys} - TdS_{sys} + \mu dN_{sys}) \end{aligned}$$

Now we compare this to our thermodynamic identity which we found were

$$d\Phi = dU - TdS - SdT - \mu dN - Nd\mu$$

We can now invoke some of the things we found earlier $dT = d\mu = dV = 0$ which gives

$$d\Phi = dU - TdS - \mu dN$$

Notice now that we have

$$dS_{tot} = -\frac{1}{T}d\Phi_{sys}$$

Since the entropy seeks to maximize itself towards equilibrium then $d\Phi_{sys}$ has to be negative and decrease. Thus proving the initial statement

Problem 6.37. Use the Maxwell distribution to calculate the average value of v^2 for the molecules in an ideal gas. Check that your answer agrees with equation 6.41.

Maxwell distribution is given by

$$\mathcal{D}(v) = \left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} 4\pi v^2 e^{-\frac{mv^2}{2kT}}$$

We can find the average square speed by integrating from 0 to infinity since nothing can have negative speed (remember there is a difference between speed and velocity)

$$\begin{aligned} \langle v^2 \rangle &= \int_0^\infty v^2 \mathcal{D}(v) dv \\ &= \int_0^\infty v^2 \left(\frac{m}{2\pi kT}\right)^{\frac{3}{2}} 4\pi v^2 e^{-\frac{mv^2}{2kT}} dv \end{aligned}$$

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$$= 4\pi \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} \int_0^{\infty} v^4 e^{-\frac{mv^2}{2kT}} dv$$

From Schaum's Formulas and Tables we can find that the solution to the integral is

$$\int_0^{\infty} x^m e^{-ax^2} dx = \frac{\Gamma\left(\frac{m+1}{2}\right)}{2a^{\frac{m+1}{2}}}$$

Where

$$\Gamma\left(x + \frac{1}{2}\right) = \frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2x-1)}{2^x} \sqrt{\pi}$$

We can solve the integral now

$$\begin{aligned} \int_0^{\infty} v^4 e^{-\frac{mv^2}{2kT}} dv &= \frac{\Gamma\left(\frac{4+1}{2}\right)}{2\left(\frac{m}{2kT}\right)^{\frac{4+1}{2}}} \\ &= \left(\frac{2kT}{m}\right)^{\frac{5}{2}} \frac{\Gamma\left(2 + \frac{1}{2}\right)}{2} \\ &= \left(\frac{2kT}{m}\right)^{\frac{5}{2}} \frac{1}{2} \cdot \frac{1 \cdot 3}{2^2} \sqrt{\pi} \\ &= \left(\frac{2kT}{m}\right)^{\frac{5}{2}} \frac{3}{2^3} \sqrt{\pi} \end{aligned}$$

Inserting this result we get

$$\begin{aligned} \langle v^2 \rangle &= 4\pi \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} \left(\frac{2kT}{m} \right)^{\frac{5}{2}} \frac{3}{2^3} \sqrt{\pi} \\ &= 3 \frac{4}{2^3} \frac{\pi \sqrt{\pi}}{\pi^{\frac{3}{2}}} \cdot \left(\frac{m}{2kT} \right)^{\frac{3}{2}} \left(\frac{2kT}{m} \right)^{\frac{5}{2}} \\ &= \frac{3}{2} \cdot \frac{2kT}{m} \\ &= \frac{3kT}{m} \end{aligned}$$

This is in agreement with equation 6.41

Problem E1**Problem E1**

Consider two nuclei in a magnetic field B . One nucleus has spin $s = 1$ with the component $s_z = -1, 0, 1$ along the B field, and its energy in the three spin states is $-\epsilon_1, 0$, and ϵ_1 , respectively. The other nucleus has spin $s' = 1/2$ with the component $s'_z = -1/2, 1/2$ along the B field, and its energy in the two spin states is $-\epsilon_2$ and ϵ_2 , respectively. The total energy of the system is therefore

$$\epsilon_{\text{tot}} = s_z \epsilon_1 + 2s'_z \epsilon_2.$$

We have $\epsilon_1 > 0$ and $\epsilon_2 > 0$, and the two nuclei are in thermal equilibrium with a reservoir at temperature T . The system is illustrated in figure 1.

Question a:

Give the partition function Z of the system.

The partition function is given by

$$Z = \sum_E d(E) e^{-\beta E}$$

For two distinguishable particles then

$$Z_{\text{tot}} = Z_1 Z_2$$

The degeneracy of all energy level is zero so we just multiply the partition functions of each nucleus with each other

$$Z = (1 + e^{\beta \epsilon_1} + e^{-\beta \epsilon_1})(e^{\beta \epsilon_2} + e^{-\beta \epsilon_2})$$

Question b:

Calculate the average energy $E(T) = \overline{\epsilon_{\text{tot}}}$ of the system as a function of T . Find $\lim_{T \rightarrow 0} E(T)$ and $\lim_{T \rightarrow \infty} E(T)$, and give a physical interpretation of these limits.

We can now take the average energy of the system by the usual method

$$\begin{aligned} \bar{E} &= -\frac{1}{Z} \frac{\partial Z}{\partial \beta} \\ &= -\frac{\frac{\partial}{\partial \beta} (1 + e^{\beta \epsilon_1} + e^{-\beta \epsilon_1})(e^{\beta \epsilon_2} + e^{-\beta \epsilon_2})}{(1 + e^{\beta \epsilon_1} + e^{-\beta \epsilon_1})(e^{\beta \epsilon_2} + e^{-\beta \epsilon_2})} \\ &= -\frac{(e^{\beta \epsilon_2} + e^{-\beta \epsilon_2}) \frac{\partial}{\partial \beta} (1 + e^{\beta \epsilon_1} + e^{-\beta \epsilon_1}) + (1 + e^{\beta \epsilon_1} + e^{-\beta \epsilon_1}) \frac{\partial}{\partial \beta} (e^{\beta \epsilon_2} + e^{-\beta \epsilon_2})}{(1 + e^{\beta \epsilon_1} + e^{-\beta \epsilon_1})(e^{\beta \epsilon_2} + e^{-\beta \epsilon_2})} \\ &= -\frac{(e^{\beta \epsilon_2} + e^{-\beta \epsilon_2})(\epsilon_1 e^{\beta \epsilon_1} - \epsilon_1 e^{-\beta \epsilon_1}) + (1 + e^{\beta \epsilon_1} + e^{-\beta \epsilon_1})(\epsilon_2 e^{\beta \epsilon_2} - \epsilon_2 e^{-\beta \epsilon_2})}{(1 + e^{\beta \epsilon_1} + e^{-\beta \epsilon_1})(e^{\beta \epsilon_2} + e^{-\beta \epsilon_2})} \\ &= \frac{(e^{\beta \epsilon_2} + e^{-\beta \epsilon_2})(\epsilon_1 e^{-\beta \epsilon_1} - \epsilon_1 e^{\beta \epsilon_1}) + (1 + e^{\beta \epsilon_1} + e^{-\beta \epsilon_1})(\epsilon_2 e^{-\beta \epsilon_2} - \epsilon_2 e^{\beta \epsilon_2})}{(1 + e^{\beta \epsilon_1} + e^{-\beta \epsilon_1})(e^{\beta \epsilon_2} + e^{-\beta \epsilon_2})} \end{aligned}$$

$$\begin{aligned}
&= \frac{(e^{\beta\epsilon_2} + e^{-\beta\epsilon_2})(\epsilon_1 e^{-\beta\epsilon_1} - \epsilon_1 e^{\beta\epsilon_1})}{(1 + e^{\beta\epsilon_1} + e^{-\beta\epsilon_1})(e^{\beta\epsilon_2} + e^{-\beta\epsilon_2})} + \frac{(1 + e^{\beta\epsilon_1} + e^{-\beta\epsilon_1})(\epsilon_2 e^{-\beta\epsilon_2} - \epsilon_2 e^{\beta\epsilon_2})}{(1 + e^{\beta\epsilon_1} + e^{-\beta\epsilon_1})(e^{\beta\epsilon_2} + e^{-\beta\epsilon_2})} \\
&= \frac{\epsilon_1(e^{-\beta\epsilon_1} - e^{\beta\epsilon_1})}{1 + e^{\beta\epsilon_1} + e^{-\beta\epsilon_1}} + \frac{\epsilon_2(e^{-\beta\epsilon_2} - e^{\beta\epsilon_2})}{e^{\beta\epsilon_2} + e^{-\beta\epsilon_2}}
\end{aligned}$$

I cannot reduce it any further so this is the answer I would give. I now find the limits again by looking at which function in each denominator and numerator that grows the fastest

$$\lim_{T \rightarrow 0} \bar{E} = \lim_{\beta \rightarrow \infty} \bar{E} = \lim_{\beta \rightarrow \infty} \left[\frac{\epsilon_1(e^{-\beta\epsilon_1} - e^{\beta\epsilon_1})}{1 + e^{\beta\epsilon_1} + e^{-\beta\epsilon_1}} + \frac{\epsilon_2(e^{-\beta\epsilon_2} - e^{\beta\epsilon_2})}{e^{\beta\epsilon_2} + e^{-\beta\epsilon_2}} \right]$$

Here it is the positively growing functions that dominate so

$$\lim_{T \rightarrow 0} \bar{E} = \lim_{\beta \rightarrow \infty} \left[\frac{\epsilon_1(-e^{\beta\epsilon_1})}{e^{\beta\epsilon_1}} + \frac{\epsilon_2(-e^{\beta\epsilon_2})}{e^{\beta\epsilon_2}} \right] = -\epsilon_1 - \epsilon_2$$

So this means that both nuclei are in their ground state, which makes sense for $T = 0$

Now for the other more trivial limit

$$\begin{aligned}
\lim_{T \rightarrow \infty} \bar{E} &= \lim_{\beta \rightarrow 0} \bar{E} = \lim_{\beta \rightarrow 0} \left[\frac{\epsilon_1(e^{-\beta\epsilon_1} - e^{\beta\epsilon_1})}{1 + e^{\beta\epsilon_1} + e^{-\beta\epsilon_1}} + \frac{\epsilon_2(e^{-\beta\epsilon_2} - e^{\beta\epsilon_2})}{e^{\beta\epsilon_2} + e^{-\beta\epsilon_2}} \right] \\
&= \frac{\epsilon_1(1 - 1)}{1 + 1 + 1} + \frac{\epsilon_2(1 - 1)}{1 + 1} = 0
\end{aligned}$$

So this means that all energy levels are equally occupied which makes sense for a system of maximum entropy when $T = \infty$

We now take into account that the two spins interact with each other, so that the energy decreases by the amount U , when they are parallel, see figure 1. Thus, the energy of the system is $\epsilon_1 + \epsilon_2 - U$ when both spins are parallel with the B field, and the energy is $-\epsilon_1 - \epsilon_2 - U$ when both spins are anti-parallel to the B field. The energy of all other states is unchanged.

Question c:

Calculate the probability P that the two spins are parallel. Find $\lim_{U \rightarrow \infty} P$, and give a physical interpretation of this limit.

Now is the annoying part. We have to expand our partition function from earlier

$$\begin{aligned}
Z &= (1 + e^{\beta\epsilon_1} + e^{-\beta\epsilon_1})(e^{\beta\epsilon_2} + e^{-\beta\epsilon_2}) \\
&= e^{\beta\epsilon_2} + e^{\beta(\epsilon_1 + \epsilon_2)} + e^{-\beta(\epsilon_1 - \epsilon_2)} + e^{-\beta\epsilon_2} + e^{\beta(\epsilon_1 - \epsilon_2)} + e^{-\beta(\epsilon_1 + \epsilon_2)}
\end{aligned}$$

We now need to alter then energy of the system for the two cases where they are parallel such that our new partition function becomes

$$Z = e^{\beta\epsilon_2} + e^{\beta(\epsilon_1 + \epsilon_2 + U)} + e^{-\beta(\epsilon_1 - \epsilon_2)} + e^{-\beta\epsilon_2} + e^{\beta(\epsilon_1 - \epsilon_2)} + e^{-\beta(\epsilon_1 + \epsilon_2 - U)}$$

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The probability that the spins are parallel then becomes

$$P = \frac{e^{\beta(\epsilon_1 + \epsilon_2 + U)} + e^{-\beta(\epsilon_1 + \epsilon_2 - U)}}{e^{\beta\epsilon_2} + e^{\beta(\epsilon_1 + \epsilon_2 + U)} + e^{-\beta(\epsilon_1 - \epsilon_2)} + e^{-\beta\epsilon_2} + e^{\beta(\epsilon_1 - \epsilon_2)} + e^{-\beta(\epsilon_1 + \epsilon_2 - U)}}$$

In the limit we find that

$$\lim_{U \rightarrow \infty} P = \frac{e^{\beta(\epsilon_1 + \epsilon_2 + U)} + e^{-\beta(\epsilon_1 + \epsilon_2 - U)}}{e^{\beta\epsilon_2} + e^{\beta(\epsilon_1 + \epsilon_2 + U)} + e^{-\beta(\epsilon_1 - \epsilon_2)} + e^{-\beta\epsilon_2} + e^{\beta(\epsilon_1 - \epsilon_2)} + e^{-\beta(\epsilon_1 + \epsilon_2 - U)}}$$

Notice that out of all of them that $e^{\beta(\epsilon_1 + \epsilon_2 + U)}$ grows the fastest in both the denominator and numerator so

$$\lim_{U \rightarrow \infty} P = \frac{e^{\beta(\epsilon_1 + \epsilon_2 + U)}}{e^{\beta(\epsilon_1 + \epsilon_2 + U)}} = 1$$

When the interaction energy U approaches infinity. Then it becomes so favorable that the nuclei will always want to be parallel.

Problem E2**Problem E2**

A system, which is in fact a defect in a crystal, is in thermal equilibrium with a heat reservoir with temperature T . The defect is characterized by its polarization vector $\vec{p}$, which has a constant length p and 8 possible directions given by the normal vectors in an xyz -coordinate system,

$$\vec{p} = (\pm 1, \pm 1, \pm 1) \frac{p}{\sqrt{3}}. \quad (1)$$

In an electric field pointing in the $(1, 1, 1)$ direction, i.e. $\vec{E} = (1, 1, 1) \cdot E/\sqrt{3}$, the energy of the defect is

$$E(\vec{p}) = -\vec{p} \cdot \vec{E}. \quad (2)$$

Question a:

Show that the defect has 4 different energies, and determine the degeneracy of each level.

The energy levels are given by

$$\begin{aligned} E(\vec{p}) &= -\vec{p} \cdot \vec{E} = -\begin{pmatrix} \pm 1 \\ \pm 1 \\ \pm 1 \end{pmatrix} \frac{p}{\sqrt{3}} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \frac{E}{\sqrt{3}} \\ &= -\frac{Ep}{3} (\pm 1 \pm 1 \pm 1) \end{aligned}$$

The case with $[+ + +]$ has an energy level of

$$E = -Ep$$

The case with $[- - -]$ has an energy level of

$$E = Ep$$

The 3 cases with $[+ + -, + - +, - + +]$ has an energy level of

$$E = -\frac{Ep}{3}$$

The 3 cases with $[+ - -, - + -, - - +]$ has an energy level of

$$E = \frac{Ep}{3}$$

Thus we have 4 energy levels. Two of them only singly degenerate and the other two are triply degenerate

Question b:

Show that the partition function for one defect can be written as

$$Z(T) = 8 \cosh^3(x) \quad (3)$$

with $x = pE/(3kT)$.

The partition function is

$$\begin{aligned} Z &= \sum_E d(E) e^{-\beta E} \\ &= e^{\beta Ep} + e^{-\beta Ep} + 3e^{\frac{\beta Ep}{3}} + 3e^{-\frac{\beta Ep}{3}} \end{aligned}$$

Let $x = \frac{pE}{3kT}$ then

$$\begin{aligned} Z &= e^{3x} + e^{-3x} + 3e^x + 3e^{-x} \\ &= (e^x + e^{-x})^3 \\ &= 8 \cosh^3(x) \end{aligned}$$

Where the hyperbolic cosine function is

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

Question c:

Determine the average energy $\bar{E}$ of one defect as a function of T and sketch it. Calculate $\bar{E}$ in the limits $T \rightarrow 0$ and $T \rightarrow \infty$.

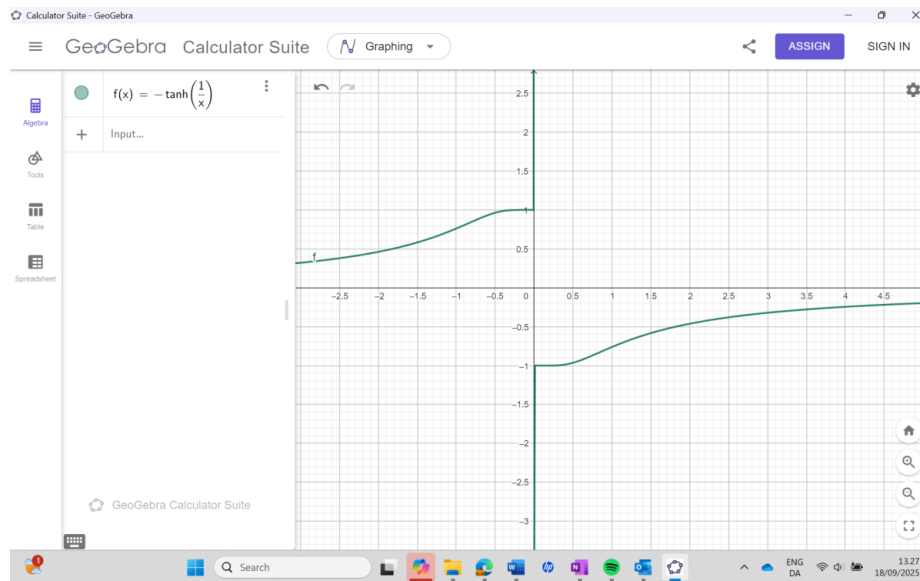
The average energy is given by

$$\begin{aligned} \bar{E} &= -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{1}{Z} \frac{\partial x}{\partial \beta} \frac{\partial Z}{\partial x} = -\frac{pE}{3Z} \frac{\partial Z}{\partial x} \\ &= -\frac{pE}{24 \cosh^3(x)} 8 \cdot 3 \cosh^2(x) \sinh(x) \\ &= -pE \tanh(x) \\ &= -pE \frac{e^x - e^{-x}}{e^x + e^{-x}} \end{aligned}$$

Now we can find the limits relatively easily. Use the same method as I've explained since week 1

$$\begin{aligned} \lim_{T \rightarrow 0} \bar{E} &= \lim_{x \rightarrow \infty} \bar{E} = -pE \\ \lim_{T \rightarrow \infty} \bar{E} &= \lim_{x \rightarrow 0} \bar{E} = 0 \end{aligned}$$

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Question d:

Determine the heat capacity C_V of one defect. Sketch also this function of T and comment on the limits $T \rightarrow 0$ and $T \rightarrow \infty$.

$$\begin{aligned} C_V &= \left(\frac{\partial E}{\partial T} \right)_{V,N} = \frac{\partial x}{\partial T} \frac{\partial E}{\partial x} = -\frac{pE}{3kT^2} \cdot (-pE \operatorname{sech}^2(x)) \\ &= \frac{p^2 E^2}{3kT^2} \operatorname{sech}^2(x) \\ &= \frac{p^2 E^2}{3kT^2} \operatorname{sech}^2\left(\frac{pE}{3kT}\right) \\ &= \frac{p^2 E^2}{3kT^2} \cdot \frac{2}{e^{\frac{pE}{3kT}} + e^{-\frac{pE}{3kT}}} \end{aligned}$$

Now we can find the limits relatively easily. Use the same method as I've explained since week 1

$$\lim_{T \rightarrow 0} C_V = \lim_{T \rightarrow 0} \frac{p^2 E^2}{3kT^2} \cdot \frac{2}{e^{\frac{pE}{3kT}} + e^{-\frac{pE}{3kT}}} = \lim_{T \rightarrow 0} \frac{p^2 E^2}{3kT^2} \cdot \frac{2}{e^{\frac{pE}{3kT}}} = \lim_{T \rightarrow 0} \frac{2p^2 E^2}{3k} \cdot \frac{T^{-2}}{e^{\frac{pE}{3kT}}}$$

It is not trivial here which function grows fastest. The T^{-2} tends to infinity and the exponential function tends to 0, so we have to find out which dominates with good old L'hôpital!

$$\lim_{T \rightarrow 0} \frac{T^{-2}}{e^{\frac{pE}{3kT}}} \stackrel{LH}{=} \lim_{T \rightarrow 0} \frac{-2T^{-3}}{-\frac{pE}{3kT^2} e^{\frac{pE}{3kT}}} = \lim_{T \rightarrow 0} \frac{T^{-1}}{\frac{pE}{3k} e^{\frac{pE}{3kT}}} \stackrel{LH}{=} \lim_{T \rightarrow 0} \frac{2}{\left(\frac{pE}{3k}\right)^2 e^{\frac{pE}{3kT}}} = 0$$

Therefore

$$\lim_{T \rightarrow 0} C_V = 0$$

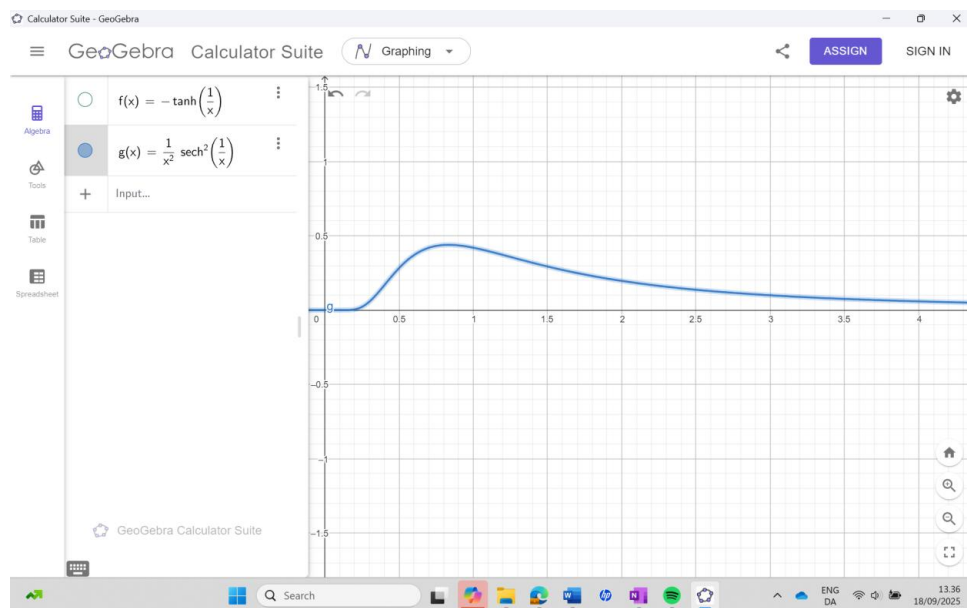
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The other limit is trivial

$$\lim_{T \rightarrow \infty} C_V = 0$$

When $T \approx 0$ then intuitively the system cannot supply enough energy to make the defect jump to the next energy level, which would result in zero heat capacity

When $T = \infty$ then intuitively the system is already at maximum entropy. That is, all states are equally occupied. If you heat up the system even more, then it won't be able to take up more energy and therefore it will have zero heat capacity



Question e:

Determine the entropy S of one defect as a function of T . Comment on the limits $T \rightarrow 0$ and $T \rightarrow \infty$.

We can use Helmholtz free energy

$$F = U - TS = -kT \ln(Z)$$

$$\Updownarrow$$

$$S = \frac{U}{T} + k \ln(Z)$$

$$= -\frac{pE}{T} \tanh(x) + k \ln(8 \cosh^3(x))$$

$$= -3kx \tanh(x) + k \ln(8 \cosh^3(x))$$

$$= -3kx \tanh(x) + 3k \ln(2 \cosh(x))$$

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$$= -3kx \frac{e^x - e^{-x}}{e^x + e^{-x}} + 3k \ln(e^x + e^{-x}), \quad x = \frac{pE}{3kT}$$

We can now find the limits

$$\lim_{T \rightarrow 0} S = \lim_{x \rightarrow \infty} S = -3kx \frac{e^x}{e^x} + 3k \ln(e^x) = 0$$

$$\lim_{T \rightarrow \infty} S = \lim_{x \rightarrow 0} S = 3k \ln(2) = k \ln(8)$$

When the temperature is zero then the defect is in ground state with multiplicity one, so $S = k \ln(\Omega) = k \ln(1) = 0$

When the temperature is infinite then the defect has maximum entropy and all states are occupied with equal probability such that we have $\Omega = 8$

Problem E3**Problem E3**

We consider the chemical reaction $A + B \rightleftharpoons AB$. The gases of A -atoms, B -atoms, and AB -molecules can be considered ideal, and they are confined in a container of volume V at constant temperature T . The total number of A -atoms and B -atoms are N_A^0 and N_B^0 so that there are $N_A = N_A^0 - N_{AB}$ and $N_B = N_B^0 - N_{AB}$ free A -atoms and B -atoms in the container. Here, N_{AB} denotes the number of AB -molecules. Let Z_i be the single-particle partition function given by Schroeder Eq. (6.84), where $i \in \{A, B, AB\}$.

Question a:

Calculate the free energy $F = U - TS$ of the mixture.

We have three distinguishable types of atoms which means that the partition function can be written as

$$Z_{tot} = Z_A Z_B Z_{AB}$$

The atoms of each types are however indistinguishable from each other. And we assume that $Z_i \gg N_i$ then

$$Z_{tot} = \frac{Z_A^{N_A} Z_B^{N_B} Z_{AB}^{N_{AB}}}{N_A! N_B! N_{AB}!}$$

The free energy is given by

$$F = -kT \ln(Z)$$

$$\Downarrow$$

$$\frac{F}{kT} = \ln(N_A!) + \ln(N_B!) + \ln(N_{AB}!) - N_A \ln(Z_A) - N_B \ln(Z_B) - N_{AB} \ln(Z_{AB})$$

We can use Stirling's approximations

$$\ln(N!) \approx N \ln(N) - N$$

Then we get

$$\begin{aligned} \frac{F}{kT} = & N_A \ln(N_A) - N_A + N_B \ln(N_B) - N_B + N_{AB} \ln(N_{AB}) - N_{AB} - N_A \ln(Z_A) - N_B \ln(Z_B) \\ & - N_{AB} \ln(Z_{AB}) \end{aligned}$$

We cannot do anymore about this expression

Question b:

Use that F is minimum for constant volume V and temperature T to show that in equilibrium

$$\frac{n_{AB}}{n_A n_B} = V \frac{Z_{AB}}{Z_A Z_B} \equiv K(T), \quad (4)$$

where $n_i = N_i/V$. Equation (4) is called the law of mass action.

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We have only one variable to determine equilibrium and that is N_{AB} which means that

$$\left(\frac{\partial F}{\partial N_{AB}} \right)_{V,T} = 0$$

We insert $N_A = N_A^0 - N_{AB}$ and $N_B = N_B^0 - N_{AB}$ then we get

$$\begin{aligned} \frac{F}{kT} = & (N_A^0 - N_{AB}) \ln(N_A^0 - N_{AB}) - (N_A^0 - N_{AB}) + (N_B^0 - N_{AB}) \ln(N_B^0 - N_{AB}) - (N_B^0 - N_{AB}) \\ & + N_{AB} \ln(N_{AB}) - N_{AB} - (N_A^0 - N_{AB}) \ln(Z_A) - (N_B^0 - N_{AB}) \ln(Z_B) - N_{AB} \ln(Z_{AB}) \end{aligned}$$

Now we take the derivative

$$\left(\frac{\partial F}{\partial N_{AB}} \right)_{V,T} = \left(\frac{\partial \frac{F}{kT}}{\partial N_{AB}} \right)_{V,T} = 0$$

To make life a little easier notice that

$$\begin{aligned} ((k-x) \ln(k-x))' &= -\frac{k-x}{k-x} - \ln(k-x) = -1 - \ln(k-x) \\ (x \ln(x))' &= 1 + \ln(x) \end{aligned}$$

Then we get

$$\begin{aligned} \frac{\partial \frac{F}{kT}}{\partial N_{AB}} &= -1 - \ln(N_A^0 - N_{AB}) + 1 - 1 - \ln(N_B^0 - N_{AB}) + 1 + 1 + \ln(N_{AB}) - 1 + \ln(Z_A) + \ln(Z_B) \\ &\quad - \ln(Z_{AB}) \\ &= -\ln(N_A^0 - N_{AB}) - \ln(N_B^0 - N_{AB}) + \ln(N_{AB}) + \ln(Z_A) + \ln(Z_B) - \ln(Z_{AB}) \\ &= \ln \left(\frac{N_{AB}}{(N_A^0 - N_{AB})(N_B^0 - N_{AB})} \right) + \ln \left(\frac{Z_A Z_B}{Z_{AB}} \right) \\ &= \ln \left(\frac{N_{AB}}{N_A N_B} \right) - \ln \left(\frac{Z_{AB}}{Z_A Z_B} \right) = 0 \\ &\quad \Updownarrow \\ &\ln \left(\frac{N_{AB}}{N_A N_B} \right) = \ln \left(\frac{Z_{AB}}{Z_A Z_B} \right) \\ &\quad \Updownarrow \\ &\frac{N_{AB}}{N_A N_B} = \frac{Z_{AB}}{Z_A Z_B} \\ &\quad \Updownarrow \\ &\frac{\frac{N_{AB}}{V}}{\frac{N_A}{V} \cdot \frac{N_B}{V}} = \frac{n_{AB}}{n_A n_B} = \frac{V Z_{AB}}{Z_A Z_B} \end{aligned}$$

Question c:

We now neglect any internal degrees of freedom, i.e. $Z_{\text{int}} = 1$ in Eq. (6.84). Discuss what happens if T increases. Discuss what happens if the ratio $m_{AB}/(m_A m_B)$ increases, where m_i is the mass of species i .

The partition function can be separated into translational motion and internal motion (rotation and vibration)

$$Z_1 = Z_{tr} Z_{int}$$

Where

$$Z_{tr} = \frac{V}{v_Q} = V \left(\frac{h}{\sqrt{2\pi m k T}} \right)^{-3}$$

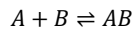
Therefore

$$\begin{aligned} \frac{V Z_{AB}}{Z_A Z_B} &= \frac{V^2 \left(\frac{h}{\sqrt{2\pi m_{AB} k T}} \right)^{-3}}{V \left(\frac{h}{\sqrt{2\pi m_A k T}} \right)^{-3} V \left(\frac{h}{\sqrt{2\pi m_B k T}} \right)^{-3}} \\ &= \frac{\left(\frac{1}{\sqrt{m_{AB}}} \right)^{-3}}{\left(\frac{1}{\sqrt{m_A}} \right)^{-3} \left(\frac{1}{\sqrt{m_B}} \right)^{-3}} \\ &= \frac{\left(\frac{1}{\sqrt{m_A}} \right)^3 \left(\frac{1}{\sqrt{m_B}} \right)^3}{\left(\frac{1}{\sqrt{m_{AB}}} \right)^3} \\ &= \left(\frac{\sqrt{m_{AB}}}{\sqrt{m_A} \sqrt{m_B}} \right)^3 \left(\frac{h}{\sqrt{2\pi k T}} \right)^3 \\ &= \left(\frac{m_{AB}}{m_A m_B} \right)^{\frac{3}{2}} \left(\frac{h}{\sqrt{2\pi k T}} \right)^3 \end{aligned}$$

We know have this relationship

$$\frac{n_{AB}}{n_A n_B} = \left(\frac{m_{AB}}{m_A m_B} \right)^{\frac{3}{2}} \left(\frac{h}{\sqrt{2\pi k T}} \right)^3$$

We have the reaction



This is kind of like Le Chateliers principle. When the temperature increases then the reaction shifts towards the left. If I had to give an intuitive explanation then the translational energy

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causes the AB molecules to break up into A and B. This kind of makes sense if you have things flying insanely fast into each other they tend to break instead of binding to each other.

When $\frac{m_{AB}}{m_A m_B}$ increases then the reaction shifts towards the right. If the mass of AB is a lot larger than A and B then we could think of it as the overall energy being converted into mass. They slow down and won't break at each collision. This is all hand-wavy and you could probably find a better intuitive explanation for this

Problem E4**Problem E4**

We consider an ideal gas of N identical atoms in a volume V and at temperature T . In addition to the translational degrees of freedom, each atom has two internal energy levels with energies ϵ_1 and ϵ_2 . We have $\epsilon_2 > \epsilon_1$.

Question a:

Find the partition function Z_1 for a single atom in the gas and the partition function Z for all N atoms. Include the fact that the particles are indistinguishable only through the factor $N!$.

The partition function for the internal energy levels is

$$Z_{int} = \sum_E d(E) e^{-\beta E} = e^{-\beta \epsilon_1} + e^{-\beta \epsilon_2}$$

The partition function for the translational energy levels is

$$Z_{tr} = \frac{V}{v_Q} = V \left(\frac{\sqrt{2\pi m k T}}{h} \right)^3$$

Therefore the partition function for single atom in the gas is

$$Z_1 = Z_{tr} Z_{int} = V \left(\frac{\sqrt{2\pi m k T}}{h} \right)^3 (e^{-\beta \epsilon_1} + e^{-\beta \epsilon_2})$$

For indistinguishable atoms the total partition function is

$$Z_{total} = \frac{1}{N!} Z_1^N = \frac{1}{N!} \left[V \left(\frac{\sqrt{2\pi m k T}}{h} \right)^3 (e^{-\beta \epsilon_1} + e^{-\beta \epsilon_2}) \right]^N$$

Question b:

Calculate the energy $U(T)$ and the heat capacity $C_V(T)$ of the gas, and sketch the two functions. What are $U(T)$ and $C_V(T)$ in the limits $T \ll (\epsilon_2 - \epsilon_1)/k$ and $T \gg (\epsilon_2 - \epsilon_1)/k$?

The energy can be found by the formula

$$U = -\frac{\partial \ln(Z)}{\partial \beta} = -N \frac{\partial \ln(Z_{int})}{\partial \beta} - N \frac{\partial \ln(Z_{tr})}{\partial \beta} = U_{int} + \frac{3}{2} N k T$$

Where

$$U_{int} = -\frac{N}{Z_{int}} \frac{\partial Z_{int}}{\partial \beta} = \frac{N \epsilon_1 e^{-\beta \epsilon_1} + N \epsilon_2 e^{-\beta \epsilon_2}}{e^{-\beta \epsilon_1} + e^{-\beta \epsilon_2}}$$

Therefore

$$U = \frac{N \epsilon_1 e^{-\beta \epsilon_1} + N \epsilon_2 e^{-\beta \epsilon_2}}{e^{-\beta \epsilon_1} + e^{-\beta \epsilon_2}} + \frac{3}{2} N k T$$

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$$= \frac{N\epsilon_1 + N\epsilon_2 e^{-\beta(\epsilon_2 - \epsilon_1)}}{1 + e^{-\beta(\epsilon_2 - \epsilon_1)}} + \frac{3}{2} NkT$$

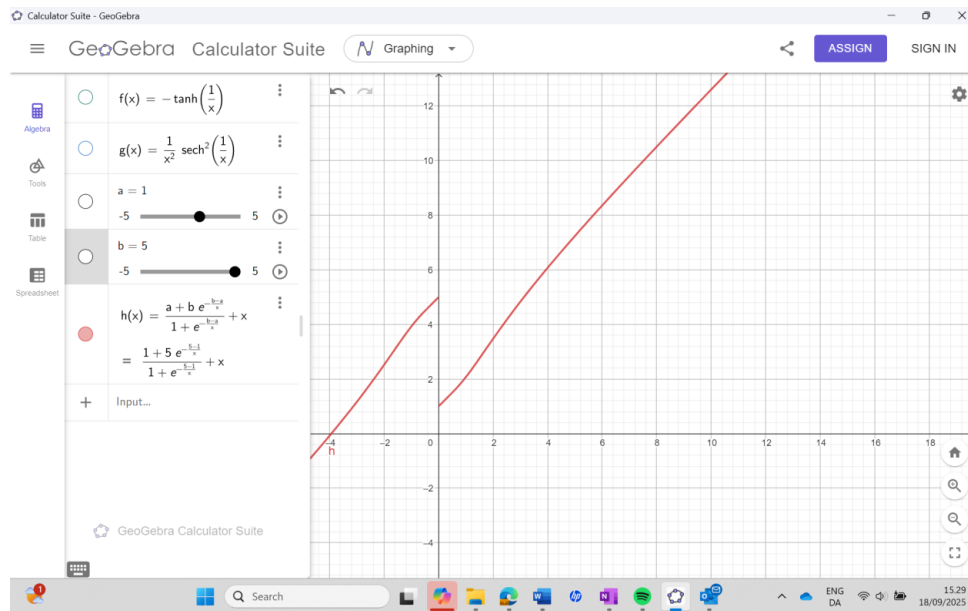
$$= \frac{N\epsilon_1 + N\epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}}{1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}} + \frac{3}{2} NkT$$

In the limit where $kT \ll (\epsilon_2 - \epsilon_1)$ then

$$U \approx N\epsilon_1 + \frac{3}{2} NkT$$

In the limit where $kT \gg (\epsilon_2 - \epsilon_1)$ then

$$U \approx \frac{N(\epsilon_1 + \epsilon_2)}{2} + \frac{3}{2} NkT$$



Now calculating the heat capacity

$$C_V = \left(\frac{\partial U}{\partial T} \right)_{V,N}$$

$$= \frac{\partial}{\partial T} \left[\frac{N\epsilon_1 + N\epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}}{1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}} + \frac{3}{2} NkT \right]$$

$$= \frac{3}{2} Nk + \frac{1}{1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}} \frac{\partial}{\partial T} \left[N\epsilon_1 + N\epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right] + \left(N\epsilon_1 + N\epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right) \frac{\partial}{\partial T} \left[\frac{1}{1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}} \right]$$

$$\begin{aligned}
&= \frac{3}{2}Nk + \frac{N\epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}}{1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}} \frac{\partial}{\partial T} \left[-\frac{(\epsilon_2 - \epsilon_1)}{kT} \right] - \frac{\left(N\epsilon_1 + N\epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right)}{\left(1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right)^2} \frac{\partial}{\partial T} \left[1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right] \\
&= \frac{3}{2}Nk + \frac{N\epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}}{1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}} \frac{(\epsilon_2 - \epsilon_1)}{kT^2} - \frac{e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \left(N\epsilon_1 + N\epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right)}{\left(1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right)^2} \frac{\partial}{\partial T} \left[-\frac{(\epsilon_2 - \epsilon_1)}{kT} \right] \\
&= \frac{3}{2}Nk + \frac{N\epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}}{1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}} \frac{(\epsilon_2 - \epsilon_1)}{kT^2} - \frac{e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \left(N\epsilon_1 + N\epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right) (\epsilon_2 - \epsilon_1)}{\left(1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right)^2 kT^2} \\
&= \frac{3}{2}Nk + \frac{N(\epsilon_2 - \epsilon_1)}{kT^2} \left[\frac{\epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}}{1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}} - \frac{e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \left(\epsilon_1 + \epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right)}{\left(1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right)^2} \right] \\
&= \frac{3}{2}Nk + \frac{N(\epsilon_2 - \epsilon_1)}{kT^2} \left[\frac{\epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \left(1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right) - e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \left(\epsilon_1 + \epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right)}{\left(1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right)^2} \right] \\
&= \frac{3}{2}Nk + \frac{N(\epsilon_2 - \epsilon_1)}{kT^2} e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \left[\frac{\left(\epsilon_2 + \epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right) - \left(\epsilon_1 + \epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right)}{\left(1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right)^2} \right] \\
&= \frac{3}{2}Nk + \frac{N(\epsilon_2 - \epsilon_1)}{kT^2} e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \left[\frac{(\epsilon_2 - \epsilon_1)}{\left(1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right)^2} \right] \\
&= \frac{3}{2}Nk + \frac{N(\epsilon_2 - \epsilon_1)^2}{kT^2} \left[\frac{e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}}{\left(1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right)^2} \right] \\
&= \frac{3}{2}Nk + Nk \left(\frac{\epsilon_2 - \epsilon_1}{kT} \right)^2 \left[\frac{e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}}{\left(1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right)^2} \right]
\end{aligned}$$

In the limit where $kT \ll (\epsilon_2 - \epsilon_1)$ then

$$C_V \approx \frac{3}{2}Nk + Nk \left(\frac{\epsilon_2 - \epsilon_1}{kT} \right)^2 \left[e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right]$$

This limit we calculated in Problem E2

$$\lim_{T \rightarrow 0} \frac{T^{-2}}{e^{\frac{pE}{3kT}}} \stackrel{LH}{=} 0$$

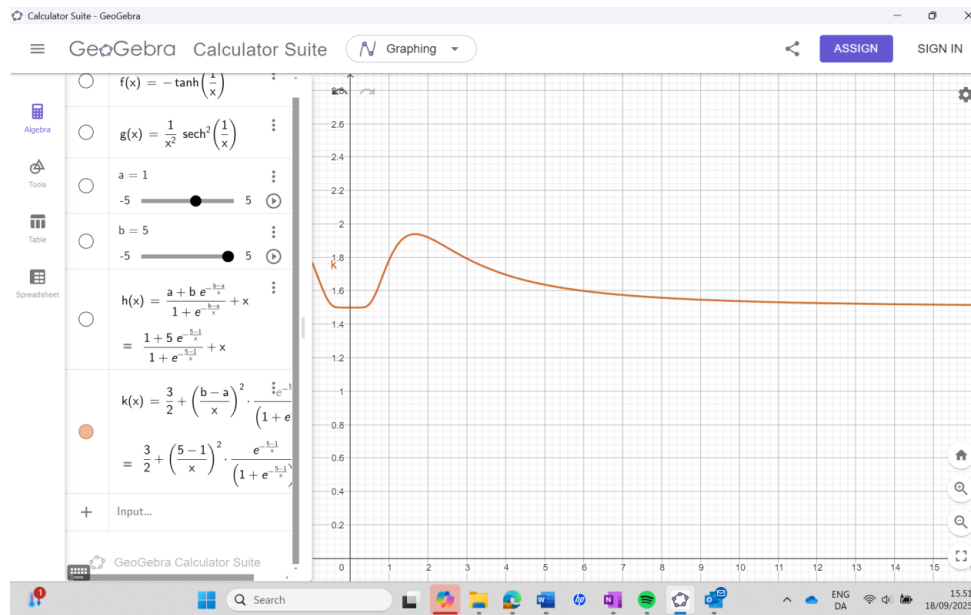
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Therefore

$$C_V \approx \frac{3}{2}Nk$$

In the limit where $kT \gg (\epsilon_2 - \epsilon_1)$ then

$$C_V \approx \frac{3}{2}Nk$$



Question c:

Calculate the entropy $S(T)$ of the gas, and show that it can be written as $S = S_{\text{trans}} + S_{\text{int}}$. Determine S_{int} for $T \ll (\epsilon_2 - \epsilon_1)/k$ and $T \gg (\epsilon_2 - \epsilon_1)/k$.

To determine the entropy we use

$$F = U - TS = -kT \ln(Z)$$

$$\Updownarrow$$

$$S = \frac{U}{T} + k \ln(Z)$$

The energy we can write as $U = U_{\text{int}} + U_{\text{tr}}$ and

$$Z = \frac{1}{N!} Z_1^N = \frac{1}{N!} (Z_{\text{tr}} Z_{\text{int}})^N$$

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Then we get

$$S = \frac{U_{tr}}{T} + k \ln \left(\frac{Z_{tr}^N}{N!} \right) + \frac{U_{int}}{T} + k \ln(Z_{int}^N) = S_{tr} + S_{int}$$

We find using Stirling's approximation

$$\begin{aligned} S_{tr} &= \frac{3}{2} Nk + k \ln \left(\frac{Z_{tr}^N}{N!} \right) \\ &= \frac{3}{2} Nk + Nk \ln(Z_{tr}) - Nk \ln(N) + Nk \\ &= \frac{5}{2} Nk + Nk \ln \left(\frac{Z_{tr}}{N} \right) \end{aligned}$$

We know that

$$Z_{tr} = \frac{V}{v_Q} = V \left(\frac{\sqrt{2\pi m k T}}{h} \right)^3$$

Then

$$S_{tr} = \frac{5}{2} Nk + Nk \ln \left(\frac{V}{N} \left(\frac{\sqrt{2\pi m k T}}{h} \right)^3 \right)$$

This is Sackur-Tetrode equation for the one with a keen eye

Now we find

$$\begin{aligned} S_{int} &= \frac{U_{int}}{T} + k \ln(Z_{int}^N) \\ &= \frac{1}{T} \left(\frac{N\epsilon_1 + N\epsilon_2 e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}}{1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}} \right) + Nk \ln(e^{-\beta\epsilon_1} + e^{-\beta\epsilon_2}) \\ &= \left(\frac{\frac{N\epsilon_1}{T} + \frac{N\epsilon_2}{T} e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}}{1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}} \right) + Nk \ln \left(e^{-\beta\epsilon_1} \left(1 + e^{-\frac{\epsilon_2 - \epsilon_1}{kT}} \right) \right) \\ &= \left(\frac{\frac{N\epsilon_1}{T} + \frac{N\epsilon_2}{T} e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}}{1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}} \right) + Nk \ln \left(1 + e^{-\frac{\epsilon_2 - \epsilon_1}{kT}} \right) - \frac{Nk\epsilon_1}{kT} \\ &\quad \Updownarrow \\ S_{int} &= \left(\frac{\frac{\epsilon_1}{kT} + \frac{\epsilon_2}{kT} e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}}{1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}} \right) + \ln \left(1 + e^{-\frac{\epsilon_2 - \epsilon_1}{kT}} \right) - \frac{\epsilon_1}{kT} \end{aligned}$$

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$$\begin{aligned}
&= \left(\frac{\frac{\epsilon_1}{kT} + \frac{\epsilon_2}{kT} e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} - \frac{\epsilon_1}{kT} \left(1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \right)}{1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}} \right) + \ln \left(1 + e^{-\frac{\epsilon_2 - \epsilon_1}{kT}} \right) \\
&= \left(\frac{\frac{\epsilon_2}{kT} e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} - \frac{\epsilon_1}{kT} e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}}{1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}} \right) + \ln \left(1 + e^{-\frac{\epsilon_2 - \epsilon_1}{kT}} \right) \\
&= \left(\frac{e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}} \left(\frac{\epsilon_2 - \epsilon_1}{kT} \right)}{1 + e^{-\frac{(\epsilon_2 - \epsilon_1)}{kT}}} \right) + \ln \left(1 + e^{-\frac{\epsilon_2 - \epsilon_1}{kT}} \right)
\end{aligned}$$

In the limit where $kT \ll (\epsilon_2 - \epsilon_1)$ then we can use

$$\lim_{T \rightarrow 0} \frac{T^{-2}}{e^{\frac{1}{T}}} \stackrel{LH}{=} \lim_{T \rightarrow 0} \frac{T^{-1}}{e^{\frac{1}{T}}} = 0$$

Then we get

$$S \approx 0$$

In the limit where $kT \gg (\epsilon_2 - \epsilon_1)$ then

$$\begin{aligned}
\frac{S}{Nk} &\approx \left(\frac{1 \left(\frac{\epsilon_2 - \epsilon_1}{kT} \right)}{1 + 1} \right) + \ln(1 + 1) = \ln(2) \\
&\Updownarrow \\
S &= Nk \ln(2)
\end{aligned}$$

Problem 7.28. Consider a free Fermi gas in two dimensions, confined to a square area $A = L^2$.

- Find the Fermi energy (in terms of N and A), and show that the average energy of the particles is $\epsilon_F/2$.
- Derive a formula for the density of states. You should find that it is a constant, independent of ϵ .
- Explain how the chemical potential of this system should behave as a function of temperature, both when $kT \ll \epsilon_F$ and when T is much higher.
- Because $g(\epsilon)$ is a constant for this system, it is possible to carry out the integral 7.53 for the number of particles analytically. Do so, and solve for μ as a function of N . Show that the resulting formula has the expected qualitative behavior.
- Show that in the high-temperature limit, $kT \gg \epsilon_F$, the chemical potential of this system is the same as that of an ordinary ideal gas.

- a) From quantum mechanics we know the solution of the energy levels for an infinite square well in higher dimensions are generalized as

$$\epsilon = \frac{h^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2 + \dots)$$

But we only have 2 dimensions so the energy levels that we can play around with are

$$\epsilon = \frac{h^2}{8mL^2} (n_x^2 + n_y^2) = \frac{h^2}{8mL^2} |\vec{n}|^2$$

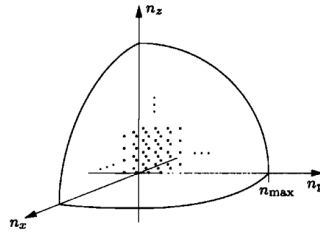
Our Fermi level is the maximum value of $|\vec{n}|$ that is occupied

$$\epsilon_F = \frac{h^2}{8mL^2} n_{\max}^2 = \frac{h^2 n_{\max}^2}{8mA}$$

Since we have a Fermi gas then each particle will be able to occupy two states

We know have the following picture but in a 2D plane for the occupied states.

Figure 7.10. Each triplet of integers (n_x, n_y, n_z) represents a pair of definite-energy electron states (one with each spin orientation). The set of all independent states fills the positive octant of n -space.



Instead of occupying $1/8$ of the sphere in the 3D case, we now occupy $1/4$ of the circle in a 2D case. We can say that the number of particles is exactly 2 times the number of states within our quarter circle because it is a Fermi gas. And the circle has radius n_{max} . Again assume that the roughness at the edge is negligent because of a large n_{max} analogous to the derivation in the book

$$N = 2 \cdot \frac{\pi n_{max}^2}{4} = \frac{\pi n_{max}^2}{2}$$

$$\Updownarrow$$

$$n_{max} = \sqrt{\frac{2N}{\pi}}$$

Inserting into our expression for Fermi energy

$$\epsilon_F = \frac{h^2}{8mA} \cdot \frac{2N}{\pi} = \frac{h^2 N}{4\pi mA}$$

The total energy of our Fermi gas is the sum of all states which we can approximate by an integral

$$U = 2 \sum_{n_x} \sum_{n_y} \epsilon(\vec{n}) = 2 \iint \epsilon(\vec{n}) dn_x dn_y = 2 \iint \frac{h^2}{8mA} (n_x^2 + n_y^2) dn_x dn_y$$

We can calculate the integral using circular coordinates to calculate the area of the quarter circle. Converting cartesian coordinates to circular coordinates we use that

$$r = \sqrt{n_x^2 + n_y^2}$$

Then we get the energy

$$\begin{aligned} U &= 2 \int_0^{n_{max}} \int_0^{\frac{\pi}{2}} \frac{h^2}{8mA} (r^2) r d\phi dr \\ &= \frac{h^2}{4mA} \int_0^{\frac{\pi}{2}} d\phi \int_0^{n_{max}} r^3 dr \\ &= \frac{h^2 \pi}{8mA} \left[\frac{r^4}{4} \right]_0^{n_{max}} \\ &= \frac{h^2 \pi n_{max}^4}{32mA} \\ &= \frac{h^2 \pi}{32mA} \left(\frac{2N}{\pi} \right)^2 \\ &= \frac{h^2 N^2}{8\pi mA} \end{aligned}$$

The average energy is therefore

$$\bar{E} = \frac{U}{N} = \frac{h^2 N}{8\pi mA} = \frac{\epsilon_F}{2}$$

Which is what we wanted to show

b) Now we derive a formula for the density of states

The density of states, g , is described by counting all of the states, which in our simplified example is simply calculating the area and doubling it because it is a Fermi gas

$$\int_{\epsilon_1}^{\epsilon_2} g(\epsilon) d\epsilon = 2 \int_0^{\frac{\pi}{2}} d\phi \int_{r_1}^{r_2} r dr = \pi \int_{r_1}^{r_2} r dr$$

We have the expression of our energy

$$\epsilon = \frac{h^2}{8mL^2} (n_x^2 + n_y^2) = \frac{h^2 r^2}{8mA} \Leftrightarrow r = \frac{\sqrt{8mA\epsilon}}{h}$$

Now we do integration by substitution

$$\frac{dr}{d\epsilon} = \frac{\sqrt{8mA}}{2h} \epsilon^{-\frac{1}{2}} \Leftrightarrow dr = \frac{\sqrt{8mA}}{2h} \epsilon^{-\frac{1}{2}} d\epsilon$$

Then we get

$$\begin{aligned} \int_{\epsilon_1}^{\epsilon_2} g(\epsilon) d\epsilon &= \pi \int_{r_1}^{r_2} \frac{\sqrt{8mA\epsilon}}{h} \frac{\sqrt{8mA}}{2h} \epsilon^{-\frac{1}{2}} d\epsilon \\ &= \int_{\epsilon_2}^{\epsilon_1} \frac{4\pi mA}{h^2} d\epsilon \end{aligned}$$

Notice that it doesn't matter what r_1 and r_2 is. As long as we have a function that can describe the states as a function of ϵ , so I've taken the freedom to change r_i to ϵ_i which gives a density of states as

$$g(\epsilon) = \frac{4\pi mA}{h^2}, \quad \epsilon > 0$$

Notice that it is impossible for $\epsilon = \frac{h^2}{8mL^2} (n_x^2 + n_y^2) < 0$, which means that

$$g(\epsilon) = 0, \quad \epsilon \leq 0$$

Well most of the time it doesn't make much sense to calculate anything with $\epsilon < 0$ since we often define the lowest possible energy as the zero-point energy but you get the point

c) Now we explore the chemical potential as function of temperature

In cases where $T \neq 0$ we have to resort to the Sommerfeld expansion

$$N = \int_0^\infty g(\epsilon) \bar{n}_{FD}(\epsilon) d\epsilon$$

$$= \frac{4\pi mA}{h^2} \int_0^\infty \frac{1}{e^{\frac{(\epsilon-\mu)}{kT}} + 1} d\epsilon$$

We know that the Fermi-Dirac distribution is symmetric around μ . The proof for it is here

$$\frac{1}{e^{\frac{(\epsilon-\mu)}{kT}} + 1} = 1 - \frac{1}{1 + e^{\frac{(\epsilon-\mu)}{kT}}}$$

Insert $\epsilon = \mu \pm \Delta$ and we see

$$\frac{1}{e^{\frac{\Delta}{kT}} + 1} = 1 - \frac{1}{1 + e^{\frac{\Delta}{kT}}}$$

This means that the probability for a state at $\mu + \Delta$ to be occupied is the same as the probability for a state $\mu - \Delta$ to be unoccupied. Now we can say some smart stuff just by looking at this expression

$$N = \int_0^\infty g(\epsilon) \bar{n}_{FD}(\epsilon) d\epsilon$$

Our Fermi-Dirac distribution is symmetric in the sense that an equal amount of particles will have transferred from the states below μ to the states above μ . As the temperature increases then more particles will transfer from lower energy states to higher energy states. This needs to be balanced with $g(\epsilon)$ and N . Since N is fixed and $g(\epsilon)$ in this case is constant then the integral $\int_0^\infty g(\epsilon) \bar{n}_{FD}(\epsilon) d\epsilon$ will luckily also be constant no matter how we change the temperature since both the density of states and distribution function is symmetric around ϵ_F and μ . In the end this means that $\mu = \epsilon_F$ at $T = 0$ as given in the book. But since we do not magically create or destroy any particles from existence thanks to $g(\epsilon)$ being constant then $\mu = \epsilon_F$ for sufficiently low temperatures $kT \ll \epsilon_F$.

However we have an issue at $\epsilon = 0$. At some point the Fermi-Dirac distribution also wants to take particles from states below the zero-point energy line and puts them at states far above ϵ_F . But we have that $g(\epsilon) = 0$ for $\epsilon \leq 0$, which is also why the lower limit for our integral begins at zero. This basically means that the distribution function will create new particles out of thin air! We cannot have that since N should be constant. We therefore have to shift the distribution function towards lower energies, which we can do by decreasing μ . So for larger temperatures then μ stops being constant and tends to decrease

d) We solve the integral analytically

$$N = \frac{4\pi mA}{h^2} \int_0^\infty \frac{1}{e^{\frac{(\epsilon-\mu)}{kT}} + 1} d\epsilon$$

We can solve the integral using Schaum's solution

$$\mathbf{17.25.7.} \quad \int \frac{dx}{p + qe^{ax}} = \frac{x}{p} - \frac{1}{ap} \ln(p + qe^{ax})$$

$$\int_0^\infty \frac{1}{e^{\frac{(\epsilon-\mu)}{kT}} + 1} d\epsilon = \int_0^\infty \frac{1}{e^{\frac{\epsilon}{kT}} e^{-\frac{\mu}{kT}} + 1} d\epsilon$$

Now we define $p = 1$, $q = e^{-\frac{\mu}{kT}}$ and $a = \frac{1}{kT}$ then we have

$$\begin{aligned} \int_0^\infty \frac{1}{e^{a\epsilon} q + p} d\epsilon &= \left[\frac{\epsilon}{p} - \frac{1}{ap} \ln(p + qe^{a\epsilon}) \right]_0^\infty \\ &= \left[\epsilon - kT \ln \left(1 + e^{-\frac{\mu}{kT}} e^{\frac{\epsilon}{kT}} \right) \right]_0^\infty \\ &= \left[\epsilon - kT \ln \left(1 + e^{\frac{\epsilon-\mu}{kT}} \right) \right]_0^\infty \\ &= \lim_{\epsilon \rightarrow \infty} \left[\epsilon - kT \ln \left(1 + e^{\frac{\epsilon-\mu}{kT}} \right) \right] - \left[0 - kT \ln \left(1 + e^{\frac{0-\mu}{kT}} \right) \right] \\ &= \lim_{\epsilon \rightarrow \infty} \left[\epsilon - kT \ln \left(e^{\frac{\epsilon-\mu}{kT}} \right) \right] - \left[-kT \ln \left(1 + e^{-\frac{\mu}{kT}} \right) \right] \\ &= \lim_{\epsilon \rightarrow \infty} \left[\epsilon - kT \cdot \left(\frac{\epsilon - \mu}{kT} \right) \right] + kT \ln \left(1 + e^{-\frac{\mu}{kT}} \right) \\ &= \mu + kT \ln \left(1 + e^{-\frac{\mu}{kT}} \right) \\ &= \mu + kT \ln \left(e^{-\frac{\mu}{kT}} \left(e^{\frac{\mu}{kT}} + 1 \right) \right) \\ &= \mu + kT \ln \left(e^{-\frac{\mu}{kT}} \right) + kT \ln \left(e^{\frac{\mu}{kT}} + 1 \right) \\ &= kT \ln \left(e^{\frac{\mu}{kT}} + 1 \right) \end{aligned}$$

Therefore

$$N = \frac{4\pi mA}{h^2} \cdot kT \ln \left(e^{\frac{\mu}{kT}} + 1 \right)$$

⇕

$$\ln \left(e^{\frac{\mu}{kT}} + 1 \right) = \frac{N h^2}{4\pi m k T A}$$

⇕

$$\frac{\mu}{kT} = e^{\frac{N h^2}{4 \pi m k T A}} - 1$$

⇕

$$\mu = kT \ln \left(e^{\frac{N h^2}{4 \pi m k T A}} - 1 \right) = kT \ln \left(e^{\frac{\epsilon_F}{kT}} - 1 \right)$$

Okay that took a while, so let us look at the limit $kT \ll \epsilon_F$ then

$$\mu \approx kT \ln \left(e^{\frac{\epsilon_F}{kT}} \right) = \epsilon_F$$

This makes sense as expected. μ is constant and equal to the Fermi level. Now how does the function behave for large T. At some point the logarithmic function dips below zero when $e^{\frac{\epsilon_F}{kT}} = 2$ or $kT = \frac{\epsilon_F}{\ln(2)}$ and then it will continue to decrease as kT increases because we just multiply an increasingly negative number with a positive number. So it behaves as expected

Using the approximation $e^x \approx 1 + x$, $|x| \ll 1$ for $kT \gg \epsilon_F$ then

$$\mu \approx kT \ln \left(\frac{\epsilon_F}{kT} \right)$$

This is another alternative argument for why the function decreases for large temperatures.



e) From Schroeder the Helmholtz energy is

$$F = -kT \ln(Z) = -kT \ln(Z_{tr}Z_{int})$$

The translational motion for a 2D ideal gas is

$$Z_{tr} = \int_0^\infty \int_0^\infty e^{-\frac{\hbar^2}{8mkT} \left(\frac{n_x^2}{L_x^2} + \frac{n_y^2}{L_y^2} \right)} dn_x dn_y = \frac{2\pi mkT}{h^2} L_x L_y = \frac{A}{a_Q}$$

Therefore from Schroeder we can adapt the equation for a 3D gas to a 2D gas by replacing volume and quantum volume by an area and the quantum area

$$\mu = \left(\frac{\partial F}{\partial N} \right)_{T,V} = -kT \ln \left(\frac{AZ_{int}}{Na_Q} \right) = -kT \ln \left(\frac{2\pi mkTA}{h^2 N} Z_{int} \right)$$

Our system in the high temperature limit $kT \gg \epsilon_F$ is given by

$$\mu \approx kT \ln \left(\frac{\epsilon_F}{kT} \right) = kT \ln \left(\frac{h^2 N}{4\pi mkTA} \right) = -kT \ln \left(\frac{4\pi mkTA}{h^2 N} \right)$$

In our case we therefore have $Z_{int} = 2$, which makes sense because of the two degenerate spin states

Problem F1

Problem F1

Consider a system described by two single particle levels with energy ϵ and $-\epsilon$. The system is in equilibrium with a reservoir with which it can exchange both energy and particles. The temperature of the reservoir is T and its chemical potential is μ .

Consider first the case where the particles are fermions.

Question a:

Write down the grand partition function for the system.

The grand partition function is given by

$$\mathcal{Z} = \sum_s e^{-\frac{(E(s) - \mu N(s))}{kT}} = e^{-\frac{(E(s) - \mu N(s))}{kT}}$$

We have in total 4 states. Let me list them up for you

$$\{(N, E)\} = \{(0, 0), (1, \epsilon), (1, -\epsilon), (2, 0)\}$$

So therefore

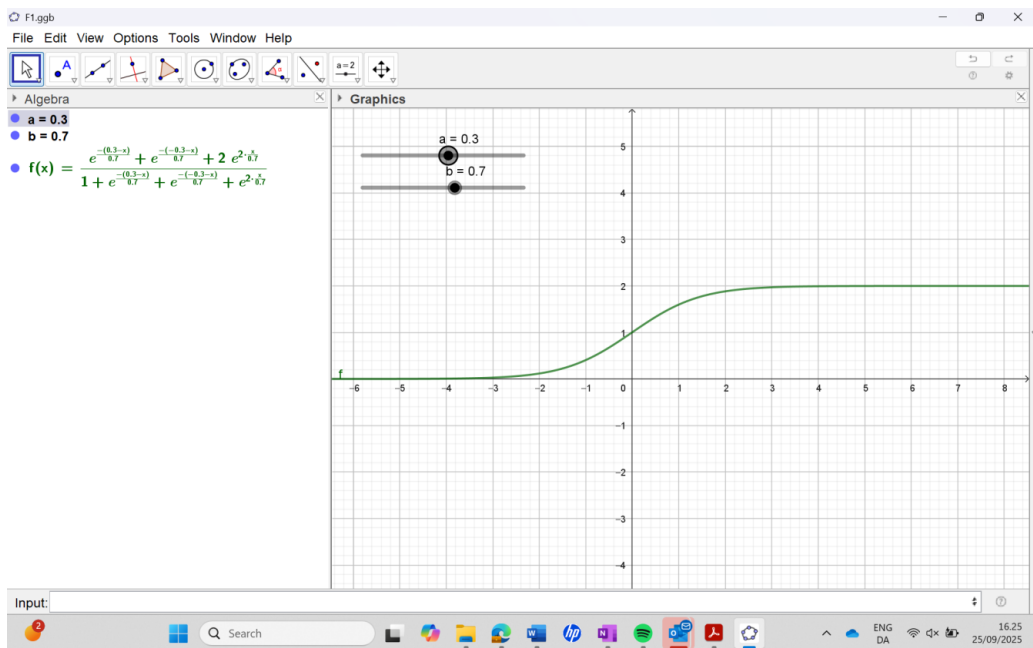
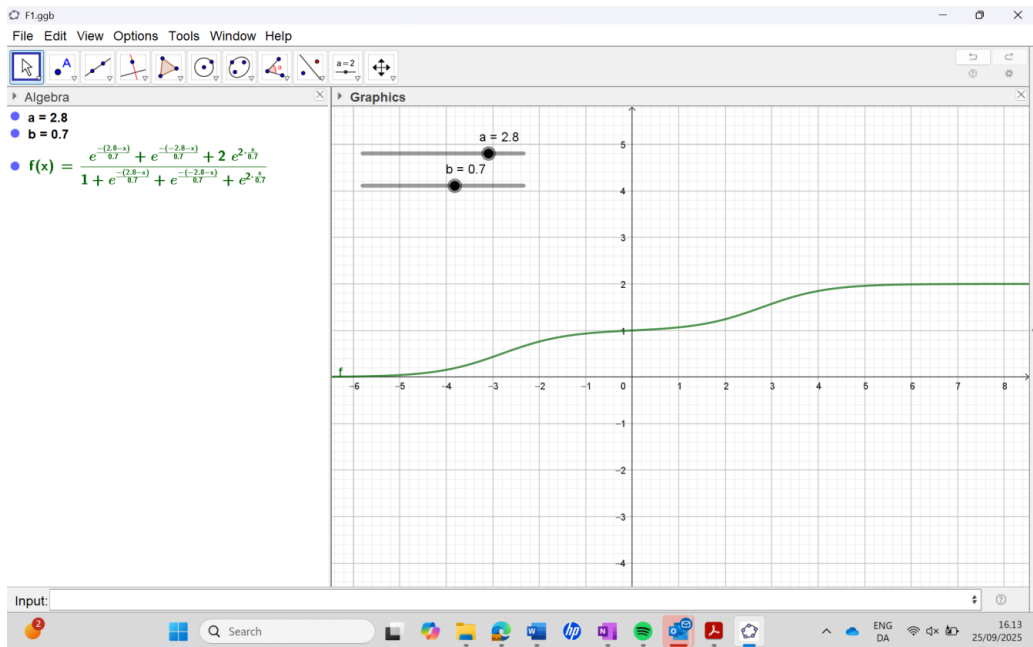
$$\mathcal{Z} = 1 + e^{-\frac{(\epsilon - \mu)}{kT}} + e^{-\frac{(-\epsilon - \mu)}{kT}} + e^{\frac{2\mu}{kT}}$$

Question b:

Give an expression for the average number of particles N in the system. Sketch N as a function of μ for fixed temperature $T > 0$.

The average number of particles in the system is

$$\begin{aligned} \bar{N} &= \frac{kT}{\mathcal{Z}} \frac{\partial \mathcal{Z}}{\partial \mu} = \frac{kT}{\mathcal{Z}} \frac{\partial}{\partial \mu} \left(1 + e^{-\frac{(\epsilon - \mu)}{kT}} + e^{-\frac{(-\epsilon - \mu)}{kT}} + e^{\frac{2\mu}{kT}} \right) \\ &= \frac{kT}{\mathcal{Z}} \left(\frac{1}{kT} e^{-\frac{(\epsilon - \mu)}{kT}} + \frac{1}{kT} e^{-\frac{(-\epsilon - \mu)}{kT}} + \frac{2}{kT} e^{\frac{2\mu}{kT}} \right) \\ &= \frac{1}{\mathcal{Z}} \left(e^{-\frac{(\epsilon - \mu)}{kT}} + e^{-\frac{(-\epsilon - \mu)}{kT}} + 2e^{\frac{2\mu}{kT}} \right) \\ &= \frac{e^{-\frac{(\epsilon - \mu)}{kT}} + e^{-\frac{(-\epsilon - \mu)}{kT}} + 2e^{\frac{2\mu}{kT}}}{1 + e^{-\frac{(\epsilon - \mu)}{kT}} + e^{-\frac{(-\epsilon - \mu)}{kT}} + e^{\frac{2\mu}{kT}}} \end{aligned}$$



Consider next the case where the particles are bosons.

Question c:

Write down the grand partition function for the system.

For bosons we can describe the grand partition function for a system with k states as

$$Z = \prod_k \left(\sum_{n=0}^{\infty} e^{-\frac{n(E_k - \mu)}{kT}} \right) = \prod_k \left(\frac{1}{1 - e^{-\frac{(E_k - \mu)}{kT}}} \right)$$

Think about why this is true. We have to take into account all of the possible combinations of the number of bosons in each of the two states. We need a double sum of (m, n) with energies E_m and E_n . But then I can always split the exponential function to only one dependent on m and one dependent on n .

$$Z = \sum_m \sum_n e^{-\frac{mE_m + nE_n - m\mu - n\mu}{kT}} = \sum_m \sum_n e^{-\frac{m(E_m - \mu)}{kT}} e^{-\frac{n(E_n - \mu)}{kT}} = \sum_m e^{-\frac{m(E_m - \mu)}{kT}} \sum_n e^{-\frac{n(E_n - \mu)}{kT}}$$

Generalizing it gives me

$$Z = \prod_k \left(\sum_{n=0}^{\infty} e^{-\frac{n(E_k - \mu)}{kT}} \right)$$

And then I can apply the geometric series $\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}$ to obtain

$$Z = \prod_k \left(\sum_{n=0}^{\infty} e^{-\frac{n(E_k - \mu)}{kT}} \right) = \prod_k \left(\frac{1}{1 - e^{-\frac{(E_k - \mu)}{kT}}} \right)$$

The result I get from this is

$$Z = \frac{1}{1 - e^{-\frac{(\epsilon - \mu)}{kT}}} \cdot \frac{1}{1 - e^{-\frac{(-\epsilon - \mu)}{kT}}}$$

Question d:

Give an expression for the average number of particles N in the system. Sketch N as a function of μ for fixed temperature $T > 0$. What is the maximum number of particles in the system and for which value of μ is it obtained?

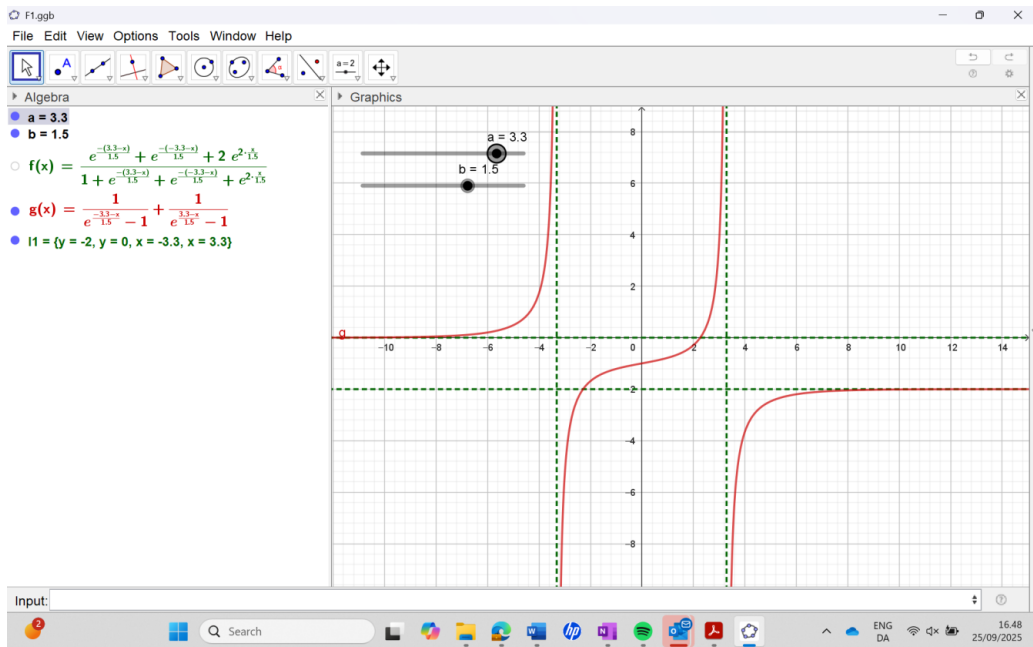
$$\begin{aligned} \bar{N} &= \frac{kT}{Z} \frac{\partial Z}{\partial \mu} = \frac{kT}{Z} \frac{\partial}{\partial \mu} \left(\frac{1}{1 - e^{-\frac{(\epsilon - \mu)}{kT}}} \cdot \frac{1}{1 - e^{-\frac{(-\epsilon - \mu)}{kT}}} \right) \\ &= \frac{kT}{Z} \frac{1}{1 - e^{-\frac{(\epsilon - \mu)}{kT}}} \frac{\partial}{\partial \mu} \left(\frac{1}{1 - e^{-\frac{(\epsilon - \mu)}{kT}}} \right) + \frac{kT}{Z} \frac{1}{1 - e^{-\frac{(-\epsilon - \mu)}{kT}}} \frac{\partial}{\partial \mu} \left(\frac{1}{1 - e^{-\frac{(-\epsilon - \mu)}{kT}}} \right) \end{aligned}$$

$$\begin{aligned}
&= -\frac{kT}{Z} \frac{1}{1 - e^{-\frac{(\epsilon-\mu)}{kT}}} \frac{1}{\left(1 - e^{-\frac{(-\epsilon-\mu)}{kT}}\right)^2} \frac{\partial}{\partial \mu} \left(1 - e^{-\frac{(-\epsilon-\mu)}{kT}}\right) - \frac{kT}{Z} \frac{1}{1 - e^{-\frac{(-\epsilon-\mu)}{kT}}} \frac{1}{\left(1 - e^{-\frac{(\epsilon-\mu)}{kT}}\right)^2} \frac{\partial}{\partial \mu} \left(1 - e^{-\frac{(\epsilon-\mu)}{kT}}\right) \\
&= -\frac{kT}{Z} \frac{1}{1 - e^{-\frac{(\epsilon-\mu)}{kT}}} \frac{-e^{-\frac{(-\epsilon-\mu)}{kT}}}{\left(1 - e^{-\frac{(-\epsilon-\mu)}{kT}}\right)^2} \frac{\partial}{\partial \mu} \left(-\frac{(-\epsilon-\mu)}{kT}\right) - \frac{kT}{Z} \frac{1}{1 - e^{-\frac{(-\epsilon-\mu)}{kT}}} \frac{-e^{-\frac{(\epsilon-\mu)}{kT}}}{\left(1 - e^{-\frac{(\epsilon-\mu)}{kT}}\right)^2} \frac{\partial}{\partial \mu} \left(-\frac{(\epsilon-\mu)}{kT}\right) \\
&= -\frac{kT}{Z} \frac{1}{1 - e^{-\frac{(\epsilon-\mu)}{kT}}} \frac{-e^{-\frac{(-\epsilon-\mu)}{kT}}}{\left(1 - e^{-\frac{(-\epsilon-\mu)}{kT}}\right)^2} \frac{1}{kT} - \frac{kT}{Z} \frac{1}{1 - e^{-\frac{(-\epsilon-\mu)}{kT}}} \frac{-e^{-\frac{(\epsilon-\mu)}{kT}}}{\left(1 - e^{-\frac{(\epsilon-\mu)}{kT}}\right)^2} \frac{1}{kT} \\
&= \frac{1}{Z} \frac{1}{1 - e^{-\frac{(\epsilon-\mu)}{kT}}} \frac{e^{-\frac{(-\epsilon-\mu)}{kT}}}{\left(1 - e^{-\frac{(-\epsilon-\mu)}{kT}}\right)^2} + \frac{1}{Z} \frac{1}{1 - e^{-\frac{(-\epsilon-\mu)}{kT}}} \frac{e^{-\frac{(\epsilon-\mu)}{kT}}}{\left(1 - e^{-\frac{(\epsilon-\mu)}{kT}}\right)^2} \\
&= \frac{1}{\left(\frac{1}{1 - e^{-\frac{(\epsilon-\mu)}{kT}}} \cdot \frac{1}{1 - e^{-\frac{(-\epsilon-\mu)}{kT}}}\right)} \frac{1}{1 - e^{-\frac{(\epsilon-\mu)}{kT}}} \frac{e^{-\frac{(-\epsilon-\mu)}{kT}}}{\left(1 - e^{-\frac{(-\epsilon-\mu)}{kT}}\right)^2} \\
&\quad + \frac{1}{\left(\frac{1}{1 - e^{-\frac{(\epsilon-\mu)}{kT}}} \cdot \frac{1}{1 - e^{-\frac{(-\epsilon-\mu)}{kT}}}\right)} \frac{1}{1 - e^{-\frac{(-\epsilon-\mu)}{kT}}} \frac{e^{-\frac{(\epsilon-\mu)}{kT}}}{\left(1 - e^{-\frac{(\epsilon-\mu)}{kT}}\right)^2} \\
&= \frac{e^{-\frac{(-\epsilon-\mu)}{kT}}}{1 - e^{-\frac{(-\epsilon-\mu)}{kT}}} + \frac{e^{-\frac{(\epsilon-\mu)}{kT}}}{1 - e^{-\frac{(\epsilon-\mu)}{kT}}} \\
&= \frac{1}{e^{\frac{(-\epsilon-\mu)}{kT}} - 1} + \frac{1}{e^{\frac{(\epsilon-\mu)}{kT}} - 1}
\end{aligned}$$

The maximum number of particles is found in the limit where $\mu \rightarrow -\epsilon$ from the negative side then the term $e^{\frac{(-\epsilon-\mu)}{kT}}$ tends to infinity but which infinity. Because $\mu < -\epsilon$ then $\frac{(-\epsilon-\mu)}{kT} > 0$ which means that $e^{\frac{(-\epsilon-\mu)}{kT}} > 1$. And therefore $\left(e^{\frac{(-\epsilon-\mu)}{kT}} - 1\right) > 0$ and $\frac{1}{e^{\frac{(-\epsilon-\mu)}{kT}} - 1} > 0$

When $\mu \rightarrow -\epsilon$ then $e^{\frac{(-\epsilon-\mu)}{kT}} \rightarrow 1$ which means that $\frac{1}{e^{\frac{(-\epsilon-\mu)}{kT}} - 1} \rightarrow +\infty$. Therefore an infinite number of bosons can fit in our system

If you plot it you get a mess. But you have to realize that only the physically reasonable part of the graph is from $-\infty$ to $-\epsilon$. The rest is just artifacts that has unphysical meaning



Problem F2

Problem F2

Consider a system consisting of two indistinguishable bosons, each of which can be in three quantum states: Two with energy $-\epsilon$ and one with energy ϵ , where $\epsilon > 0$, see figure 1. The system is in equilibrium with a reservoir having the temperature T .

Question a:

Determine the possible states of the 2-particle system, and write down its partition function Z .

The total number of possible states is $\frac{n^k}{k!}$. (This is from a modified binomial distribution, where we replace $\frac{n!}{(n-k)!}$ with n^k because bosons can occupy the same states). So the number of possible states is $\frac{3^2}{2!} = 6$. We can count that there are 3 states with energy -2ϵ because we just order the two bosonic particles in the two lowest states. Then there are 2 states with energy 0 because we order one particle in ϵ and then the other in one of the two $-\epsilon$ energy states. Finally there is one state with energy 2ϵ so

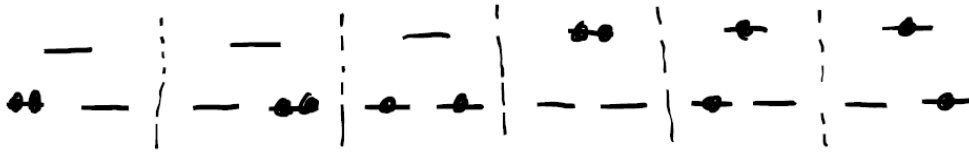
$$Z = \sum_s e^{-\beta E(s)} = 3e^{2\epsilon\beta} + 2 + e^{-2\epsilon\beta}$$

We now take into account that the two bosons interact with energy U , when they are in the same quantum state. Consequently the energy is $2\epsilon + U$, when both particles are in the upper quantum state and $-2\epsilon + U$, when one of the lower states contain both particles.

Question b:

Write down the partition function for the system. Give an expression for the energy $E(T)$ as a function of the temperature.

We have two -2ϵ energy states where the bosons are in the same state. And one where they are both in the 2ϵ state



So we modify the partition function

$$Z = \sum_s e^{-\beta E(s)} = 2e^{\beta(2\epsilon+U)} + e^{2\epsilon\beta} + 2 + e^{-\beta(2\epsilon+U)}$$

The energy for the system is

$$\begin{aligned}\bar{E} &= -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{2(2\epsilon - U)e^{\beta(2\epsilon - U)} + 2\epsilon e^{2\epsilon\beta} - (2\epsilon + U)e^{-\beta(2\epsilon + U)}}{2e^{\beta(2\epsilon - U)} + e^{2\epsilon\beta} + 2 + e^{-\beta(2\epsilon + U)}} \\ &= \frac{2(U - 2\epsilon)e^{\beta(2\epsilon - U)} - 2\epsilon e^{2\epsilon\beta} + (2\epsilon + U)e^{-\beta(2\epsilon + U)}}{2e^{\beta(2\epsilon - U)} + e^{2\epsilon\beta} + 2 + e^{-\beta(2\epsilon + U)}}\end{aligned}$$

Question c:

Give an expression for the entropy $S(T)$ of the system as a function of the temperature. Compute $\lim_{T \rightarrow \infty} S(T)$ and $\lim_{T \rightarrow 0} S(T)$ for $U < 0$ and for $U > 0$. Give a physical interpretation.

We can find the entropy using Helmholtz relation

$$F = U - TS = -kT \ln(Z)$$

$$\Updownarrow$$

$$S = \frac{U}{T} + k \ln(Z)$$

$$\begin{aligned}&= \frac{1}{T} \left(\frac{2(U - 2\epsilon)e^{\beta(2\epsilon - U)} - 2\epsilon e^{2\epsilon\beta} + (2\epsilon + U)e^{-\beta(2\epsilon + U)}}{2e^{\beta(2\epsilon - U)} + e^{2\epsilon\beta} + 2 + e^{-\beta(2\epsilon + U)}} \right) + k \ln(2e^{\beta(2\epsilon - U)} + e^{2\epsilon\beta} + 2 + e^{-\beta(2\epsilon + U)}) \\ &= k \left(\frac{2(U - 2\epsilon)\beta e^{\beta(2\epsilon - U)} - 2\epsilon\beta e^{2\epsilon\beta} + (2\epsilon + U)\beta e^{-\beta(2\epsilon + U)}}{2e^{\beta(2\epsilon - U)} + e^{2\epsilon\beta} + 2 + e^{-\beta(2\epsilon + U)}} \right) + k \ln(2e^{\beta(2\epsilon - U)} + e^{2\epsilon\beta} + 2 + e^{-\beta(2\epsilon + U)})\end{aligned}$$

We cannot do anything else with this. Now finding the limits

$$\lim_{T \rightarrow \infty} S = \lim_{\beta \rightarrow 0} S = 0 + k \ln(2 + 1 + 2 + 1) = k \ln(6)$$

This limit would be the same for $U > 0$ and $U < 0$. Nice this reflects that all 6 possible states are equal and we have multiplicity 6

The other limit if we let $U > 0$ then we find the fastest growing function out of all of them

$$\lim_{T \rightarrow 0} S = \lim_{\beta \rightarrow \infty} S = \lim_{\beta \rightarrow \infty} k \left(\frac{-2\epsilon\beta e^{2\epsilon\beta}}{e^{2\epsilon\beta}} \right) + k \ln(e^{2\epsilon\beta}) = \lim_{\beta \rightarrow \infty} -2\epsilon\beta k + 2\epsilon\beta k = 0$$

This makes sense because we are in the ground state. We increased the energy of the two situations where the bosons are in the same state and we are left with only 1 true ground state, which results in $S = k \ln(1) = 0$

The final limit but with $U < 0$ then we find the fastest growing function

$$\begin{aligned}
\lim_{T \rightarrow 0} S &= \lim_{\beta \rightarrow \infty} S = \lim_{\beta \rightarrow \infty} k \left(\frac{2(U - 2\epsilon)\beta e^{\beta(2\epsilon - U)}}{2e^{\beta(2\epsilon - U)}} \right) + k \ln(2e^{\beta(2\epsilon - U)}) \\
&= \lim_{\beta \rightarrow \infty} k(U - 2\epsilon)\beta + k\beta(2\epsilon - U) + k \ln(2) = k \ln(2)
\end{aligned}$$

This makes sense because we have lowered the energy of the two situations where the bosons are in the same state and we are left with 2 true ground state, which results in $S = k \ln(2)$

Problem F3

Problem F3

An electron gas at temperature T and chemical potential μ is in equilibrium with a surface, where the electrons can bind to a number of binding points. The electron gas can be seen as a reservoir for the surface. A binding point can bind maximally one electron, which either has spin $\uparrow$ or spin $\downarrow$. When an electron is bound to a binding point it vibrates, which is described by that it sits in a one-dimensional harmonic oscillator with the energies $\epsilon_j = jhf$, where $j = 0, 1, 2, \dots$

Question a:

Show that the grand partition function $\mathcal{Z}$ for a single binding point can be written as

$$\mathcal{Z} = 1 + e^{\beta\mu} g(\beta hf),$$

where $\beta = 1/(kT)$, and specify the function $g(x)$.

The grand partition function is given by

$$\mathcal{Z} = \sum_{n=0}^{\infty} e^{-\frac{n(\epsilon-\mu)}{kT}}$$

It applies for fermions like electrons that they cannot occupy the same state. A single binding point has $N = 1$ with energy $jh f$ and $N = 0$ with energy 0. Note that the fact that there are two possible configurations for each state $N = 1$ gives

$$\mathcal{Z} = 1 + 2 \sum_{j=0}^{\infty} e^{-\frac{(jh f - \mu)}{kT}} = 1 + 2e^{\beta\mu} \sum_{j=0}^{\infty} e^{-j\beta hf} = 1 + \frac{2e^{\beta\mu}}{1 - e^{-\beta hf}}$$

Notice that I used the geometric series $\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}$

Question b:

Give an expression for the average vibrational energy $E(T)$ of bound electrons in a given binding point.

Hint: You can use without proof that $\sum_{j=0}^{\infty} j \cdot x^j = x \cdot (1 - x)^{-2}$.

$$\begin{aligned} \bar{E} &= -\frac{1}{\mathcal{Z}} \left(\frac{\partial \mathcal{Z}}{\partial \beta} \right)_{\beta\mu} = -\frac{1}{1 + \frac{2e^{\beta\mu}}{1 - e^{-\beta hf}}} \left(\frac{\partial}{\partial \beta} \left(1 + \frac{2e^{\beta\mu}}{1 - e^{-\beta hf}} \right) \right) \\ &= -\frac{1}{1 + \frac{2e^{\beta\mu}}{1 - e^{-\beta hf}}} \left(2e^{\beta\mu} \frac{\partial}{\partial \beta} \left(\frac{1}{1 - e^{-\beta hf}} \right) + \frac{1}{1 - e^{-\beta hf}} \frac{\partial}{\partial \beta} (2e^{\beta\mu}) \right) \end{aligned}$$

Note that $\beta\mu$ is constant in the partial derivative so the second term is zero

$$\begin{aligned}
\bar{E} &= -\frac{1}{1 + \frac{2e^{\beta\mu}}{1 - e^{-\beta hf}}} \left(2e^{\beta\mu} \frac{\partial}{\partial \beta} \left(\frac{1}{1 - e^{-\beta hf}} \right) \right) \\
&= -\frac{1}{1 + \frac{2e^{\beta\mu}}{1 - e^{-\beta hf}}} \left(-\frac{2e^{\beta\mu}}{(1 - e^{-\beta hf})^2} \frac{\partial}{\partial \beta} (1 - e^{-\beta hf}) \right) \\
&= -\frac{1}{1 + \frac{2e^{\beta\mu}}{1 - e^{-\beta hf}}} \left(\frac{2e^{\beta\mu} e^{-\beta hf}}{(1 - e^{-\beta hf})^2} \frac{\partial}{\partial \beta} (-\beta hf) \right) \\
&= -\frac{1}{1 + \frac{2e^{\beta\mu}}{1 - e^{-\beta hf}}} \left(-\frac{2hf e^{-\beta(hf - \mu)}}{(1 - e^{-\beta hf})^2} \right) \\
&= \frac{1}{1 + \frac{2e^{\beta\mu}}{1 - e^{-\beta hf}}} \left(\frac{2hf e^{-\beta(hf - \mu)}}{(1 - e^{-\beta hf})^2} \right) \\
&= \frac{2hf e^{-\beta(hf - \mu)}}{(1 - e^{-\beta hf} + 2e^{\beta\mu})(1 - e^{-\beta hf})}
\end{aligned}$$

Question c:

Give an expression for the average number of electrons $N_b(T)$ bound to a binding point. Determine N_b for $T = 0$ and for $T \gg \epsilon_F/k$, where $\epsilon_F = h^2(3n/\pi)^{2/3}/(8m)$ is the Fermi energy for the electron gas with density $n > 0$, and m is the mass of the electron.

The average number of particles is

$$\begin{aligned}
N_b &= \frac{kT}{Z} \frac{\partial Z}{\partial \mu} = \frac{kT}{1 + \frac{2e^{\beta\mu}}{1 - e^{-\beta hf}}} \frac{\partial}{\partial \mu} \left(1 + \frac{2e^{\beta\mu}}{1 - e^{-\beta hf}} \right) \\
&= \frac{kT}{1 + \frac{2e^{\beta\mu}}{1 - e^{-\beta hf}}} \cdot \frac{2\beta e^{\beta\mu}}{1 - e^{-\beta hf}} \\
&= \frac{2kT\beta e^{\beta\mu}}{1 - e^{-\beta hf} + 2e^{\beta\mu}} \\
&= \frac{2e^{\beta\mu}}{1 - e^{-\beta hf} + 2e^{\beta\mu}}
\end{aligned}$$

When $T = 0$ then $\beta \rightarrow \infty$ and $\mu \rightarrow \epsilon_F$

$$N_b = \lim_{\beta \rightarrow \infty} \frac{2e^{\beta\mu}}{2e^{\beta\mu}} = 1$$

When $kT \gg \epsilon_F$ then we are back to Boltzmann distribution. From Boltzmann statistics we can describe the chemical potential as

$$\mu = -kT \ln \left(\frac{VZ_{int}}{Nv_Q} \right)$$

This gives

$$\begin{aligned} N_b &= \frac{2e^{\beta kT \ln \left(\frac{Nv_Q}{VZ_{int}} \right)}}{1 - e^{-\beta hf} + 2e^{\beta kT \ln \left(\frac{Nv_Q}{VZ_{int}} \right)}} \\ &= \frac{2 \frac{Nv_Q}{VZ_{int}}}{1 - e^{-\beta hf} + 2 \frac{Nv_Q}{VZ_{int}}} \end{aligned}$$

Since we have 2 different spin states then the electronic contribution $Z_{elec} = Z_{int} = 2$ and

$$N_b = \frac{\frac{Nv_Q}{V}}{1 - e^{-\beta hf} + \frac{Nv_Q}{V}}$$

Problem F4

Problem F4

In this problem, we consider a quantum well with four different single particle states: One state has energy 0, two states have energy ϵ , and one state has energy 2ϵ . The energy spectrum is illustrated in Fig. 1.

Question a:

Write down the partition function Z_1 for a single particle sitting in this quantum well. Calculate the average energy E_1 and sketch it as a function of the temperature T . Calculate $\lim_{T \rightarrow 0} E_1$ and $\lim_{T \rightarrow \infty} E_1$, and interpret the results physically.

The partition function is

$$Z_1 = \sum_E d(E) e^{-\beta E} = 1 + 2e^{-\beta\epsilon} + e^{-2\beta\epsilon}$$

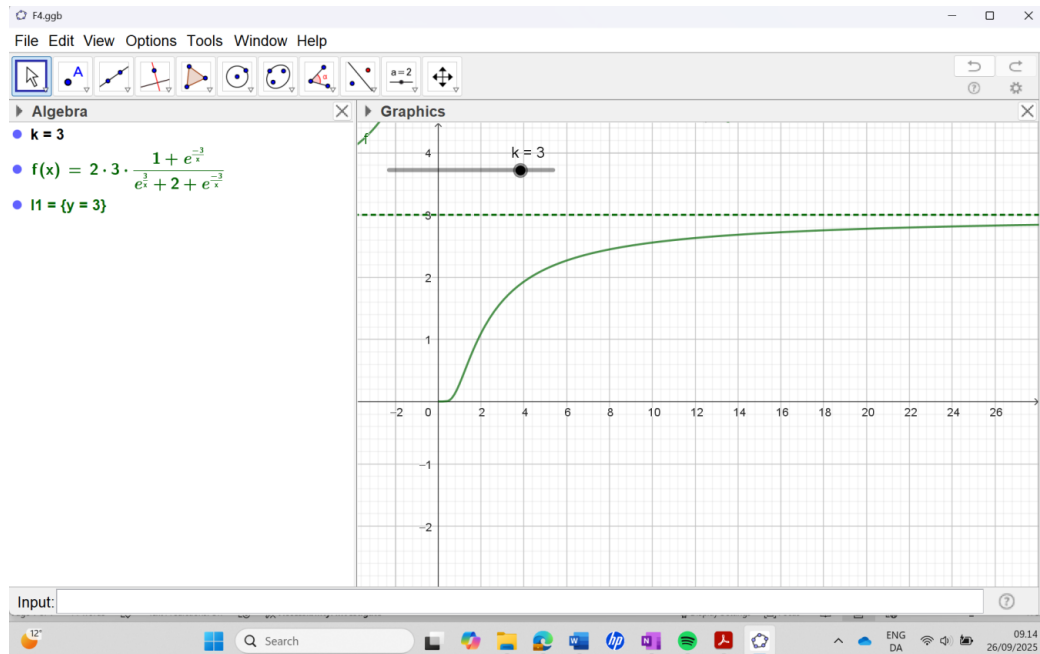
The average energy is

$$\begin{aligned} \bar{E} &= -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{-2\epsilon e^{-\beta\epsilon} - 2\epsilon e^{-2\beta\epsilon}}{1 + 2e^{-\beta\epsilon} + e^{-2\beta\epsilon}} = \frac{2\epsilon e^{-\beta\epsilon} + 2\epsilon e^{-2\beta\epsilon}}{1 + 2e^{-\beta\epsilon} + e^{-2\beta\epsilon}} \\ &= \frac{2\epsilon + 2\epsilon e^{-\beta\epsilon}}{e^{\beta\epsilon} + 2 + e^{-\beta\epsilon}} \end{aligned}$$

The limits are

$$\begin{aligned} \lim_{T \rightarrow 0} E &= \lim_{\beta \rightarrow \infty} E = \lim_{\beta \rightarrow \infty} \frac{2\epsilon}{e^{\beta\epsilon} + 2 + e^{-\beta\epsilon}} = 0 \\ \lim_{T \rightarrow \infty} E &= \lim_{\beta \rightarrow 0} E = \frac{2\epsilon + 2\epsilon}{1 + 2 + 1} = \frac{4\epsilon}{4} = \epsilon \end{aligned}$$

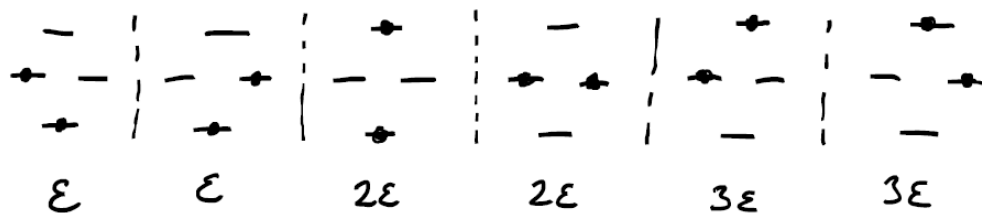
When $T=0$ then the particle is in the ground state. When $T = \infty$ then the particle is equally spread between all states and we get the average energy of all of them



Question b:

Now consider two identical non-interacting fermions in the quantum well. Find the allowed quantum states and their energies for these two particles, and write down their partition function.

The Fermions cannot exist in the state so we have $\frac{4!}{2!(4-2)!} = 6$ possible states



This gives

$$Z_2 = 2e^{-\beta\epsilon} + 2e^{-2\beta\epsilon} + 2e^{-3\beta\epsilon}$$

Question c:

Calculate the average energy E_2 of the two particles and sketch it as a function of T .
 Calculate $\lim_{T \rightarrow 0} E_2$ and $\lim_{T \rightarrow \infty} E_2$, and interpret the results physically.

The average energy is

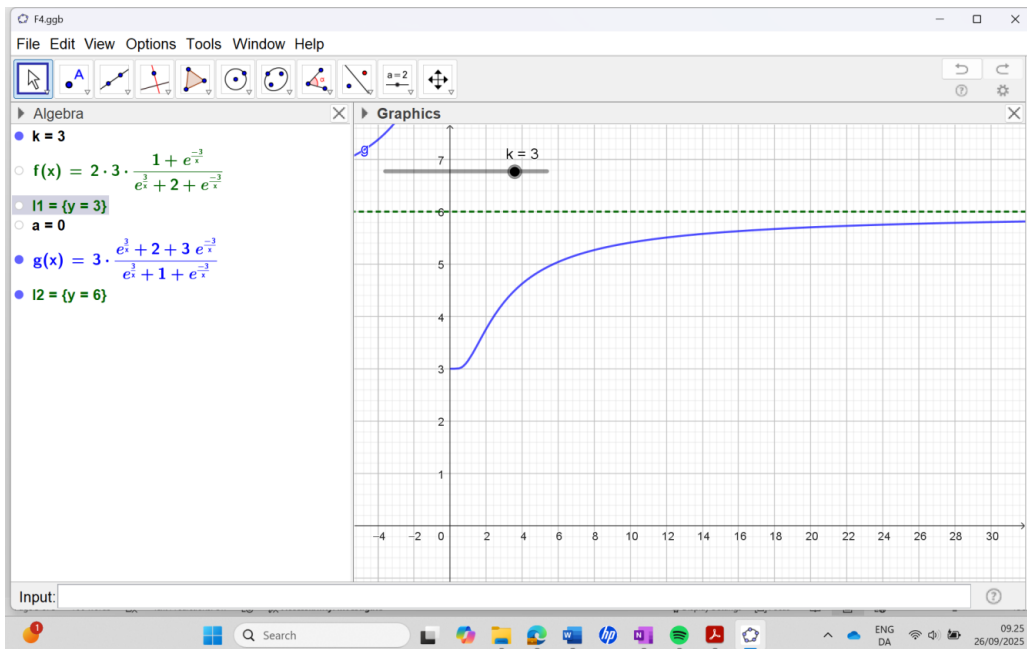
$$\bar{E}_2 = -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{-2\epsilon e^{-\beta\epsilon} - 4\epsilon e^{-2\beta\epsilon} - 6\epsilon e^{-3\beta\epsilon}}{2e^{-\beta\epsilon} + 2e^{-2\beta\epsilon} + 2e^{-3\beta\epsilon}} = \frac{\epsilon e^{\beta\epsilon} + 2\epsilon + 3\epsilon e^{-\beta\epsilon}}{e^{\beta\epsilon} + 1 + e^{-\beta\epsilon}}$$

The limits are

$$\lim_{T \rightarrow 0} E = \lim_{\beta \rightarrow \infty} E = \lim_{\beta \rightarrow \infty} \frac{\epsilon e^{\beta\epsilon}}{e^{\beta\epsilon}} = \epsilon$$

$$\lim_{T \rightarrow \infty} E = \lim_{\beta \rightarrow 0} E = \frac{\epsilon + 2\epsilon + 3\epsilon}{1 + 1 + 1} = \frac{6\epsilon}{3} = 2\epsilon$$

Same reasoning. When $T=0$ then the particle is in the ground state. When $T = \infty$ then the particle is equally spread between all states and we get the average energy of all of them



Question d:

Calculate the entropy S_2 of the two particles and sketch it as a function of T . Calculate $\lim_{T \rightarrow 0} S_2$ and $\lim_{T \rightarrow \infty} S_2$, and interpret the results physically.

The entropy we find with the help of Helmholtz relation

$$F = U - TS = -kT \ln(Z)$$

$$\Updownarrow$$

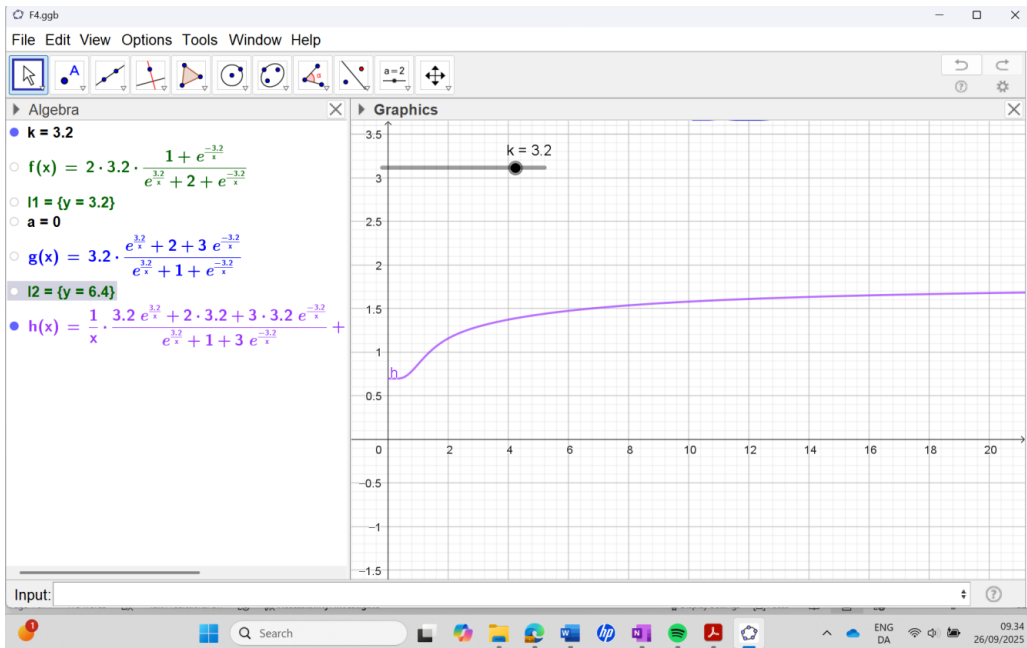
$$S = \frac{U}{T} + k \ln(Z)$$

$$= \frac{1}{T} \left(\frac{\epsilon e^{\beta\epsilon} + 2\epsilon + 3\epsilon e^{-\beta\epsilon}}{e^{\beta\epsilon} + 1 + e^{-\beta\epsilon}} \right) + k \ln(2e^{-\beta\epsilon} + 2e^{-2\beta\epsilon} + 2e^{-3\beta\epsilon})$$

$$= k\beta \left(\frac{\epsilon e^{\beta\epsilon} + 2\epsilon + 3\epsilon e^{-\beta\epsilon}}{e^{\beta\epsilon} + 1 + e^{-\beta\epsilon}} \right) + k \ln(e^{-\beta\epsilon} + e^{-2\beta\epsilon} + e^{-3\beta\epsilon}) + k \ln(2)$$

$$\Updownarrow$$

$$\frac{S}{k} = \beta \left(\frac{\epsilon e^{\beta\epsilon} + 2\epsilon + 3\epsilon e^{-\beta\epsilon}}{e^{\beta\epsilon} + 1 + e^{-\beta\epsilon}} \right) + \ln(e^{-\beta\epsilon} + e^{-2\beta\epsilon} + e^{-3\beta\epsilon}) + \ln(2)$$



Now calculating the limits

$$\lim_{T \rightarrow 0} S = \lim_{\beta \rightarrow \infty} S = \lim_{\beta \rightarrow \infty} k\beta \left(\frac{\epsilon e^{\beta\epsilon}}{e^{\beta\epsilon}} \right) + k \ln(e^{-\beta\epsilon}) + k \ln(2) = \lim_{\beta \rightarrow \infty} k\beta\epsilon - k\beta\epsilon + k \ln(2) = k \ln(2)$$

$$\lim_{T \rightarrow \infty} S = \lim_{\beta \rightarrow 0} S = k\beta \left(\frac{\epsilon + 2\epsilon + 3\epsilon}{1 + 1 + 1} \right) + k \ln(1 + 1 + 1) + k \ln(2) = k \ln(6)$$

We have to think about the entropy in terms of $S = k \ln(\Omega)$. When $T = 0$ then we are in the ground state and only the 2 lowest states are occupied giving us $k \ln(2)$. When $T = \infty$ then we are occupying all states equally, which therefore gives us $k \ln(6)$

Problem G1

Problem G1

This problem concerns the Maxwell speed distribution in one dimension (1D) and in two dimensions (2D). Consider an ideal gas consisting of particles with mass m . The energy of a particle moving with velocity $\vec{v}$ is $mv^2/2$, where $v = |\vec{v}|$ is the speed. We consider the classical regime where the particles obey Boltzmann statistics. The probability that a particle has a speed between v and $v + dv$ with dv infinitesimal is given by $\mathcal{D}(v)dv$, where $\mathcal{D}(v)$ is the distribution function.

Question a:

Focus first on the 1D case, where the velocity $\vec{v}$ of the particles is a one-dimensional vector. Argue that the distribution function $\mathcal{D}_{1D}(v)$ in 1D is given by

$$\mathcal{D}_{1D}(v) = ae^{-mv^2/(2kT)}$$

and give a value for a . What is the most likely speed $v_{\max}$ of a particle in the gas? What is the average speed $\bar{v}$ of the particles?

The translational energy for a particle is

$$E_{tr} = \frac{1}{2}mv^2$$

The distribution function must be

$$\mathcal{D}(v)dv \propto \text{degeneracy} \cdot \text{Boltzmann factor}$$

The Boltzmann factor is the probability of a particle having velocity v

$$e^{-\beta E(v)} = e^{-\frac{mv^2}{2kT}}$$

The degeneracy is the number of vectors with speed, v , is 2 because you have $\vec{v}$ and $-\vec{v}$ with the speed, v so

$$\mathcal{D}(v) = C \cdot 2 \cdot e^{-\frac{mv^2}{2kT}} = ke^{-\frac{mv^2}{2kT}}$$

The 2 doesn't matter much because it has no v dependency. For practicality, I will choose to absorb it into the unknown constant

We have to fulfil the condition that

$$\int_0^\infty \mathcal{D}(v) dv = k \int_0^\infty e^{-\frac{mv^2}{2kT}} dv = 1$$

From Schaums

$$\int_0^\infty x^m e^{-ax^2} dx = \frac{\Gamma\left(\frac{m+1}{2}\right)}{2a^{\frac{m+1}{2}}}$$

We have $m = 0$ and $a = \frac{m}{2kT}$ then

$$\int_0^\infty e^{-\frac{mv^2}{2kT}} dv = \frac{\Gamma\left(\frac{0+1}{2}\right)}{2\left(\frac{m}{2kT}\right)^{\frac{0+1}{2}}} = \frac{\sqrt{\pi}}{2\sqrt{\frac{m}{2kT}}} = \sqrt{\frac{\pi kT}{2m}}$$

Then we have

$$k\sqrt{\frac{\pi kT}{2m}} = 1 \Leftrightarrow k = \sqrt{\frac{2m}{\pi kT}}$$

Therefore

$$\mathcal{D}(v) = \sqrt{\frac{2m}{\pi kT}} e^{-\frac{mv^2}{2kT}}$$

The mean velocity is

$$\bar{v} = \int_0^\infty v \mathcal{D}(v) dv = \sqrt{\frac{2m}{\pi kT}} \frac{\Gamma\left(\frac{1+1}{2}\right)}{2\left(\frac{m}{2kT}\right)^{\frac{1+1}{2}}} = \sqrt{\frac{2m}{\pi kT}} \frac{kT}{m} = \sqrt{\frac{2kT}{\pi m}}$$

The maximum velocity is calculated by finding the max of the distribution function

$$\frac{d\mathcal{D}}{dv} = \sqrt{\frac{2m}{\pi kT}} \cdot \left(-\frac{m}{2kT}\right) 2v_{\max} e^{-\frac{mv_{\max}^2}{2kT}} = -\sqrt{\frac{2kT}{\pi m}} v_{\max} e^{-\frac{mv_{\max}^2}{2kT}} = 0$$

$$\Updownarrow$$

$$v_{\max} = 0$$

Question b:

Argue that the distribution function for the 2D case has the form

$$\mathcal{D}_{2D}(v) = bv^l e^{-mv^2/(2kT)}$$

and give a value for b and l . What is the most likely speed $v_{\max}$ of a particle in the gas?

What is the average speed $\bar{v}$ of the particles?

Same argumentation but this time the degeneracy is the circumference of a circle so

$$\mathcal{D}(v) = C \cdot 2\pi v \cdot e^{-\frac{mv^2}{2kT}}$$

$$\int_0^\infty \mathcal{D}(v) dv = \int_0^\infty C \cdot 2\pi v \cdot e^{-\frac{mv^2}{2kT}} dv = 2\pi C \frac{\Gamma\left(\frac{1+1}{2}\right)}{2\left(\frac{m}{2kT}\right)^{\frac{1+1}{2}}} = 2\pi C \frac{kT}{m} = 1$$

$$\Updownarrow$$

$$C = \frac{m}{2\pi kT}$$

This gives

$$\mathcal{D}(v) = \frac{mv}{kT} e^{-\frac{mv^2}{2kT}}$$

The mean velocity is

$$\begin{aligned} \bar{v} &= \int_0^\infty v \mathcal{D}(v) dv = \frac{m}{kT} \frac{\Gamma\left(\frac{2+1}{2}\right)}{2\left(\frac{m}{2kT}\right)^{\frac{2+1}{2}}} = \frac{m}{kT} \frac{\Gamma\left(1+\frac{1}{2}\right)}{2\left(\frac{m}{2kT}\right)^{\frac{3}{2}}} = \frac{m}{kT} \frac{\frac{\sqrt{\pi}}{2}}{2\left(\frac{m}{2kT}\right)^{\frac{3}{2}}} = \left(\frac{m}{kT}\right)^{1-\frac{3}{2}} \frac{\sqrt{\pi} \cdot 2^{\frac{3}{2}}}{2 \cdot 2} = \left(\frac{m}{kT}\right)^{-\frac{1}{2}} \sqrt{\pi} \cdot 2^{\frac{3}{2}-2} \\ &= \sqrt{\frac{\pi kT}{2m}} \end{aligned}$$

The maximum velocity is calculated by finding the max of the distribution function

$$\frac{d\mathcal{D}}{dv} = \frac{m}{kT} e^{-\frac{mv_{max}^2}{2kT}} + \frac{mv_{max}}{kT} e^{-\frac{mv_{max}^2}{2kT}} \cdot \left(-\frac{2mv_{max}}{2kT}\right) = \frac{m}{kT} e^{-\frac{mv_{max}^2}{2kT}} - \left(\frac{mv_{max}}{kT}\right)^2 e^{-\frac{mv_{max}^2}{2kT}} = 0$$

$$\Updownarrow$$

$$\frac{m}{kT} - \left(\frac{mv_{max}}{kT}\right)^2 = 0$$

$$\Updownarrow$$

$$\left(\frac{mv_{max}}{kT}\right)^2 = \frac{m}{kT}$$

$$\Updownarrow$$

$$v_{max} = \sqrt{\frac{kT}{m}}$$

Consider now a gas with density n (particles per volume) in a cylinder with transverse area A . The cylinder is so thin that the gas can be considered one-dimensional, which means that the particles only move along the cylinder axis as illustrated in Fig. 1. At time t_0 , the cylinder is opened at one end so that the gas particles start to stream out, see Fig. 1. In the following, you can regard the density n inside the cylinder as constant, which is a good approximation for short times.

Question c:

How many particles per unit time, dN/dt , stream out of the end of the cylinder?

Hint: Consider first how many particles with speed between v and $v + dv$ escape in the time interval dt .

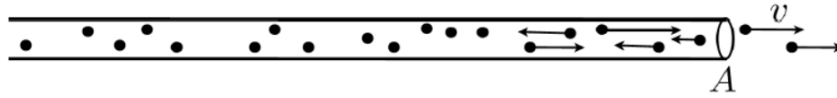


Fig. 1: An ideal classical gas in a cylinder with transverse area A . The cylinder is so thin, that gas can be considered 1D and the particles only move left or right as indicated by the arrows. After the right end of the cylinder is opened, the particles stream out of the cylinder.

Okay let's gather the facts. The speed of our particles are given by

$$\mathcal{D}(v) = \sqrt{\frac{2m}{\pi kT}} e^{-\frac{mv^2}{2kT}}$$

Since $\int_0^\infty \mathcal{D}(v) dv = 1$ then the probability of a particle having a speed between v and $v+dv$ is $\mathcal{D}(v) dv$. To understand this intuitively why that is then you should interpret the area under $\mathcal{D}(v)$ between v and $v+dv$ as being an infinitesimally small rectangle with width dv . This is usually how a Riemann sum works. Therefore the infinitesimally small probability that a particle has a speed between v and $v+dv$ is the area of the rectangle which is $\mathcal{D}(v) dv$. Thus the total number of particles in the entire cylinder with a speed between v and $v+dv$ is $dN_{tot} = N_{tot} \mathcal{D}(v) dv$. If you integrate it then you will get N_{tot} so this does not violate anything.

Using this fact then here is my reasoning

1. In a time dt then only particles within a volume $Avdt$ will be able to escape
2. The number of particles within that volume is $nAvdt$.
3. The number of particles moving towards the exit of the cylinder is half of that $\frac{nAv}{2} dt$

4. The number of particles with speed between v and $v+dv$ from those particles is $\frac{nAv}{2} dt \mathcal{D}(v) dv$

Therefore, the number of particles in the cylinder lost due to particles with a certain speed v is

$$dN_v = -\frac{nAv}{2} dt \mathcal{D}(v) dv$$

$\Updownarrow$

$$\frac{dN_v}{dt} = -\frac{nAv}{2} \mathcal{D}(v) dv$$

Now the total number of particles lost in the cylinder in a time dt can be found by summing over all speeds

$$\begin{aligned} \frac{dN}{dt} &= -\int_0^\infty \frac{nAv}{2} \mathcal{D}(v) dv = -\int_0^\infty \frac{nAv}{2} \sqrt{\frac{2m}{\pi kT}} e^{-\frac{mv^2}{2kT}} dv = -\frac{nA}{2} \sqrt{\frac{2m}{\pi kT}} \int_0^\infty v e^{-\frac{mv^2}{2kT}} dv \\ &= -\frac{nA}{2} \sqrt{\frac{2m}{\pi kT}} \frac{\Gamma\left(\frac{1+1}{2}\right)}{2 \left(\frac{m}{2kT}\right)^{\frac{1+1}{2}}} = -\frac{nA}{2} \sqrt{\frac{2m}{\pi kT}} \frac{kT}{m} = -nA \sqrt{\frac{kT}{2\pi m}} \end{aligned}$$

We can do a little sanity check. From TØ week 1 we found the 3D case

$$\frac{dN}{dt} = -\frac{nA}{2} \sqrt{\frac{kT}{m}}$$

And a quick comparison shows that

$$\frac{1}{2} > \frac{1}{\sqrt{2\pi}}$$

Does it make sense that the 3D case diffuses faster than the 1D case? Yes since there are more degrees of freedom in the 3D case. From literature it is also well known that diffusion via a random walk model is faster in 3D than in 1D

Problem G2

Problem G2

Consider the situation illustrated in figure 1. Two boxes, A and B , sit in a perfectly isolated container, such that they are isolated from the surroundings. Box A has temperature T_A , which can be considered constant, since it has a very high heat capacity. Box B has temperature $T_B(t)$ at time $t \geq 0$. The container is shaped as a box with an inner cross section area L^2 , and box A and B have the same cross section area L^2 , so that they fit perfectly in the container. Both boxes emit electromagnetic radiation as perfect black bodies from the side, which is turned towards the cavity of the container. The walls of the container are perfect mirrors, which reflect all electromagnetic radiation.

Question a:

Give an expression for the power P_e of the electromagnetic radiation that block B emits at time $t \geq 0$. Give an expression for the power P_a of the electromagnetic radiation that block B absorbs at time $t \geq 0$. Use this to derive an expression for the change per unit time of the energy of the block, $dU_B(t)/dt$.

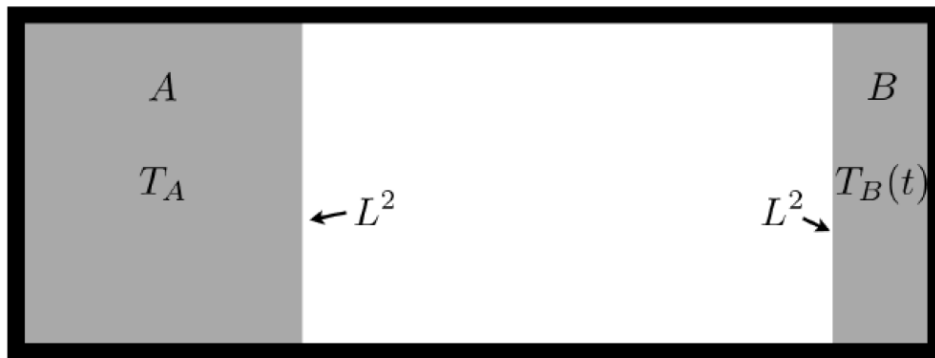


Figure 1

The power emitted by block B per unit area is

$$\frac{P_e}{A} = e_B \sigma T_B^4$$

The power absorbed by block B per unit area is

$$\frac{P_a}{A} = e_A \sigma T_A^4$$

The energy change of block B per unit time is

$$\Delta P_B = \frac{dU_B}{dt} = L^2 e_A \sigma T_A^4 - L^2 e_B \sigma T_B^4$$

They are perfect black boxes so $e = 1$ and we get

$$\frac{dU_B}{dt} = L^2 \sigma (T_A^4 - T_B^4)$$

Block B is an isolated crystal with N atoms, and its energy can be described by use of Debye-theory for quantized lattice vibrations called phonons. Its Debye temperature is T_D , where $T_B(t=0) \ll T_D \ll T_A$, and the sound velocity in the block is c_s .

Question b:

Give an expression for the change per unit time of the temperature of block B , dT_B/dt , for $T_B \ll T_D$ and for $T_B \gg T_D$.

The heat capacity is defined as

$$C_V = \left(\frac{\partial U}{\partial T} \right)_{V,N}$$

Since volume and number of particles is already constant in the system then we can change the soft differential to a hard

$$C_{V,B} = \frac{dU_B}{dT_B}$$

We can therefore see what happens if we do this

$$\frac{\frac{dU_B}{dt}}{\frac{dT_B}{dt}} = \frac{dT_B}{dt} = \frac{L^2 \sigma (T_A^4 - T_B^4)}{C_{V,B}}$$

In the low temperature limit

$$C_{V,B} = \frac{12\pi^4}{5} Nk \left(\frac{T_B}{T_D} \right)^3, \quad T_B \ll T_D$$

In the high temperature limit

$$C_{V,B} = 3Nk, \quad T_B \gg T_D$$

Question c:

Show that the number of phonons N_p in the block B is

$$N_p = \eta T_B(t)$$

for $T_B \gg T_D$ and give an expression for η .

The mean energy for all the oscillators are

$$U = \frac{9Nk_B T^4}{T_D^3} \int_0^{\frac{T_D}{T}} \frac{x^3}{e^x - 1} dx$$

Where $x = \frac{hcn}{2LkT} = \frac{hf}{kT} = \frac{\hbar\omega}{kT}$

In order to get the number of phonons then we simply calculate $N_p = \frac{U}{\hbar\omega}$ since each vibration has a unit energy of $\hbar\omega$ according to the harmonic oscillator. Let's calculate

$$N_p = \frac{U}{\hbar\omega} = \frac{9Nk_B T_B^4}{T_D^3} \int_0^{\frac{T_D}{T_B}} \frac{x^3}{\hbar\omega(e^x - 1)} dx = \frac{9NT_B^3}{T_D^3} \int_0^{\frac{T_D}{T_B}} \frac{x^2}{(e^x - 1)} dx$$

As pointed out in TØ you have to divide by $\hbar\omega$ within the integral because in the book we have

$$\langle E \rangle = 3 \sum_{n_x} \sum_{n_y} \sum_{n_z} \epsilon \bar{n}_{pl}(\epsilon)$$

We just modify it to sum over the average occupation of each phonon mode

$$N_p = 3 \sum_{n_x} \sum_{n_y} \sum_{n_z} \bar{n}_{pl}(\epsilon)$$

We are in the limit $T_B \gg T_D$ which allows us to use a Taylor approximation $e^x \approx 1 + x$ and the integral is simpler now

$$N_p = \frac{9NT_B^3}{T_D^3} \int_0^{\frac{T_D}{T_B}} \frac{x^2}{1 + x - 1} dx = \frac{9NT_B^3}{T_D^3} \int_0^{\frac{T_D}{T_B}} x dx = \frac{9NT_B^3}{T_D^3} \left[\frac{x^2}{2} \right]_0^{\frac{T_D}{T_B}} = \frac{9NT_B}{2T_D}$$

Problem G3

Problem G3

A cube with side length L , temperature T_1 and emissivity e sends out electromagnetic radiation.

Question a:

Compute the total energy per time (power) P of the electromagnetic radiation from the cube.

The power emitted is given by the formula

$$\frac{P}{A} = e\sigma T^4$$

The total area of the cube is $6L^2$ so

$$P = 6L^2 e\sigma T_1^4$$

The cube is a crystal consisting of N ions and N electrons, and its total energy $E(T)$ is given by the sum of the vibrational energy of the ions and the kinetic energy of the electrons. The vibrational energy of the ions can be described by use of the Debye theory for quantized crystal vibrations called phonons with sound velocity c_s . It is the case that $\hbar c_s \pi n_{\max}/L \gg kT_1$, where $n_{\max} = (6N/\pi)^{1/3}$. The electrons can be described as an ideal Fermi gas with Fermi temperature $T_F \gg T_1$.

Question b:

Compute the change in the energy $E(T)$ of the cube when its temperature drops from T_1 to T_2 .

Let us first find an expression for the energy of the electrons. According to the book the energy of a Fermi gas at $kT \ll \epsilon_F$ is

$$U_{el} = \frac{3}{5} N \epsilon_F + \frac{\pi^2}{4} N \frac{(kT)^2}{\epsilon_F}$$

Therefore

$$\Delta U_{el} = \frac{\pi^2}{4} N \frac{k^2(T_2^2 - T_1^2)}{\epsilon_F}$$

Now we have an expression for the energy of the phonons. The condition we have is the same as

$$\frac{\hbar c_s \pi n_{\max}}{L} = \frac{\hbar c_s n_{\max}}{2L} = kT \frac{T_D}{T} = kT_D \gg kT_1 > kT_2$$

We should therefore use the expression of energy in the low temperature limit $T \ll T_D$

$$U_{phonon} = \frac{3\pi^4}{5} N k_B T \left(\frac{T}{T_D} \right)^3$$

Therefore

$$\Delta U_{phonon} = \frac{3\pi^4 N k}{5 T_D^3} (T_2^4 - T_1^4)$$

Overall the change in energy is

$$\Delta U = \Delta U_{phonon} + \Delta U_{el} = \frac{3\pi^4 N k}{5 T_D^3} (T_2^4 - T_1^4) + \frac{\pi^2}{4} N \frac{k^2 (T_2^2 - T_1^2)}{\epsilon_F}$$

Question c:

Compute the number N_p of phonons in the cube, when its temperature is T_1 .

Hint: You can possibly use without proof that $\int_0^\infty dx x^2 / (\exp x - 1) \simeq 2.40$.

Exactly the same solution as in problem G2 except we are given the value of the integral instead

The mean energy for all the oscillators are

$$U = \frac{9 N k_B T^4}{T_D^3} \int_0^{\frac{T_D}{T}} \frac{x^3}{e^x - 1} dx$$

Where $x = \frac{hcn}{2LkT} = \frac{hf}{kT} = \frac{\hbar\omega}{kT}$

In order to get the number of phonons then we simply calculate $N_p = \frac{U}{\hbar\omega}$ since each vibration has a unit energy of $\hbar\omega$ according to the harmonic oscillator. Let's calculate

$$N_p = \frac{U}{\hbar\omega} = \frac{9 N k_B T_1^4}{T_D^3} \int_0^{\frac{T_D}{T_1}} \frac{x^3}{\hbar\omega(e^x - 1)} dx = \frac{9 N T_1^3}{T_D^3} \int_0^{\frac{T_D}{T_1}} \frac{x^2}{(e^x - 1)} dx$$

We are in the low temperature limit $T_1 \ll T_D$ so the upper limit of the integral is effectively infinite and then we can use our hint

$$N_p \approx \frac{9 N T_1^3}{T_D^3} \int_0^\infty \frac{x^2}{(e^x - 1)} dx = 2.40 \frac{9 N T_1^3}{T_D^3}$$

Problem G4

Problem G4

A container with volume V_1 is filled with electromagnetic radiation at temperature T_1 . Now the container is expanded to the volume $V_2 = 2V_1$, and the temperature of the radiation is T_2 after the system is again in thermodynamic equilibrium. In the following, the heat capacity of the container can be neglected.

Question a:

Compute the final temperature T_2 of the radiation, if container and radiation are an isolated system during the expansion, such that the total energy is conserved.

The chapter about the Planck spectrum treats this situation of radiation in a box very nicely. The result is

$$\frac{U}{V} = \frac{8\pi^5 k^4 T^4}{15h^3 c^3} \Leftrightarrow U = \frac{8V\pi^5 k^4 T^4}{15h^3 c^3}$$

The total energy is conserved which means that $\Delta U = 0$. We therefore have

$$\frac{8V_1\pi^5 k^4 T_1^4}{15h^3 c^3} = \frac{8V_2\pi^5 k^4 T_2^4}{15h^3 c^3}$$

$$\Updownarrow$$

$$V_1 T_1^4 = V_2 T_2^4 = 2V_1 T_2^4$$

$$\Updownarrow$$

$$T_2 = \frac{T_1}{2^{\frac{1}{4}}}$$

Question b:

Compute the final temperature T_2 of the radiation, if the expansion is isentropic, that is the entropy is constant.

Equation 7.89 the entropy of a photon gas is given by

$$S(T) = \frac{32\pi^5 k^4 V}{45h^3 c^3} T^3$$

For $\Delta S = 0$ we therefore have

$$S_1 = S_2$$

$$\Updownarrow$$

$$\frac{32\pi^5 k^4 V_1}{45h^3 c^3} T_1^3 = \frac{32\pi^5 k^4 2V_1}{45h^3 c^3} T_2^3$$

$$\Downarrow$$

$$2T_2^3 = T_1^3$$

$$\Downarrow$$

$$T_2 = \frac{T_1}{2^{1/3}}$$

We now consider the total energy E of the photons in the container with photon energy less than $\epsilon_{\max} = \hbar\omega_{\max}$.

Question c:

Give an expression for E in the container before the expansion, where it has volume V_1 and temperature T_1 . Show that

$$E \simeq \kappa T_1$$

for $\epsilon_{\max} \ll kT_1$, and give an expression for κ .

Hint: You can use that $e^x \simeq 1 + x$ for $x \ll 1$.

The energy in the container up to $\epsilon_{\max}$ is (equation 7.83)

$$\frac{U}{V} = \frac{8\pi}{h^3 c^3} \int_0^{\epsilon_{\max}} \frac{\epsilon^3}{e^{\beta\epsilon} - 1} d\epsilon$$

Using integration by substitution

$$x = \beta\epsilon \Leftrightarrow \epsilon = xkT, \quad \frac{dx}{d\epsilon} = \beta \Leftrightarrow d\epsilon = kT dx$$

Therefore

$$\frac{U}{V} = \frac{8\pi(kT)^4}{h^3 c^3} \int_0^{\frac{\epsilon_{\max}}{kT}} \frac{x^3}{e^x - 1} dx$$

Since $\epsilon < \epsilon_{\max} \ll kT$ then $x \ll 1$ and we can use the hint that we have been given

$$\begin{aligned} \frac{U}{V} &= \frac{8\pi(kT)^4}{h^3 c^3} \int_0^{\frac{\epsilon_{\max}}{kT}} \frac{x^3}{1 + x - 1} dx = \frac{8\pi(kT)^4}{h^3 c^3} \int_0^{\frac{\epsilon_{\max}}{kT}} x^2 dx \\ &= \frac{8\pi(kT)^4}{h^3 c^3} \left[\frac{x^3}{3} \right]_0^{\frac{\epsilon_{\max}}{kT}} = \frac{8\pi(kT)^4}{3h^3 c^3} \left(\frac{\epsilon_{\max}}{kT} \right)^3 \\ &= \frac{8\pi\epsilon_{\max}^3}{3h^3 c^3} kT \end{aligned}$$

$$\Updownarrow$$

$$U=\frac{8\pi V\epsilon_{max}^3}{3h^3c^3}kT$$

Therefore

$$\kappa=\frac{8\pi V\epsilon_{max}^3}{3h^3c^3}k$$

Problem G5

Problem G5

Consider a plate in empty space. The plate has height L_x , width L_y , and thickness L_z . There is a chemical reaction taking place in the plate, and P/V is the power that this reaction produces per volume of the plate. We assume that $L_z \ll L_x$ and $L_z \ll L_y$, but $L_z P/V$ is not small. The plate is a perfect blackbody, and there is no radiation hitting the plate from outside.

Question a:

Compute the temperature T of the plate.

What we are told is just that we have a very thin plate. The power generated inside the plate must be equal to the power emitted via blackbody radiation

$$P = eA\sigma T^4$$

We can reasonably ignore the other sides of the plate because of how thin it is and approximate the total area as $A = 2L_x L_y$. Since it is a perfect blackbody then $e = 1$. The temperature is therefore

$$T = \left(\frac{P}{2L_x L_y \sigma} \right)^{\frac{1}{4}}$$

We now consider the situation in figure 1. Plate 0 is a perfect mirror, plate 1 is the plate described in question **a**, and both plate 2 and plate 3 are perfect blackbodies. There is no power produced inside the plates 0, 2, and 3. Plate 3 is in contact with a heat reservoir and is always at temperature T_3 . The height and the width of the plates are very large compared to the thickness of the plates and very large compared to the distance between the plates.

Question b:

Compute the temperature T_1 of plate 1.

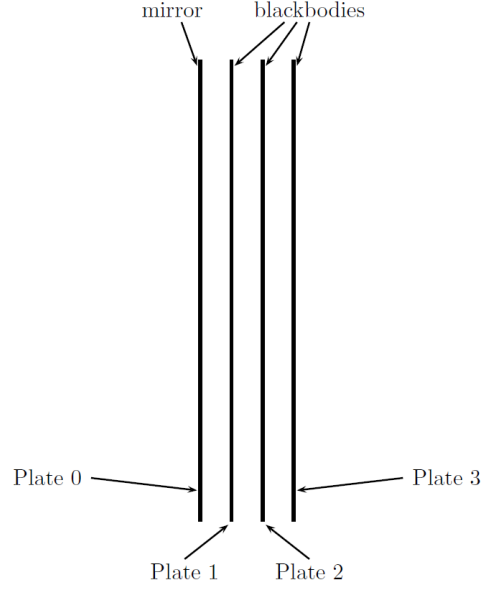


Figure 1: Four plates seen from the side. The plates are all parallel, and there is vacuum between the plates.

Let $A = L_x L_y$ be the area of only one side of each plate and let us assume that all the areas are the same. Plate 3 emits a constant power to plate 2

$$P = \sigma A T_3^4$$

Also let us assume that everything is in equilibrium such that the total power emitted from P_1 and P_2 is zero. Plate 2 emits power according to its own temperature but it also absorbs energy from plate 1 and 2. Overall the power emitted and the power absorbed should be equal

$$2\sigma A T_2^4 = \sigma A T_3^4 + \sigma A T_1^4$$

Using the same reasoning on plate 1. The power generated and absorbed should equal the power emitted

$$P_{reaction} + \sigma A T_1^4 + \sigma A T_2^4 = 2\sigma A T_1^4$$

We know have two equations and two unknowns T_1 and T_2 which we can solve for

$$2\sigma A T_2^4 = \sigma A T_3^4 + \sigma A T_1^4$$

$$\Downarrow$$

$$2T_2^4 = T_3^4 + T_1^4$$

The other equation gives

$$P_{reaction} + \sigma AT_1^4 + \sigma AT_2^4 = 2\sigma AT_1^4$$

$$\Updownarrow$$

$$T_1^4 = \frac{P_{reaction}}{\sigma A} + T_2^4 = \frac{P_{reaction}}{\sigma A} + \frac{(T_3^4 + T_1^4)}{2}$$

$$\Updownarrow$$

$$2T_1^4 = \frac{P_{reaction}}{\sigma A} + T_2^4 = \frac{2P_{reaction}}{\sigma A} + T_3^4 + T_1^4$$

$$\Updownarrow$$

$$T_1 = \left(\frac{2P_{reaction}}{\sigma A} + T_3^4\right)^{\frac{1}{4}} = \left(\frac{2P_{reaction}}{\sigma L_x L_y} + T_3^4\right)^{\frac{1}{4}}$$

S8 - Hoffmann Problems

Problem 4.6

- 6) *Einstein and Debye models*: Plot the heat capacity of Pb in the Einstein and the Debye models between 0.5 and 300 K. To do this, you need

to calculate the heat capacity in the Debye model numerically. The Debye temperature of Pb is $\Theta_D = 105$ K. Initially, use an Einstein temperature of $\Theta_E = \Theta_D$. You will see that the agreement between the two models is not particularly good. Why? Can you improve the agreement by a different choice of Θ_E ?

From the Debye model the mean energy for all oscillators is

$$\langle E \rangle = 3 \int_0^{\omega_D} \frac{g(\omega) \hbar \omega}{e^{\frac{\hbar \omega}{k_B T}} - 1} d\omega$$

Using

$$g(\omega) = \frac{V \omega^2}{2\pi^2 v^3}, \quad \omega_D^3 = 6\pi^2 \frac{N}{V} v^3$$

Then we obtain

$$\langle E \rangle = 3 \int_0^{\omega_D} \frac{\frac{3N\omega^2}{\omega_D^3} \hbar \omega}{e^{\frac{\hbar \omega}{k_B T}} - 1} d\omega = \frac{9N\hbar}{\omega_D^3} \int_0^{\omega_D} \frac{\omega^3}{e^{\frac{\hbar \omega}{k_B T}} - 1} d\omega$$

Now we differentiate to obtain the heat capacity

$$C_V = \frac{\partial \langle E \rangle}{\partial T} = \frac{9N\hbar}{\omega_D^3} \int_0^{\omega_D} \frac{e^{\frac{\hbar \omega}{k_B T}} \omega^3 \left(\frac{\hbar \omega}{k_B T^2} \right)}{\left(e^{\frac{\hbar \omega}{k_B T}} - 1 \right)^2} d\omega$$

Using integration by substitution $x = \frac{\hbar \omega}{k_B T}$, $d\omega = \frac{k_B T}{\hbar} dx$

$$\begin{aligned} C_V &= \frac{9N\hbar}{\omega_D^3} \int_0^{\frac{\hbar \omega_D}{k_B T}} \frac{e^x \left(\frac{k_B T x}{\hbar} \right)^3 \left(\frac{x}{T} \right) k_B T}{(e^x - 1)^2} \frac{k_B T}{\hbar} dx \\ &= \frac{9N\hbar}{\omega_D^3 T} \left(\frac{k_B T}{\hbar} \right)^4 \int_0^{\frac{\hbar \omega_D}{k_B T}} \frac{e^x x^4}{(e^x - 1)^2} dx \end{aligned}$$

It applies that

$$\frac{\hbar \omega_D}{k_B T} = \frac{T_D}{T}$$

Using this we obtain

$$C_V = 9Nk \left(\frac{kT}{\omega_D \hbar} \right)^3 \int_0^{\frac{T_D}{T}} \frac{e^x x^4}{(e^x - 1)^2} dx$$

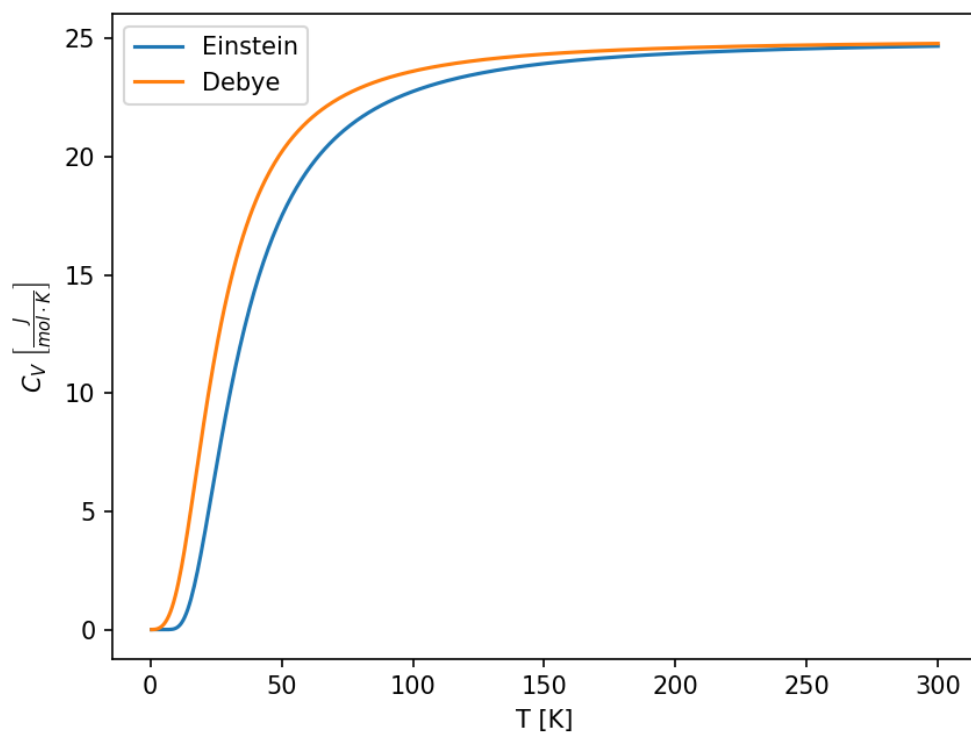
$$= 9Nk \left(\frac{T}{T_D} \right)^3 \int_0^{\frac{T_D}{T}} \frac{e^x x^4}{(e^x - 1)^2} dx$$

Nice now we have a model for the Debye model

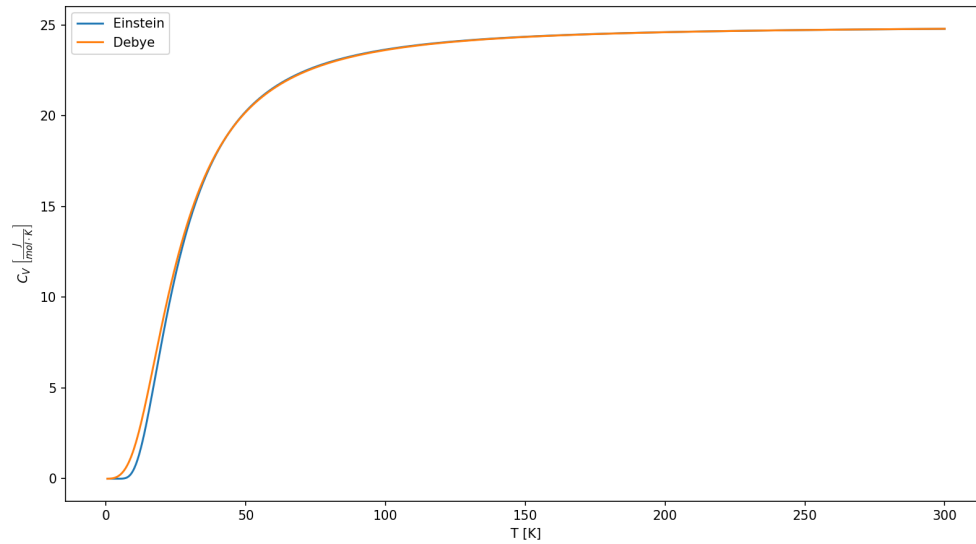
The Einstein model is simple and is just as written

$$C_V = \frac{\partial \langle E \rangle}{\partial T} = 3R \left(\frac{\Theta_E}{T} \right)^2 \frac{e^{\frac{\Theta_E}{T}}}{\left(e^{\frac{\Theta_E}{T}} - 1 \right)^2}$$

Then we obtain



Lowering the Einstein temperature to 80K brings them closer



Python code here

```
import matplotlib.pyplot as plt
import numpy as np
from scipy.integrate import quad

# Einstein Model
R = 8.314 # J/mol*K
Theta_D = 105
Theta_E = 80 # K
T = np.linspace(0.5, 300, 500)
C_Einstein = 3*R*(Theta_E/T)**2*np.exp(Theta_E/T)/(np.exp(Theta_E/T)-1)**2

# Debye Model
k = 1.381*10**(-23) #J*s
N = 6.022*10**23 #mol^-1

function = lambda x: x**4*np.exp(x)/(np.exp(x)-1)**2
integral = np.array([quad(function,0 , Theta_D/i)[0] for i in T])

C_Debye = 9*N*k*(T/Theta_D)**3*integral

fig, ax = plt.subplots()
ax.plot(T, C_Einstein)
ax.plot(T, C_Debye)
plt.ylabel('$C_v$ $\left[\frac{J}{mol \cdot K}\right]$')
```

```
plt.xlabel('T [K]')
ax.legend(['Einstein', 'Debye'])

plt.show()
```

Problem 6.3

- 3) *Free electron model:* (a) Calculate the electronic heat capacity c_e for 1 mol of copper at 300 K. Use that the Fermi energy of Cu is 7 eV and the molar volume 7.11 cm^3 . (b) Compare the result of (a) to the Dulong–Petit value of the lattice, and explain why it is so much smaller. (c) Below which temperature is c_e larger than the heat capacity c_l from the lattice? Use that the Debye temperature of Cu is 343 K. (d) Plot the sum $c = c_e + c_l$ of electronic and lattice contributions as a function of temperature T between 0.1 and 20 K. Also plot c/T as a function of T^2 . (e)* Derive the electronic heat capacity from Eq. (6.17) in a proper way by calculating the total energy and differentiating it. Hint: For this last part, assume that $k_B T \ll E_F$. In this case, the density of states $g(E)$ in the vicinity of the Fermi energy can be taken to be $g(E_F)$, independent of the energy, and the chemical potential can be taken as independent of the temperature, as usual, that is, $\mu = E_F$.

- a) The electronic heat capacity is given by

$$C_V = \frac{\pi^2}{3} k_B^2 T g(E_F)$$

The density of states at the Fermi level is given by

$$g(E_F) = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \sqrt{E_F}$$

The value of the density states is

$$g(E_F) = \frac{7.11 \text{ cm}^3}{2\pi^2} \left(\frac{2 \cdot 9.109 \cdot 10^{-31} \text{ kg}}{(1.055 \cdot 10^{-34} \text{ J} \cdot \text{s})^2} \right)^{\frac{3}{2}} \sqrt{7 \cdot 1.602 \cdot 10^{-19} \text{ C} \cdot \text{V}}$$

$$= 798755392706369266764919832736389245435904 \left(\frac{\text{s}^2}{\text{kg} \cdot \text{m}^2} \right)$$

This gives a heat capacity of

$$C_V = \frac{\pi^2}{3} \cdot \left(1.381 \cdot 10^{-23} \frac{\text{J}}{\text{K}} \right)^2 \cdot 300 \text{ K} \cdot 798755392706369266764919832736389245435904 \left(\frac{\text{s}^2}{\text{kg} \cdot \text{m}^2} \right)$$

$$\approx 0,1503491 \left(\frac{\text{J}}{\text{K}} \right)$$

b) The Dulong Petit value is

$$3R = 3 \cdot 8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} = 24,942 \frac{\text{J}}{\text{K}}$$

Most electrons cannot contribute to the heat capacity because they are trapped far below the Fermi level. Therefore the electronic heat capacity is much lower than the Dulong Petit value for a lattice

c) The heat capacity for the lattice is given by

$$C_V = \frac{12\pi^4}{5} N_A k_B \left(\frac{T}{\Theta_D} \right)^3$$

$$c_{\text{lattice}} = \frac{12\pi^4}{5} \cdot 6.022 \cdot 10^{23} \text{ mol}^{-1} \cdot 1.381 \cdot 10^{-23} \frac{\text{J}}{\text{K}} \cdot \left(\frac{1}{343 \text{ K}} \right)^3 \approx 4,817956 \cdot 10^{-5} \left(\frac{\text{J}}{\text{K}^4} \right) \cdot T^3$$

The heat capacity for the electrons is

$$c_{\text{electron}} = \frac{0,1503491 \left(\frac{\text{J}}{\text{K}} \right)}{300 \text{ K}} \approx 5,011637 \cdot 10^{-4} \left(\frac{\text{J}}{\text{K}^2} \right) T$$

The point where they are equal is given by

$$5,011637 \cdot 10^{-4} \left(\frac{\text{J}}{\text{K}^2} \right) T = 4,817956 \cdot 10^{-5} \left(\frac{\text{J}}{\text{K}^4} \right) \cdot T^3$$

$\Downarrow$

$$T = \sqrt{\frac{5,011637 \cdot 10^{-4} \left(\frac{\text{J}}{\text{K}^2} \right)}{4,817956 \cdot 10^{-5} \left(\frac{\text{J}}{\text{K}^4} \right)}} \approx 3,225213 \text{ K}$$

Below 3.2 K then the electronic heat capacity is larger than the lattice

d) Plotting

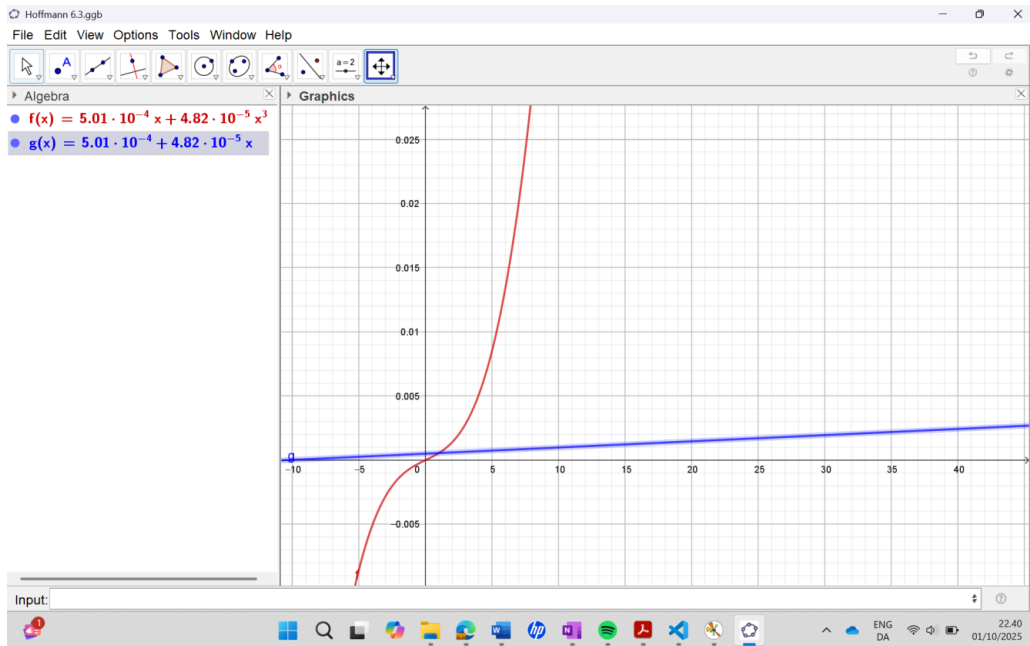
We just plot

$$c_{lattice} = \frac{12\pi^4}{5} N_A k_B \left(\frac{T}{\Theta_D} \right)^3 = 4,817956 \cdot 10^{-5} \left(\frac{J}{K^4} \right) \cdot T^3$$

$$c_{electron} = \frac{\pi^2}{3} k_B^2 T g(E_F) = 5,011637 \cdot 10^{-4} \left(\frac{J}{K^2} \right) T$$

And we also plot a linear function

$$\frac{c}{T}(x^2) = 5,011637 \cdot 10^{-4} + 4,817956 \cdot 10^{-5} \cdot x^2$$



e) Electronic heat capacity derivation

The total energy of the electron gas is given by

$$U = \int_0^{\infty} \epsilon g(\epsilon) \bar{n}_{FD}(\epsilon) d\epsilon$$

We know the Fermi-Dirac distribution

$$\bar{n}_{FD}(\epsilon) = \frac{1}{e^{\frac{(\epsilon - \mu)}{kT}} + 1}$$

We also know the density of states of the electrons

$$g(\epsilon) = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \sqrt{\epsilon}$$

Therefore only the Fermi-Dirac distribution depends on temperature. The heat capacity can be calculated as

$$C_V = \frac{\partial U}{\partial T} = \frac{\partial}{\partial T} \int_0^\infty \epsilon g(\epsilon) \bar{n}_{FD}(\epsilon) d\epsilon$$

Notice only the Fermi-Dirac distribution depends on temperature. Furthermore, in the limit $kT \ll \epsilon_F$ the distribution function is constant along most of the temperature scale except for around the Fermi level within the soft zone of $4kT$ which is very small in this case. Therefore $\frac{\partial \bar{n}_{FD}(\epsilon)}{\partial T} = 0$ for T outside of the soft zone. We therefore only need to take into account the density of states around the Fermi level which we assume is constant at $g(\epsilon_F)$. Therefore

$$\begin{aligned} C_V &= g(\epsilon_F) \int_0^\infty \epsilon \frac{\partial \bar{n}_{FD}(\epsilon)}{\partial T} d\epsilon \\ &= g(\epsilon_F) \int_0^\infty \epsilon \frac{\partial}{\partial T} \left(\frac{1}{e^{\frac{(\epsilon-\mu)}{kT}} + 1} \right) d\epsilon \\ &= -g(\epsilon_F) \int_0^\infty \frac{\epsilon}{\left(e^{\frac{(\epsilon-\mu)}{kT}} + 1 \right)^2} \frac{\partial}{\partial T} \left(e^{\frac{(\epsilon-\mu)}{kT}} + 1 \right) d\epsilon \\ &= -g(\epsilon_F) \int_0^\infty \frac{\epsilon e^{\frac{(\epsilon-\mu)}{kT}}}{\left(e^{\frac{(\epsilon-\mu)}{kT}} + 1 \right)^2} \frac{\partial}{\partial T} \left(\frac{(\epsilon-\mu)}{kT} \right) d\epsilon \\ &= g(\epsilon_F) \int_0^\infty \frac{\epsilon e^{\frac{(\epsilon-\mu)}{kT}}}{\left(e^{\frac{(\epsilon-\mu)}{kT}} + 1 \right)^2} \left(\frac{-(\epsilon-\mu)}{kT^2} \right) d\epsilon \end{aligned}$$

Using integration by substitution with

$$x = \frac{(\epsilon - \mu)}{kT} \Leftrightarrow \epsilon = xkT + \mu \Rightarrow d\epsilon = kT dx$$

Then we get

$$\begin{aligned} C_V &= g(\epsilon_F) \int_{-\frac{\mu}{kT}}^\infty \frac{(xkT + \mu)e^x}{(e^x + 1)^2} \left(\frac{x}{T} \right) kT dx \\ &= g(\epsilon_F) \int_{-\frac{\mu}{kT}}^\infty \frac{(xkT + \mu)e^x}{(e^x + 1)^2} \left(\frac{x}{T} \right) kT dx \end{aligned}$$

$$= k g(\epsilon_F) \int_{-\frac{\mu}{kT}}^{\infty} \frac{(x^2 kT + x\mu)e^x}{(e^x + 1)^2} dx$$

We also use that $\mu = \epsilon_F$ to get

$$C_V = k g(\epsilon_F) \int_{-\frac{\epsilon_F}{kT}}^{\infty} \frac{(x^2 kT + x\epsilon_F)e^x}{(e^x + 1)^2} dx$$

In the limit $kT \ll \epsilon_F$ then the lower integral limit may as well extend to infinity

$$\begin{aligned} C_V &= k g(\epsilon_F) \int_{-\infty}^{\infty} \frac{(x^2 kT + x\epsilon_F)e^x}{(e^x + 1)^2} dx \\ &= k g(\epsilon_F) \int_{-\infty}^{\infty} \frac{(x^2 kT + x\epsilon_F)e^x}{(e^x + 1)^2} dx \\ &= k g(\epsilon_F) \left[\int_{-\infty}^{\infty} \frac{(x^2 kT)e^x}{(e^x + 1)^2} dx + \int_{-\infty}^{\infty} \frac{(x\epsilon_F)e^x}{(e^x + 1)^2} dx \right] \end{aligned}$$

It applies that $\frac{e^x}{(e^x+1)^2}$ is an even function because

$$f(x) = \frac{e^x}{(e^x + 1)^2} = \frac{1}{e^{-x}(e^x + 1)(e^x + 1)} = \frac{1}{(e^x + 1)(e^{-x} + 1)} = f(-x)$$

Therefore the first integral is over an even function and the second integral is over an uneven function. We can use this to simplify the integrals

$$\begin{aligned} C_V &= k g(\epsilon_F) \int_{-\infty}^{\infty} \frac{(x^2 kT)e^x}{(e^x + 1)^2} dx \\ &= 2k^2 T g(\epsilon_F) \int_0^{\infty} \frac{x^2 e^x}{(e^x + 1)^2} dx \end{aligned}$$

This integral is very nasty and I had to look it up on Stackexchange

Consider

$$\begin{aligned}
 I(a) &= \int_0^{\infty} \frac{x}{1+e^{ax}} dx \\
 &= \int_0^{\infty} \frac{x e^{-ax}}{1+e^{-ax}} dx \\
 &= \int_0^{\infty} \sum_{k=0}^{\infty} (-1)^k x e^{-a(k+1)x} dx \\
 &= \sum_{k=0}^{\infty} (-1)^k \int_0^{\infty} x e^{-a(k+1)x} dx \\
 &= \sum_{k=0}^{\infty} \frac{(-1)^k}{a^2(k+1)^2} \\
 &= \frac{\pi^2}{12a^2}
 \end{aligned}$$

then the given integral is simply $2I'(-1)$.

Using the hint from Stackexchange I can see that we can use the following standard integral as a solution:

$$\mathbf{18.81.} \quad \int_0^{\infty} \frac{x dx}{e^x + 1} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$$

Generalizing the integral with an extra factor

$$\int_0^{\infty} \frac{x}{e^{ax} + 1} dx$$

Lets do some substitution $u = ax$ and $x = \frac{u}{a}$ and $dx = \frac{du}{a}$ then we obtain

$$f(a) = \int_0^{\infty} \frac{x}{e^{ax} + 1} dx = \int_0^{\infty} \frac{\frac{u}{a}}{e^u + 1} \frac{du}{a} = \frac{1}{a^2} \int_0^{\infty} \frac{u}{e^u + 1} du = \frac{\pi^2}{12a^2}$$

We notice that

$$\frac{df}{da} = \int_0^{\infty} \frac{d}{da} \frac{x}{e^{ax} + 1} dx = - \int_0^{\infty} \frac{x^2 e^{ax}}{(e^{ax} + 1)^2} dx$$

Similarly

$$\frac{df}{da} = \frac{d}{da} \frac{\pi^2}{12a^2} = -2 \frac{\pi^2}{12} a^{-3}$$

Therefore

$$- \int_0^{\infty} \frac{x^2 e^{ax}}{(e^{ax} + 1)^2} dx = -2 \frac{\pi^2}{12} a^{-3}$$

Setting $a = 1$ we obtain the desired solution to our integral

$$\int_0^{\infty} \frac{x^2 e^x}{(e^x + 1)^2} dx = \frac{\pi^2}{6}$$

Okay I don't expect anyone to get the solution to this integral without help from google. But using this result I obtain

$$C_V = 2k^2 T g(\epsilon_F) \int_0^{\infty} \frac{x^2 e^x}{(e^x + 1)^2} dx = \frac{\pi^2}{3} k^2 T g(\epsilon_F)$$

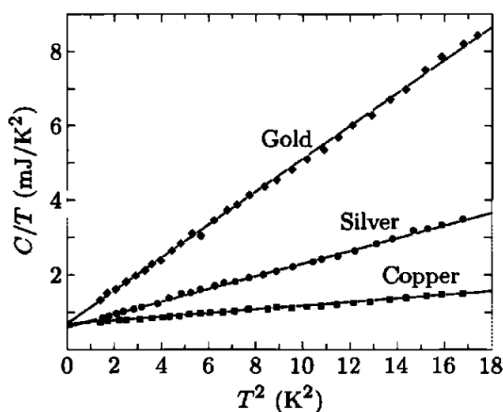
Where

$$g(\epsilon) = \frac{V}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{\frac{3}{2}} \sqrt{\epsilon}$$

And that is exactly what I wanted to prove

Problem 7.59. Explain in some detail why the three graphs in Figure 7.28 all intercept the vertical axis in about the same place, whereas their slopes differ considerably.

Figure 7.28. Low-temperature measurements of the heat capacities (per mole) of copper, silver, and gold. Adapted with permission from William S. Corak et al., *Physical Review* **98**, 1699 (1955).



The heat capacity in the low temperature limit is the electronic contribution plus the phonon contribution

$$C_V = \frac{\pi^2 N k^2 T}{2 \epsilon_F} + \frac{12 \pi^4}{5} N k_B \left(\frac{T}{T_D} \right)^3, \quad T \ll T_D$$

At relatively high temperatures the electronic contribution is very small and

$$C_V \approx \frac{12 \pi^4}{5} N k_B \left(\frac{T}{T_D} \right)^3$$

And the slopes are determined by their Debye temperature. A quick google search shows

◆ AI Overview

The Debye temperatures for gold, copper, and silver are approximately **165 K (Gold)**, **343 K (Copper)**, and **227 K (Silver)**, though values can vary slightly based on experimental conditions and theoretical models. Copper has the highest Debye temperature, indicating stronger lattice vibrations, while gold has the lowest, suggesting a more compliant lattice. 

This is in agreement with the size of slopes in the plot.

Now what about the intercept. At very low temperatures then

$$C_V \approx \frac{\pi^2 N k^2 T}{2 \epsilon_F}$$

They should therefore intercept at they y-axis if their Fermi level is approximately the same. A quick google search gives

The Fermi energies for gold, copper, and silver are approximately 5.5 eV (Gold), 7.0 eV (Copper), and 5.5 eV (Silver), respectively. These values represent the highest energy level occupied by an electron in a metal at absolute zero temperature, and they provide insight into the electronic structure and conductivity of these noble metals. 

There is not a big difference between the Fermi energies so that explains why they are intercepting at the same place

Problem H1

Problem H1

Consider a non-interacting gas of identical particles of mass m in a one-dimensional wire of length L . The single particle states are $\psi_l(x) = \sin(\pi x l/L) \sqrt{2/L}$ with energy $\epsilon_l = h^2 l^2 / (8mL^2)$, where $l = 1, 2, 3, \dots$

Question a:

Write down two equations, one for bosons and one for fermions, which determine the number of particles N in terms of the chemical potential μ and the temperature T . Give a relation between the Fermi energy ϵ_F and the density $n = N/L$ for $T = 0$.

The number of particles can be determined from the occupancy according to Fermi-Dirac statistics and Boson statistics. Remember that the occupancy is the average number of particles in a given state. Therefore, by summing over all states, we should obtain the total number of particles

$$N_{fermions} = \sum_{l=1}^{\infty} \bar{n}_{FD} = \sum_{l=1}^{\infty} \frac{1}{e^{\frac{(h^2 l^2 - \mu)}{8mL^2 kT}} + 1}$$

$$N_{bosons} = \sum_{l=1}^{\infty} \bar{n}_{BE} = \sum_{l=1}^{\infty} \frac{1}{e^{\frac{(h^2 l^2 - \mu)}{8mL^2 kT}} - 1}$$

The Fermi energy for fermions is $\epsilon_F = \frac{h^2 N^2}{8mL^2}$ since each state can have 1 particle

Bosons cannot have a Fermi energy because no state is truly every filled

Question b:

Since L is assumed large, you may convert sums over l into integrals. Show that in the classical limit $kT \gg \epsilon_F$, we have

$$\mu \approx kT \ln \left(\frac{nh}{\sqrt{2\pi m kT}} \right)$$

both for the bosons and the fermions.

When $kT \gg \epsilon_F$ then the number of states with non-zero occupancy is so large that we may as well see the occupancy as a continuous distribution instead of a discrete one. This means that

$$N_{fermions} = \sum_{l=1}^{\infty} \frac{1}{e^{\frac{(\frac{h^2 l^2}{8mL^2} - \mu)}{kT}} + 1} \approx \int_0^{\infty} \frac{1}{e^{\frac{(\frac{h^2 l^2}{8mL^2} - \mu)}{kT}} + 1} dl$$

$$N_{bosons} = \sum_{l=1}^{\infty} \frac{1}{e^{\frac{(\frac{h^2 l^2}{8mL^2} - \mu)}{kT}} - 1} \approx \int_0^{\infty} \frac{1}{e^{\frac{(\frac{h^2 l^2}{8mL^2} - \mu)}{kT}} - 1} dl$$

Now we can make an approximation using $kT \gg \epsilon_F$ this may seem counterintuitive but this approximation also tells us that

$$\frac{h^2 l^2}{8mL^2} - \mu \gg kT$$

That is because the occupation of practically all states in the system has to be small which requires the exponential to be large. When l is large then $\frac{h^2 l^2}{8mL^2} - \mu$ will naturally also be large. But what about when l is small. Remember the Sommerfeld expansion which shows that μ becomes increasingly more negative for $kT \gg \epsilon_F$. This means that $\frac{h^2 l^2}{8mL^2} - \mu$ will also be large even for small l . Therefore the ± 1 is very small compared to the exponential function in both cases which means that

$$N \approx N_{bosons} \approx N_{fermions} \approx \int_0^{\infty} \frac{1}{e^{\frac{(\frac{h^2 l^2}{8mL^2} - \mu)}{kT}}} dl = \int_0^{\infty} e^{-\frac{(\frac{h^2 l^2}{8mL^2} - \mu)}{kT}} dl = e^{\frac{\mu}{kT}} \int_0^{\infty} e^{-\frac{h^2 l^2}{8mL^2 kT}} dl$$

We can solve the integral using

$$\int_0^{\infty} x^m e^{-ax^2} dx = \frac{\Gamma\left(\frac{m+1}{2}\right)}{2a^{\frac{m+1}{2}}}$$

This gives

$$N = e^{\frac{\mu}{kT}} \frac{\Gamma\left(\frac{1}{2}\right)}{2\left(\frac{h^2}{8mL^2 kT}\right)^{\frac{1}{2}}} = \frac{e^{\frac{\mu}{kT}}}{2} \sqrt{\frac{8mL^2 kT \pi}{h^2}} = \frac{e^{\frac{\mu}{kT}} L \sqrt{2\pi m kT}}{h}$$

$\Updownarrow$

$$e^{\frac{\mu}{kT}} = \frac{Nh}{L\sqrt{2\pi m kT}} = \frac{nh}{\sqrt{2\pi m kT}}$$

$\Updownarrow$

$$\mu = kT \ln\left(\frac{nh}{\sqrt{2\pi m kT}}\right)$$

Question c:

The integral $\int_0^\infty dx (A \exp(x^2) - 1)^{-1}$ diverges for $A \rightarrow 1$. Use this to argue whether bosons in one-dimension form a Bose-Einstein condensate for any non-zero temperature $T > 0$ or not.

BEC is formed whenever the number of atoms that can be held in the excited state is capped such that they are forced in the ground state. According to our treatment for a 1D gas the number of atoms that can be held in the excited states of our system is

$$N = \int_0^\infty \frac{1}{e^{\frac{\left(\frac{h^2 l^2}{8mL^2} - \mu\right)}{kT}} - 1} dl = \int_0^\infty \frac{1}{e^{\frac{-\mu}{kT} e^{\frac{h^2 l^2}{8mL^2 kT}}} - 1} dl$$

Let $x^2 = \frac{h^2 l^2}{8mL^2 kT}$ then $\frac{dx}{dl} = \sqrt{\frac{h^2}{8mL^2 kT}}$ and

$$N = \sqrt{\frac{8mL^2 kT}{h^2}} \int_0^\infty \frac{1}{e^{\frac{-\mu}{kT} e^{x^2}} - 1} dx$$

Now we can use the same argumentation as in Schroeder. If we cut the limit somewhat greater than ϵ_0 but much less than kT then $N \approx N_{ex}$

We need to find the maximum number of atoms that can be in the excited state. The answer is when $\mu \rightarrow 0$ because then the integral diverges and the cloud can hold an infinite number of atoms. BEC requires $\mu = 0$ since the ground state (which we define as zero energy on a relative scale) should be able to hold an infinitely large number of particles. The book may be confusing in this sense because it first say that $\mu = \epsilon_0$ at $T = 0$ and then say that $\mu = 0$ later on but they forgot to say that they change the relative scale to define the ground state as the zero point energy. Now to finish the argument since a BEC needs $\mu = 0$ but the number of atoms that can be held in the excited state for a 1D gas shoots to infinity when $\mu \rightarrow 0$ then we have a contradiction! We can therefore not have a BEC for $T > 0$

Problem H2

Problem H2

We have N noninteracting, indistinguishable, spinless bosons in a cubic box. The width, length, and height of the box are all L , and the mass of one atom is m . The system is in thermal and diffusive equilibrium with a reservoir, and the reservoir has temperature T and chemical potential μ .

a) Show that the mean energy U of the system is given by

$$U = \sum_{n_x=1}^{\infty} \sum_{n_y=1}^{\infty} \sum_{n_z=1}^{\infty} \frac{f}{e^{\beta g} - 1} \quad (1)$$

where n_x , n_y , and n_z are summed over all positive integers, f is independent of T and μ , and g is independent of T . Provide expressions for f and g .

The energy is given by

$$U = \sum_{n_x} \sum_{n_y} \sum_{n_z} \epsilon(\vec{n}) \bar{n}_{BE}(\epsilon)$$

The energy states for particles in a box is

$$\epsilon = \frac{h^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2)$$

The bose-einstein distirbution is

$$\bar{n}_{BE} = \frac{1}{e^{\frac{(\epsilon-\mu)}{kT}} - 1}$$

Therefore

$$U = \sum_{n_x} \sum_{n_y} \sum_{n_z} \frac{\frac{h^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2)}{e^{\frac{\left(\frac{h^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2) - \mu\right)}{kT}} - 1}$$

Therefore

$$f = \frac{h^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2), \quad g = \epsilon - \mu$$

We now assume that the temperature T is below the temperature T_c , at which a Bose-Einstein condensate starts to appear.

b) Show that the chemical potential is given by

$$\mu = a + \frac{bT}{1 - (T/T_c)^{3/2}}, \quad (2)$$

where a and b are independent of T , and give expressions for a and b .

In a BEC the ground state becomes the most dominating such that

$$\begin{aligned} N_0 &\approx \frac{kT}{\epsilon_0 - \mu} \\ \Downarrow \\ N_0 \epsilon_0 - kT &= N_0 \mu \\ \Downarrow \\ \mu &= \epsilon_0 - \frac{kT}{N_0} \end{aligned}$$

From the book we have an equation for the number of atoms in the ground state

$$N_0 = \left(1 - \left(\frac{T}{T_c}\right)^{\frac{3}{2}}\right) N$$

This gives

$$\mu = \epsilon_0 - \frac{kT}{\left(1 - \left(\frac{T}{T_c}\right)^{\frac{3}{2}}\right) N}$$

Therefore

$$a = \epsilon_0, \quad b = -\frac{k}{N}$$

c) Compute the heat capacity for L and N fixed and T less than T_c . The result for the heat capacity should contain sums over n_x , n_y , and n_z .

The heat capacity is given by

$$C_V = \left(\frac{\partial U}{\partial T}\right)_{V,N} = \frac{\partial}{\partial T} \left(\sum_{n_x} \sum_{n_y} \sum_{n_z} \frac{\epsilon}{e^{\frac{(\epsilon - \mu)}{kT}} - 1} \right)$$

$$\begin{aligned}
&= \sum_{n_x} \sum_{n_y} \sum_{n_z} \epsilon \frac{\partial}{\partial T} \frac{1}{e^{\frac{(\epsilon-\mu)}{kT}} - 1} \\
&= \sum_{n_x} \sum_{n_y} \sum_{n_z} - \frac{\epsilon}{\left(e^{\frac{(\epsilon-\mu)}{kT}} - 1\right)^2} \frac{\partial}{\partial T} \left(e^{\frac{(\epsilon-\mu)}{kT}} - 1\right) \\
&= \sum_{n_x} \sum_{n_y} \sum_{n_z} - \frac{\epsilon e^{\frac{(\epsilon-\mu)}{kT}}}{\left(e^{\frac{(\epsilon-\mu)}{kT}} - 1\right)^2} \frac{\partial}{\partial T} \left(\frac{(\epsilon - \mu)}{kT}\right) \\
&= \sum_{n_x} \sum_{n_y} \sum_{n_z} - \frac{\epsilon e^{\frac{(\epsilon-\mu)}{kT}}}{\left(e^{\frac{(\epsilon-\mu)}{kT}} - 1\right)^2} \frac{\partial}{\partial T} \left(\frac{\left(\epsilon - \epsilon_0 + \frac{kT}{\left(1 - \left(\frac{T}{T_c}\right)^{\frac{3}{2}}\right)N} \right)}{kT} \right) \\
&= \sum_{n_x} \sum_{n_y} \sum_{n_z} - \frac{\epsilon e^{\frac{(\epsilon-\mu)}{kT}}}{\left(e^{\frac{(\epsilon-\mu)}{kT}} - 1\right)^2} \frac{\partial}{\partial T} \left(\frac{(\epsilon - \epsilon_0)}{kT} + \frac{1}{\left(1 - \left(\frac{T}{T_c}\right)^{\frac{3}{2}}\right)N} \right) \\
&= \sum_{n_x} \sum_{n_y} \sum_{n_z} - \frac{\epsilon e^{\frac{(\epsilon-\mu)}{kT}}}{\left(e^{\frac{(\epsilon-\mu)}{kT}} - 1\right)^2} \left(-\frac{(\epsilon - \epsilon_0)}{kT^2} + \frac{\partial}{\partial T} \frac{1}{\left(1 - \left(\frac{T}{T_c}\right)^{\frac{3}{2}}\right)N} \right) \\
&= \sum_{n_x} \sum_{n_y} \sum_{n_z} - \frac{\epsilon e^{\frac{(\epsilon-\mu)}{kT}}}{\left(e^{\frac{(\epsilon-\mu)}{kT}} - 1\right)^2} \left(-\frac{(\epsilon - \epsilon_0)}{kT^2} - \frac{1}{\left(1 - \left(\frac{T}{T_c}\right)^{\frac{3}{2}}\right)^2 N} \frac{\partial}{\partial T} \left(1 - \left(\frac{T}{T_c}\right)^{\frac{3}{2}}\right) \right) \\
&= \sum_{n_x} \sum_{n_y} \sum_{n_z} - \frac{\epsilon e^{\frac{(\epsilon-\mu)}{kT}}}{\left(e^{\frac{(\epsilon-\mu)}{kT}} - 1\right)^2} \left(-\frac{(\epsilon - \epsilon_0)}{kT^2} + \frac{1}{\left(1 - \left(\frac{T}{T_c}\right)^{\frac{3}{2}}\right)^2} \frac{\partial}{\partial T} \left(\frac{3}{2} \sqrt{T}\right) \right)
\end{aligned}$$

$$= \sum_{n_x} \sum_{n_y} \sum_{n_z} \frac{\epsilon e^{\frac{(\epsilon-\mu)}{kT}}}{\left(e^{\frac{(\epsilon-\mu)}{kT}} - 1\right)^2} \left(\frac{(\epsilon - \epsilon_0)}{kT^2} - \frac{1}{\left(1 - \left(\frac{T}{T_c}\right)^{\frac{3}{2}}\right)^2 NT_c^{\frac{3}{2}}} \left(\frac{3}{2}\sqrt{T}\right) \right)$$

Where for $T < T_c$

$$\epsilon = \frac{h^2}{8mL^2} (n_x^2 + n_y^2 + n_z^2)$$

$$\mu = \epsilon_0 - \frac{kT}{\left(1 - \left(\frac{T}{T_c}\right)^{\frac{3}{2}}\right) N}$$

Yes it is a very ugly expression but nothing more can be done

If you wanted to you could make the approximation that μ is approximately constant for $T \ll T_c$ and then you would simply get

$$\begin{aligned} C_V &= \sum_{n_x} \sum_{n_y} \sum_{n_z} - \frac{\epsilon e^{\frac{(\epsilon-\mu)}{kT}}}{\left(e^{\frac{(\epsilon-\mu)}{kT}} - 1\right)^2} \frac{\partial}{\partial T} \left(\frac{(\epsilon - \mu)}{kT} \right) \\ &\approx \sum_{n_x} \sum_{n_y} \sum_{n_z} \frac{\epsilon e^{\frac{(\epsilon-\mu)}{kT}}}{\left(e^{\frac{(\epsilon-\mu)}{kT}} - 1\right)^2} \left(\frac{(\epsilon - \mu)}{kT^2} \right) \end{aligned}$$

This is admittedly a lot nicer and should not be too far away from the other result. A quick and dirty graphical analysis shows that

$$\frac{1}{\left(1 - \left(\frac{T}{T_c}\right)^{\frac{3}{2}}\right)^2 NT_c^{\frac{3}{2}}} \left(\frac{3}{2}\sqrt{T}\right) \approx 0, \quad \text{for } T \ll T_c$$

Just look at the graph below and be convinced so I would say the exact results and the approximation is okay.

